Report prepared for the WHO Consultation on the Composition of Influenza Virus Vaccines for the Northern Hemisphere 2025-26

Worldwide Influenza Centre, WHO CC for Reference and Research on Influenza
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Vaccine Composition Meeting: London, 24th - 27th February 2025

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Laboratories involved in Global Influenza Surveillance

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Additional Personnel from the Francis Crick Institute

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Influenza Epidemiology in the WHO European Region: The epidemiological section of this report was prepared using data reported to the European Surveillance System (TESSy) and accessed through the European Respiratory Virus Surveillance Summary (ERVISS).

Contents

Global influenza activity

Influenza by type/subtype

Global maps for influenza viruses with collection dates from 1st September to 30th November 2024 (Figure 1) and from 1st December 2024 to 7th February 2025 (Figure 2) as deposited in GISAID, coloured by Type/subtype and downloaded on 14th February 2025. Map prepared with Microreact. Geographic markers scaled to detection proportions. Full length HA. Overview of influenza activity summarised from GISRS Influenza updates.

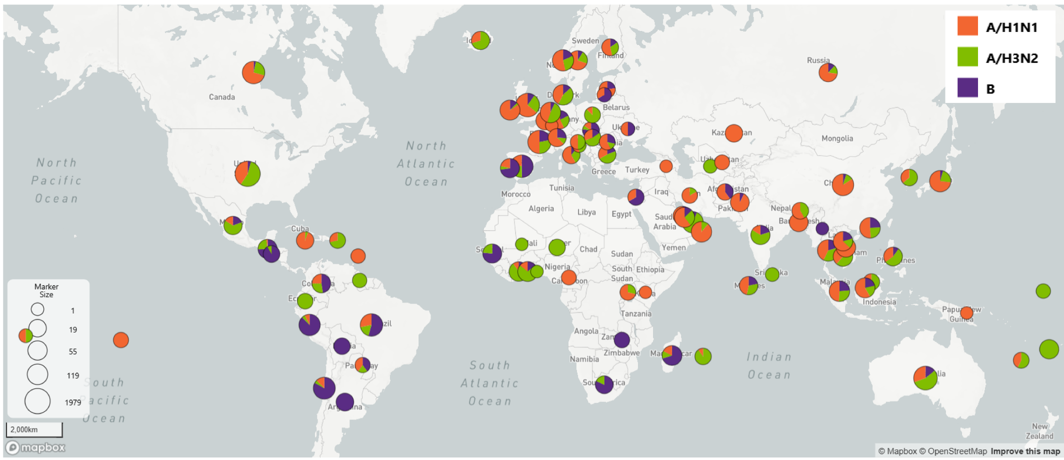


Figure 1: Global distribution of influenza virus subtypes from 1st September to 30th November 2024

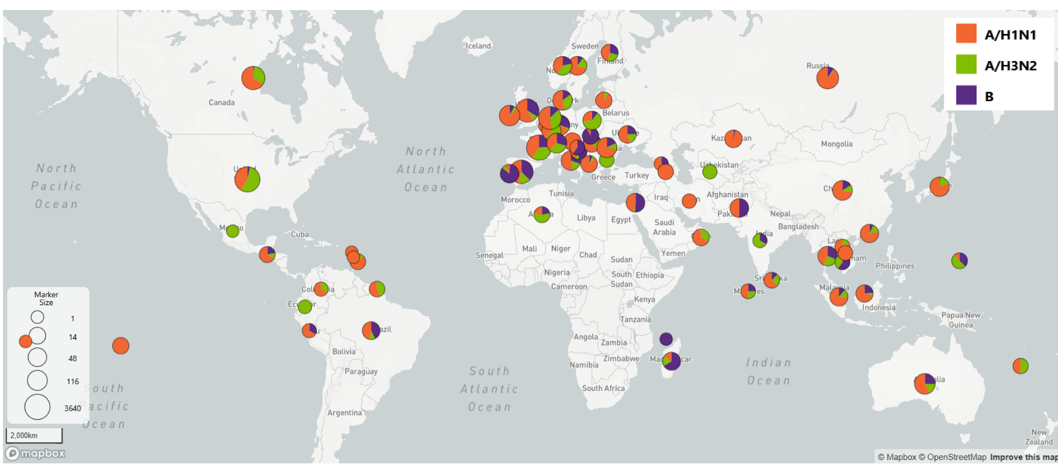


Figure 2: Global distribution of influenza virus subtypes from 1st December 2024 to 4th February 2025

As of February 2025, in the Northern hemisphere influenza activity remained elevated in many regions including North and Central America, the Caribbean, Tropical South America, Northern Africa, Europe and Asia. Countries in North America, North Africa, Europe and Western Asia reported increased activity.

In the Southern hemisphere, activity remained elevated in a few countries in Eastern Africa, South-East Asia and Oceania.

The relative proportions of A/H1N1, A/H3N2 and B/Victoria varied by geographic region with cocirculation of A/H1N1 and A/H3N2 overall and some predominance of B/Victoria. During December 2024 and January 2025

(Figure 2), subtype A/H1N1 predominated in most countries in Europe, the Caribbean and Tropical South America, Canada, South East Asia, West Asia, the Middle East and Oceania. Subtype A/H3N2 predominated in some countries: US, Poland, Bulgaria, Luxembourg and India. Some regions and countries showed predominance of B/Victoria such as Madagascar, Portugal, Slovenia, Croatia and Cambodia, as indicated by the different colours in the pie charts by country.

Shipments received at the WIC

During the current influenza season we received 1943 samples (1743 patients and 200 duplicates) in 68 packages from 56 NICs in 49 countries.

Table of shipments to the WIC

		Influenza Type		Subtype/lineage			
Country	Number of samples received	A	B	A/H1N1	A/H3N2	B/Victoria	Number of shipments
Algeria	10	1		3	4	2	1
Belgium	22			12	9	1	1
Bulgaria	29			8	17	4	1
Cameroon	50			14	36		1
Cote d'Ivoire	59			5	46	8	1
Cyprus	30			19	10	1	1
Denmark	21			4	11	6	2
Egypt	3				3		1
France	130		15	53	45	17	3
Georgia	20		2	13	1	4	1
Germany	56			25	13	18	3
Ghana	40			5	20	15	2
Greece	33	1		18	9	5	2
Hong Kong	30			10	10	10	1
Hungary	18			8	5	5	2
Iceland	32			10	22		1
Iran	19		4	10	5		1
Ireland	59		3	50	5	1	2
Israel	60			27	1	32	1
Italy	11		1	8	2		1
Kazakhstan	40		2	38			1
Kosovo	30			5	20	5	1
Latvia	10		1	7		2	1
Luxembourg	22	1	6	8	7		1
Madagascar	50		6	5	5	34	1
Mauritius	33		1	5	27		1
Montenegro	25		1	21	3		1
Morocco	20			18		2	1
Netherlands	44			12	30	2	4
Norway	81			42	21	18	2
Oman	18			7	6	5	1
Palestine	21				19	2	1
Poland	66		7	18	41		1
Portugal	21			1	4	16	1
Qatar	100		16	59	25		1
Romania	25			19	3	3	1
Russia	25				14	11	1
Senegal	30				14	16	1

(continued)

Country	Number of samples received	Influenza Type		Subtype/lineage			Number of shipments
		A	B	A/H1N1	A/H3N2	B/Victoria	
Serbia	18			18			1
Slovakia	16		6	2		8	1
Slovenia	10		1	4	3	2	1
South Sudan	21			6	11	4	1
Spain	186	4	1	35	40	106	4
Sweden	16			13	2	1	1
Switzerland	37			12	12	13	2
Togo	136			7	55	74	1
Turkmenistan	3				3		1
Ukraine	29	1		2	5	21	1
United Kingdom	78	2		51	24	1	3
TOTALS	1943	10	73	717	668	475	68

Epidemiology

WHO European Region Summary, Week 06/2025 (3–9 February 2025)

Overview

Rates of influenza-like illness (ILI) and/or acute respiratory infection (ARI) are elevated above baseline levels in 28 of 31 countries and areas of the WHO European Region reporting data this week.

Influenza percent positivity and number of ILI cases presenting to both primary and secondary care at the Regional level appears to be peaking or have peaked, though there is substantial variation at the country/area level. However, levels remain high and percent positivity among children aged less than 5 years hospitalized for influenza is still increasing. Influenza A(H1) is the dominant subtype across the Region.

RSV positivity among primary care sentinel and SARI patients remains moderate but continues to plateau after peaking in late December.

SARS-CoV-2 activity remains low at the Regional level, with some variation at the country level.

Respiratory virus activity

Rates of influenza-like illness are elevated above baseline in 26 countries and rates of acute respiratory infection are elevated in 12 countries. Across the Region, the percentage of specimens from patients presenting to sentinel ILI and ARI sites that tested positive for:

- Influenza remained above the 10% epidemic threshold at 44% compared to 47% in the prior week. The median country positivity rate for 27 countries and areas was 41% (range: 0%–95%) and 23 countries and areas reported at least 10%.

- SARS-CoV-2 was 1% compared to 2% in the prior week. The median positivity rate for 26 countries and areas was 1% (range: 0%–8%).

- RSV was 6% compared to 5% in the prior week. The median positivity rate for 24 countries and areas was 6% (range: 0%–18%).

Influenza virus detections with known type reported from sentinel primary care for the past week (n=1,852) were mainly influenza type A viruses (59%) and, among the A viruses subtyped (n=850), there was a mix of A(H1) (56%) and A(H3) (44%).

Of 35 countries and areas reporting on influenza intensity: 29 reported intensity of medium or higher.

Of 34 countries and areas reporting on influenza geographical spread: 31 reported regional or widespread distribution of influenza.

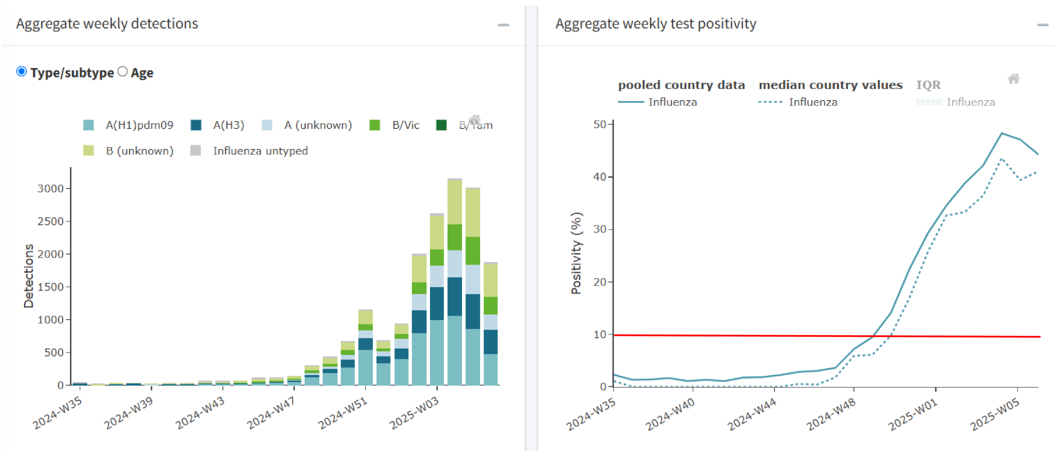


Figure 3: Aggregate weekly sentinel influenza detections and test positivity from week 35 (1 September 2024) to week 6 (3 - 9 February 2025)

Severity of infection

Across the Region, the percentage of all specimens from patients presenting to sentinel SARI sites that tested positive for:

- Influenza virus was 30% compared to 33% in the prior week. Median positivity from 14 countries and areas reporting data was 17% (range: 0%–91%). Among the SARI specimens, there were 701 influenza virus detections reported from the network this week, which were mostly influenza type A (77%). Of the 102 that were subtyped, 65% were A(H1) and 35% were A(H3).

- SARS-CoV-2 was 1% compared to 2% in the prior week. Median positivity from 12 countries and areas reporting data was 0% (range: 0%–4%).

- RSV was 10% compared to 9% in the prior week. Median positivity from 11 countries and areas reporting data was 8% (range: 0%–26%).

COVID-19 death rates remained low among the 40 countries reporting data.

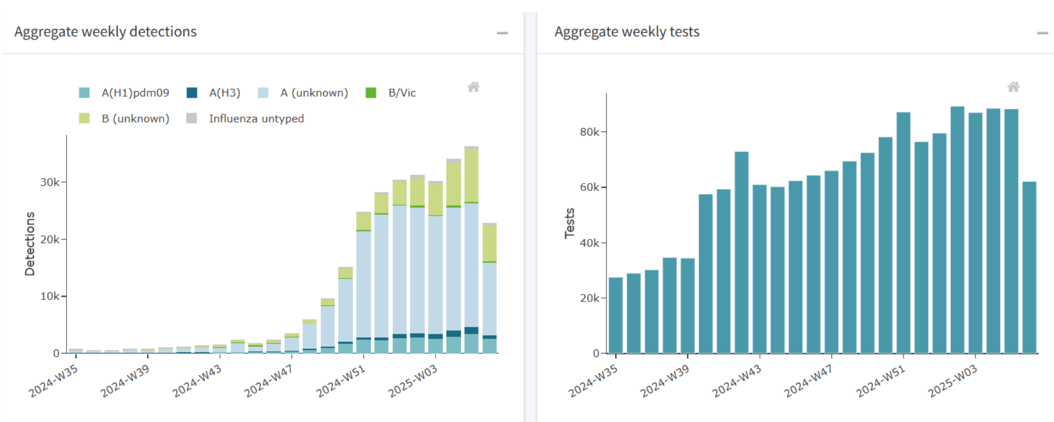


Figure 4: Aggregate weekly non-sentinel influenza detections and tests from week 35 (1 September 2024) to week 6 (3 - 9 February 2025)

Influenza genetic diversity by WHO region

Genetic diversity by Type/Lineage and clade

Global and regional genetic diversity by number of samples genetically characterised as belonging to each clade (subclade). Please note differences in X axis scale across figures.

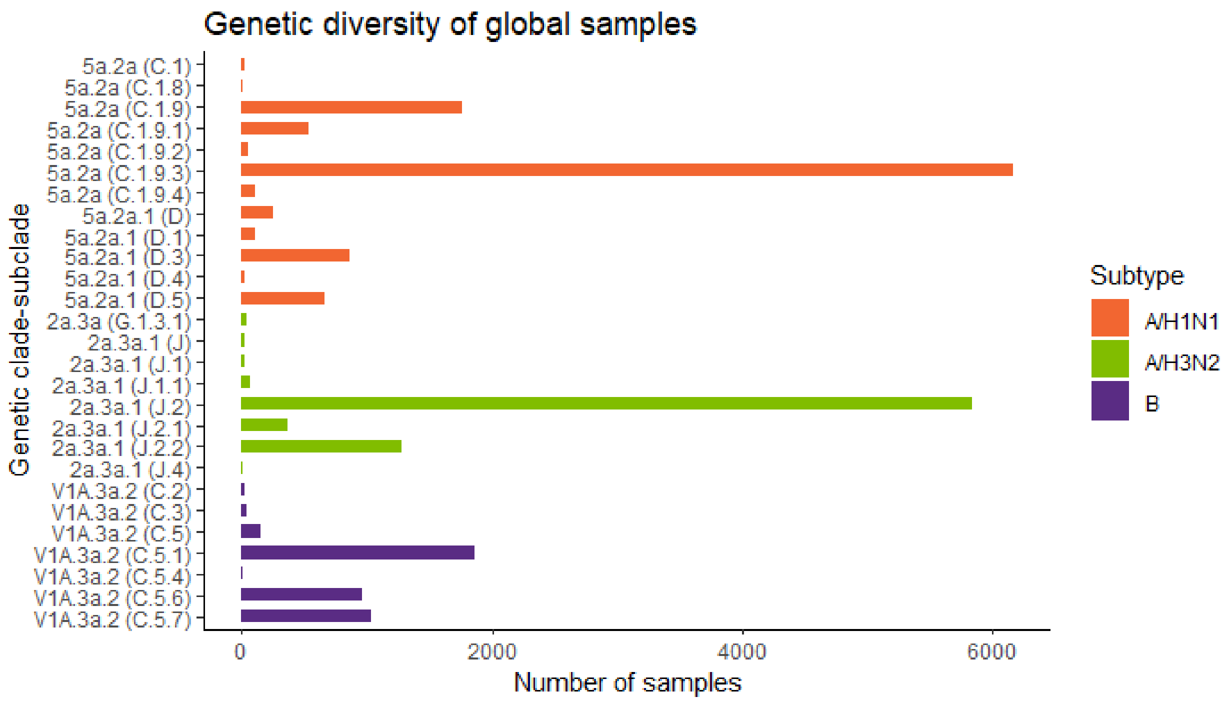
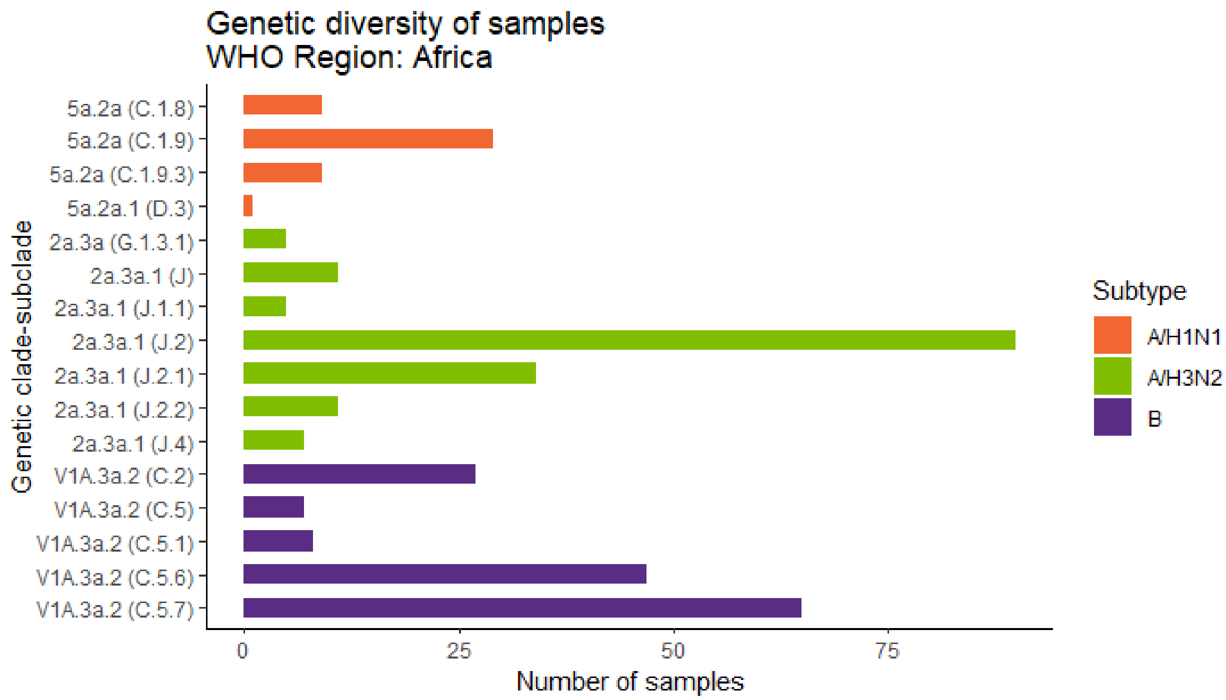
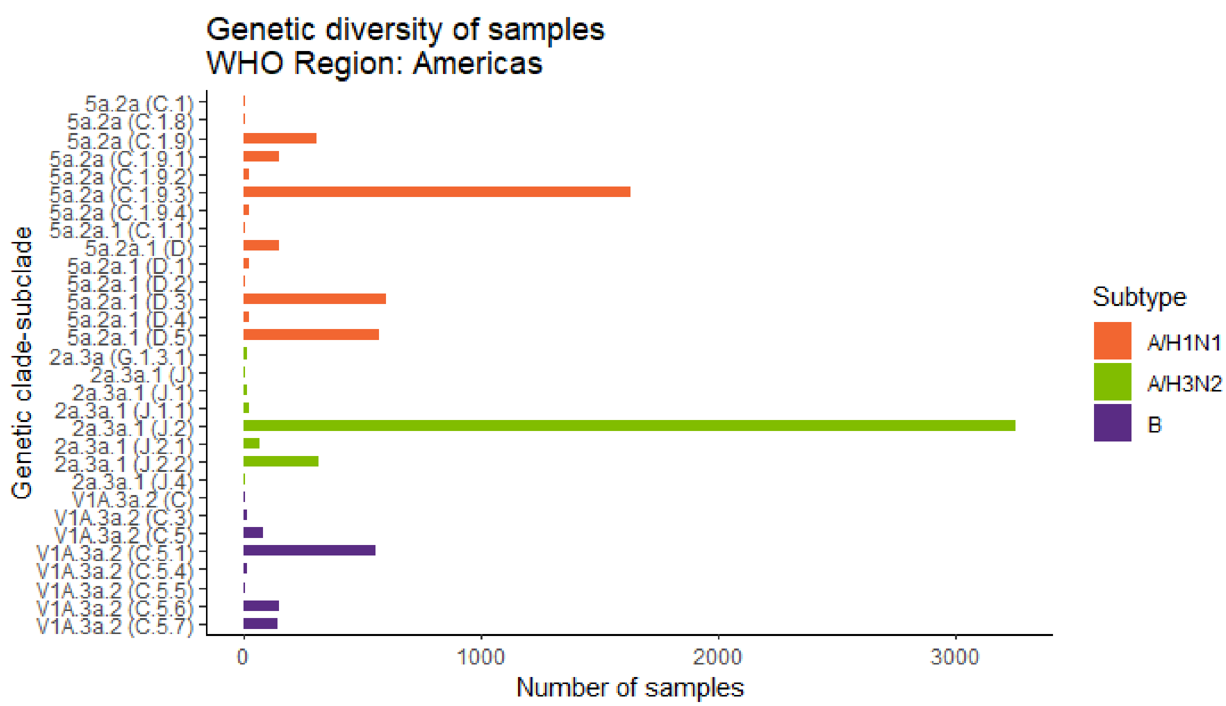
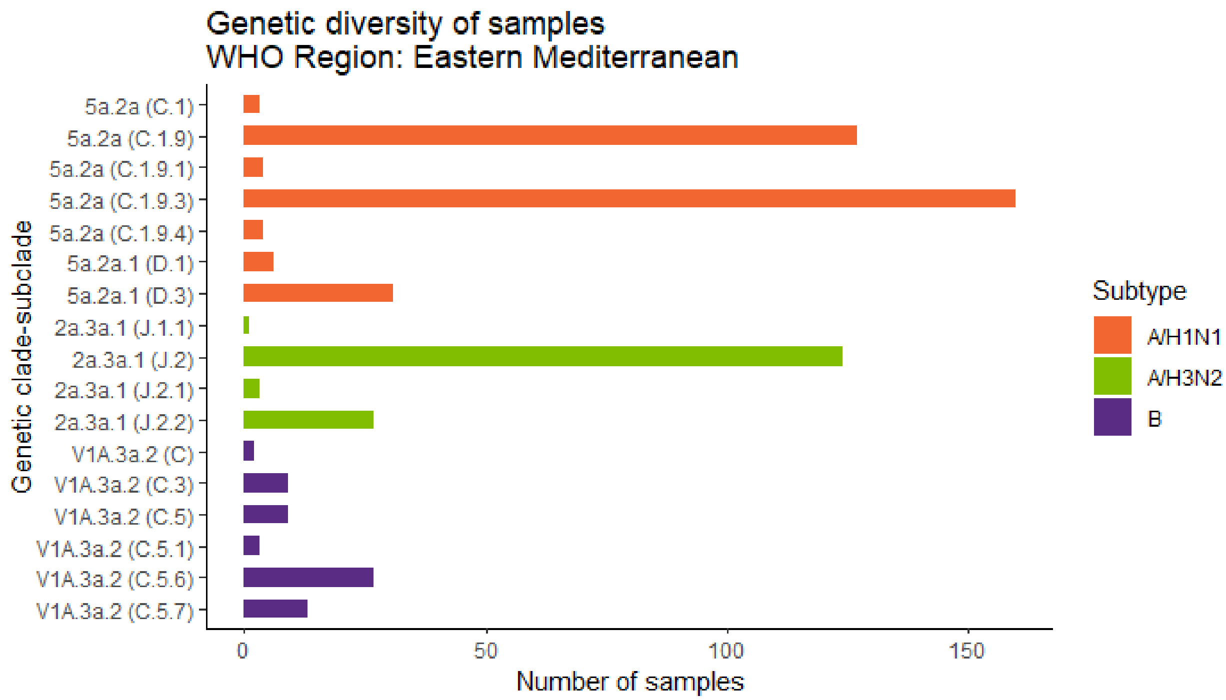
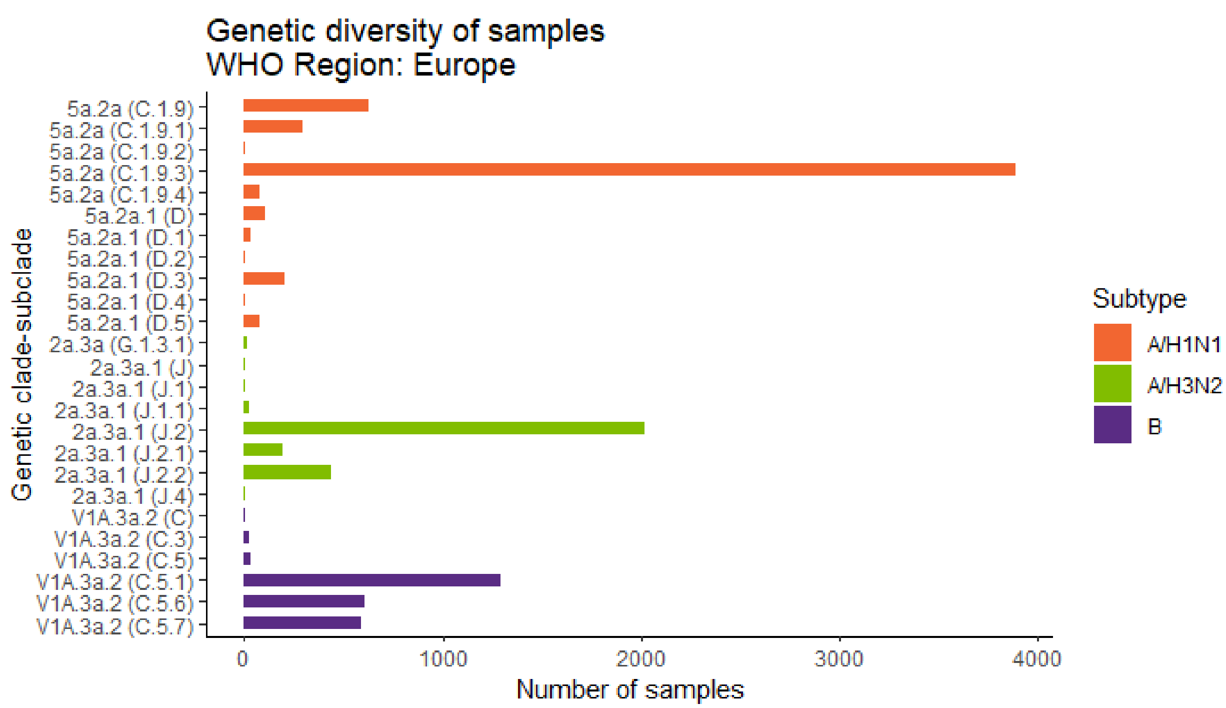


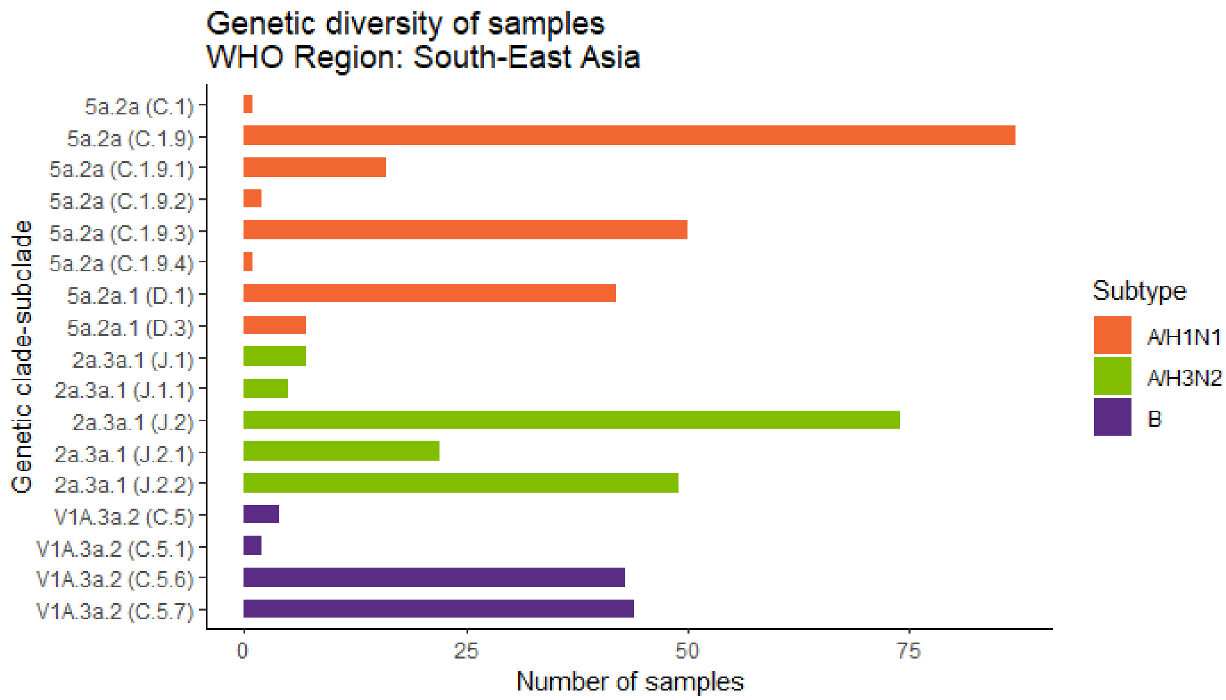
Figure 5: Clade-subclade proportions by WHO region


Genetic diversity of samples
WHO Region: Africa
Genetic clade-subclade
5a.2a (C.1.8)
5a.2a (C.1.9)
5a.2a (C.1.9.3)
5a.2a.1 (D.3)
2a.3a (G.1.3.1)
2a.3a.1 (J)
2a.3a.1 (J.1.1)
2a.3a.1 (J.2)
2a.3a.1 (J.2.1)
2a.3a.1 (J.2.2)
2a.3a.1 (J.4)
V1A.3a.2 (C.2)
V1A.3a.2 (C.5)
V1A.3a.2 (C.5.1)
V1A.3a.2 (C.5.6)
V1A.3a.2 (C.5.7)
0
25
50
75
Number of samples
Subtype
A/H1N1
A/H3N2
B

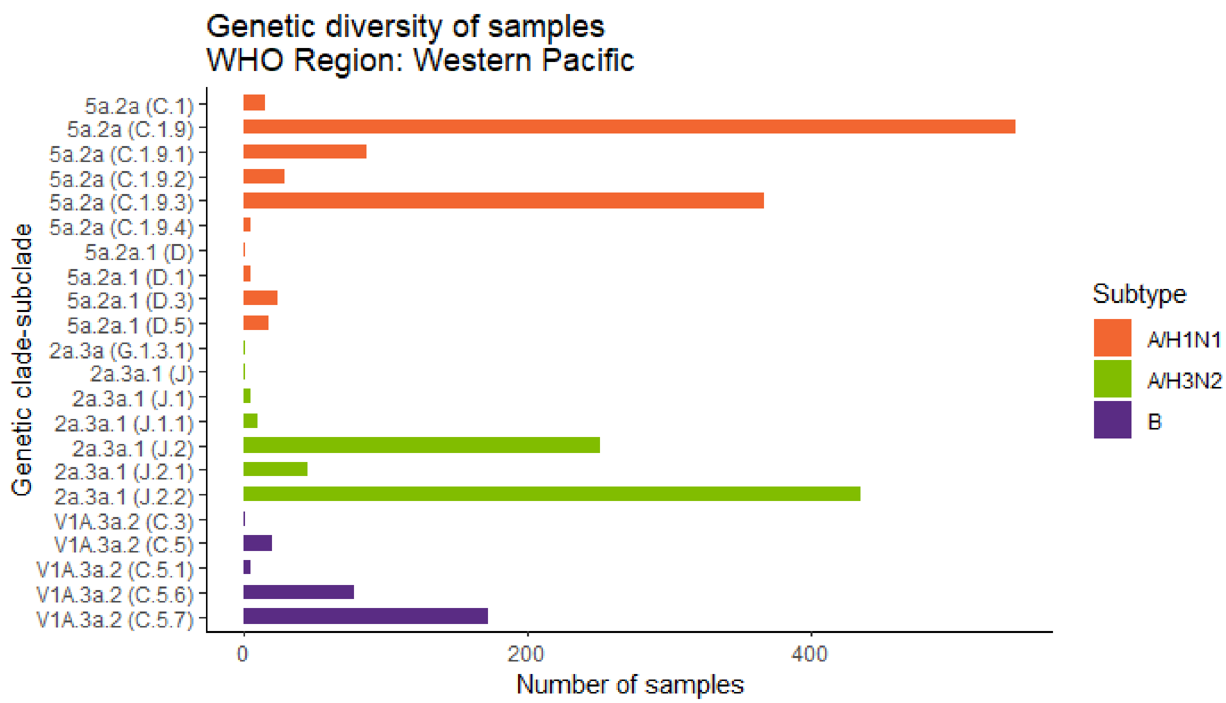

Genetic diversity of samples
WHO Region: Americas
Genetic clade-subclade
5a.2a (C.1)
5a.2a (C.1.8)
5a.2a (C.1.9)
5a.2a (C.1.9.1)
5a.2a (C.1.9.2)
5a.2a (C.1.9.3)
5a.2a (C.1.9.4)
5a.2a.1 (C.1.1)
5a.2a.1 (D)
5a.2a.1 (D.1)
5a.2a.1 (D.2)
5a.2a.1 (D.3)
5a.2a.1 (D.4)
5a.2a.1 (D.5)
2a.3a (G.1.3.1)
2a.3a.1 (J)
2a.3a.1 (J.1)
2a.3a.1 (J.1.1)
2a.3a.1 (J.2)
2a.3a.1 (J.2.1)
2a.3a.1 (J.2.2)
2a.3a.1 (J.4)
V1A.3a.2 (C)
V1A.3a.2 (C.3)
V1A.3a.2 (C.5)
V1A.3a.2 (C.5.1)
V1A.3a.2 (C.5.4)
V1A.3a.2 (C.5.5)
V1A.3a.2 (C.5.6)
V1A.3a.2 (C.5.7)
0
1000
2000
3000
Number of samples
Subtype
A/H1N1
A/H3N2
B


Genetic diversity of samples
WHO Region: Eastern Mediterranean
Genetic clade-subclade
5a.2a (C.1)
5a.2a (C.1.9)
5a.2a (C.1.9.1)
5a.2a (C.1.9.3)
5a.2a (C.1.9.4)
5a.2a.1 (D.1)
5a.2a.1 (D.3)
2a.3a.1 (J.1.1)
2a.3a.1 (J.2)
2a.3a.1 (J.2.1)
2a.3a.1 (J.2.2)
V1A.3a.2 (C)
V1A.3a.2 (C.3)
V1A.3a.2 (C.5)
V1A.3a.2 (C.5.1)
V1A.3a.2 (C.5.6)
V1A.3a.2 (C.5.7)
0
50
100
150
Number of samples
Subtype
A/H1N1
A/H3N2
B


Genetic diversity of samples
WHO Region: Europe
Genetic clade-subclade
5a.2a (C.1.9)
5a.2a (C.1.9.1)
5a.2a (C.1.9.2)
5a.2a (C.1.9.3)
5a.2a (C.1.9.4)
5a.2a.1 (D)
5a.2a.1 (D.1)
5a.2a.1 (D.2)
5a.2a.1 (D.3)
5a.2a.1 (D.4)
5a.2a.1 (D.5)
2a.3a (G.1.3.1)
2a.3a.1 (J)
2a.3a.1 (J.1)
2a.3a.1 (J.1.1)
2a.3a.1 (J.2)
2a.3a.1 (J.2.1)
2a.3a.1 (J.2.2)
2a.3a.1 (J.4)
V1A.3a.2 (C)
V1A.3a.2 (C.3)
V1A.3a.2 (C.5)
V1A.3a.2 (C.5.1)
V1A.3a.2 (C.5.6)
V1A.3a.2 (C.5.7)
0
1000
2000
3000
4000
Number of samples
Subtype
A/H1N1
A/H3N2
B



Genetic diversity of samples
WHO Region: South-East Asia
Genetic clade-subclade
5a.2a (C.1)
5a.2a (C.1.9)
5a.2a (C.1.9.1)
5a.2a (C.1.9.2)
5a.2a (C.1.9.3)
5a.2a (C.1.9.4)
5a.2a.1 (D.1)
5a.2a.1 (D.3)
2a.3a.1 (J.1)
2a.3a.1 (J.1.1)
2a.3a.1 (J.2)
2a.3a.1 (J.2.1)
2a.3a.1 (J.2.2)
V1A.3a.2 (C.5)
V1A.3a.2 (C.5.1)
V1A.3a.2 (C.5.6)
V1A.3a.2 (C.5.7)
0
25
50
75
Number of samples
Subtype
A/H1N1
A/H3N2
B



Genetic diversity of samples
WHO Region: Western Pacific
Genetic clade-subclade
5a.2a (C.1)
5a.2a (C.1.9)
5a.2a (C.1.9.1)
5a.2a (C.1.9.2)
5a.2a (C.1.9.3)
5a.2a (C.1.9.4)
5a.2a.1 (D)
5a.2a.1 (D.1)
5a.2a.1 (D.3)
5a.2a.1 (D.5)
2a.3a (G.1.3.1)
2a.3a.1 (J)
2a.3a.1 (J.1)
2a.3a.1 (J.1.1)
2a.3a.1 (J.2)
2a.3a.1 (J.2.1)
2a.3a.1 (J.2.2)
V1A.3a.2 (C.3)
V1A.3a.2 (C.5)
V1A.3a.2 (C.5.1)
V1A.3a.2 (C.5.6)
V1A.3a.2 (C.5.7)
0
200
400
Number of samples
Subtype
A/H1N1
A/H3N2
B

Influenza A/H1N1

Genetic analyses

Global distribution of A/H1N1 subclades

Global geographical distribution of influenza A/H1N1 genetic clades (subclades) viruses with collection dates from 1st November 2024 through to 7th February 2025, obtained with full HA sequences as deposited in GISAID, downloaded on 14th February 2025 and classified using Nextclade. Map prepared with Microreact. Geographic markers scaled to detection proportions.

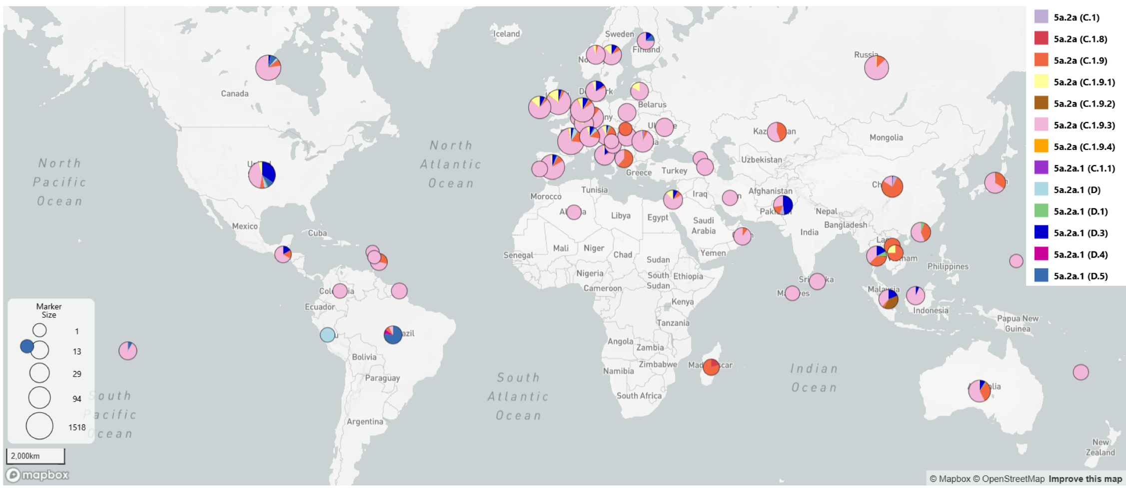


Figure 6: Global distribution of A/H1N1 genetic subclades

Global time-dependent variation of A/H1N1 subclades

Global time-dependent variation in frequencies of genetic clades-subclades of A/H1N1 viruses collected from 1st September 2024 through to 7th February 2025.

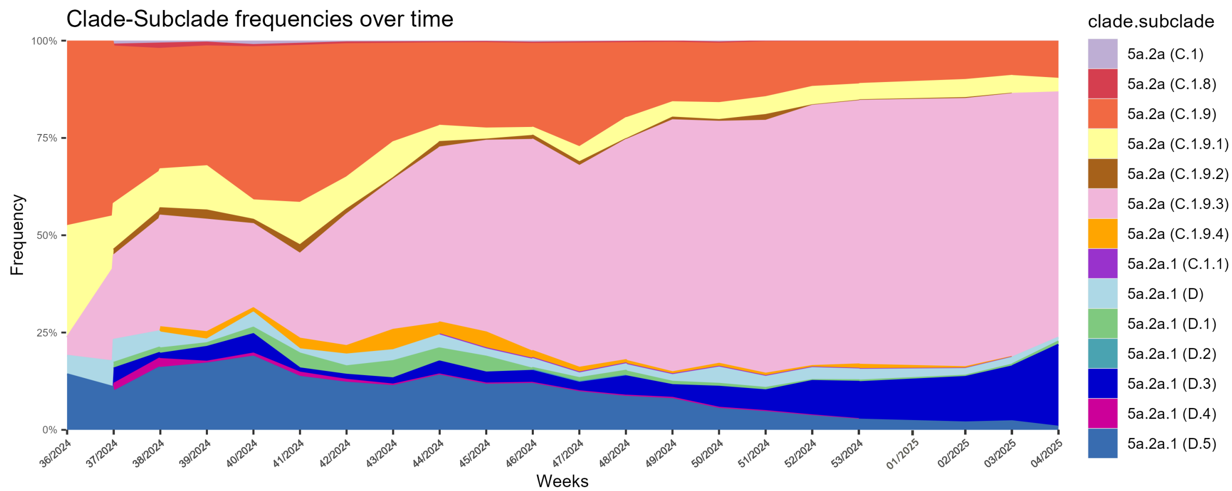


Figure 7: A/H1N1 clade-subclade frequencies over time (weeks 36/2024 to 4/2025)

Phylogenetic analysis of A/H1N1 sequences

Maximum likelihood phylogenetic tree inferred using IQTree2 with HA sequence data obtained from GISAID, collection dates from 1st November 2024 to 7th February 2025, manually curated and then downsampled using Treemmer to retain a representative tree topology of 200 sequences, with emphasis on sequences generated at the WIC. Amino acid signatures annotated with Treetime. References and CVVs are marked as Cell or Egg.

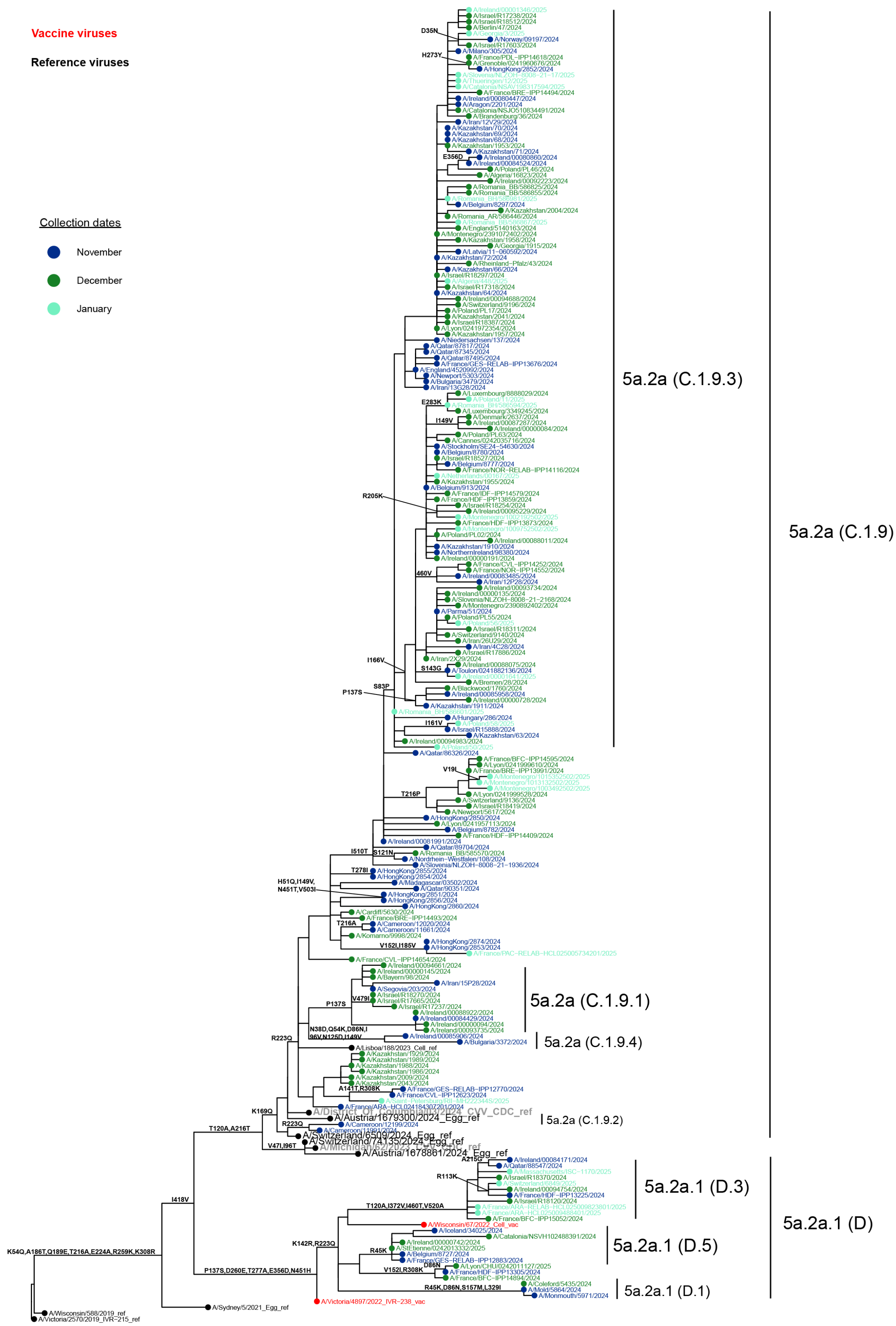


Figure 8: A/H1N1 Maximum likelihood phylogenetic tree, HA segment, collection dates from 1st November 2024 to 7th February 2025

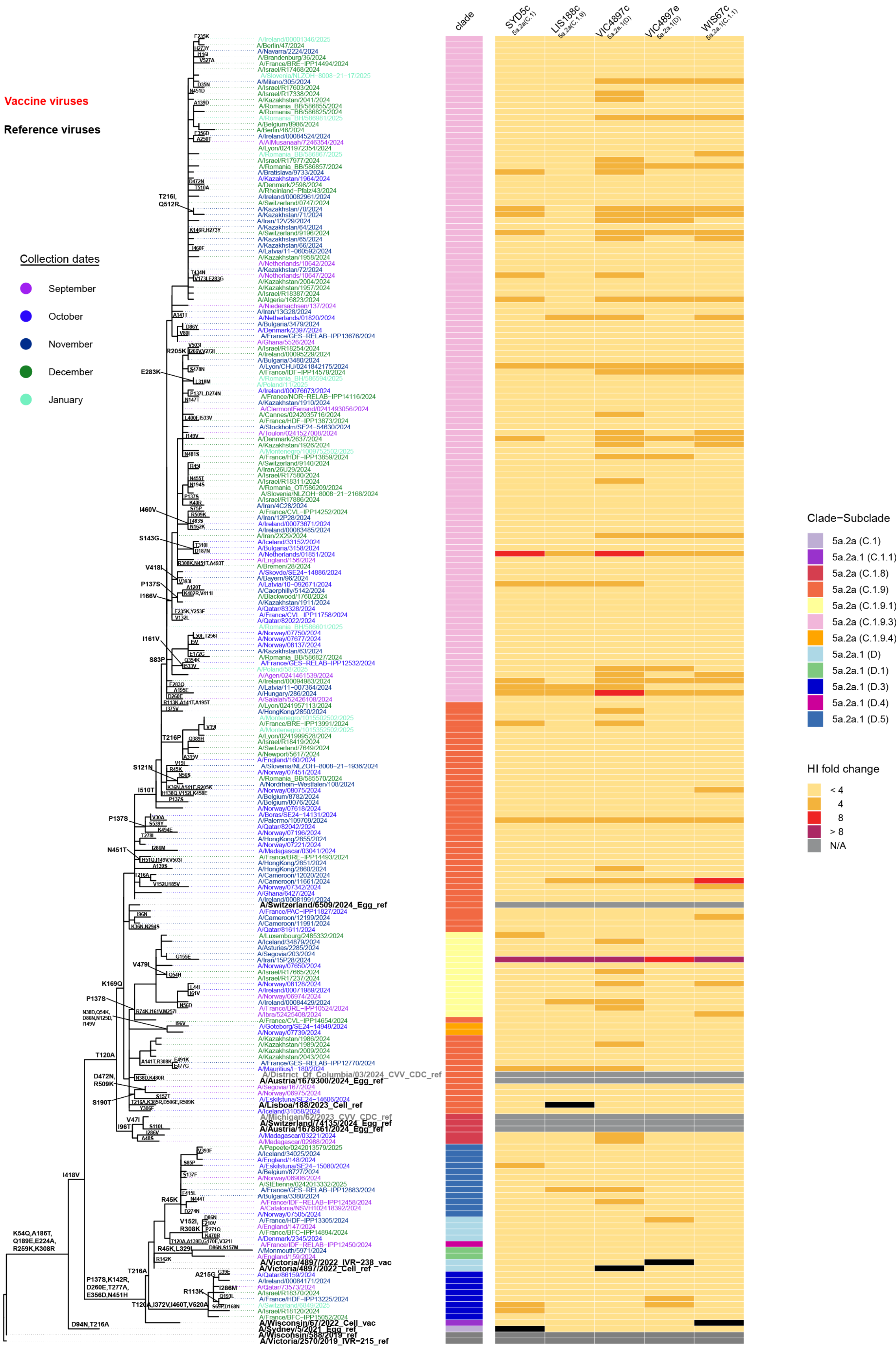


Figure 9: ML phylogenetic tree with combined heatmap representing antigenic reactivities by HI

Antigenic analyses: A/H1N1

Haemagglutination inhibition tables: A/H1N1

Table H1-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-10-18)

	Viruses	Other information	Genetic group	Collection Date	Passage / Ferret number / Genetic group	Haemagglutination inhibition titre — Post-infection ferret antiserum: A/Sydney 5/21 MDCK F46/22 5a.2a (C.1)	A/Lisboa/ 188/2023 MDCK F09/24 5a.2a (C.1.9)	A/Victoria/ 4897/2022 MDCK F05/23 5a.2a.1 (D)	IVR-238 A/Vic/4897/22 Egg F07/23 5a.2a.1 (D)	A/Wisconsin/ 67/2022 MDCK F17/23 5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	CK3/MDCK4	2560	2560	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	IAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	IAT2/MDCK3	2560	2560	5120	2560	2560
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	1280	2560	640
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	2560	2560	2560
	TEST VIRUSES									
244912	A/SouthSudan/1026/2024		5a.2a (C.1)	2024-06-21	SIAT2	5120	2560	5120	5120	2560
245017	A/Iceland/12972/2024		5a.2a (C.1.1)	2024-04-24	MDCK3	2560	2560	5120	2560	2560
244924	A/SouthSudan/1036/2024		5a.2a (C.1.9)	2024-05-17	SIAT2	2560	2560	2560	2560	2560
244922	A/SouthSudan/1038/2024		5a.2a (C.1.9)	2024-06-10	SIAT1	5120	2560	5120	2560	2560
250018	A/Ghana/3877/2024		5a.2a (C.1.9)	2024-07-09	MDCK1	5120	2560	2560	2560	1280
250017	A/Ghana/3905/2024		5a.2a (C.1.9)	2024-07-15	MDCK1	5120	2560	2560	2560	1280
245003	A/Iceland/14025/2024		5a.2a (C.1.9.2)	2024-05-05	MDCK3	2560	2560	2560	2560	1280
250016	A/Ghana/5526/2024		5a.2a (C.1.9.3)	2024-09-06	MDCK1	2560	2560	5120	2560	1280
						Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< relates to the lowest dilution of antiserum used
[1] hyperimmune sheep serum; ND =

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum

Table H1-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-11-29)

	Viruses	Other information	Genetic group	Collection Date	Passage	Haemagglutination inhibition titre – Post-infection ferret antiserum				
						A/Sydney 5/21 MDCK	A/Lisboa/ 188/2023 MDCK	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
					Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23
					Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	1280	5120	2560	2560
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	1280	2560	640
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	2560	2560
	TEST VIRUSES									
250126	A/Norway/06975/2024		5a.2a (C.1.9)	2024-09-17	MDCK1	2560	2560	2560	2560	1280
250120	A/Norway/07221/2024		5a.2a (C.1.9)	2024-10-02	MDCK1	5120	2560	2560	2560	1280
250119	A/Norway/07342/2024		5a.2a (C.1.9)	2024-10-03	MDCK1	2560	1280	2560	2560	640
250118	A/Norway/07196/2024		5a.2a (C.1.9)	2024-10-06	MDCK1	2560	2560	2560	2560	1280
250115	A/Norway/07451/2024		5a.2a (C.1.9)	2024-10-09	MDCK1	2560	2560	2560	2560	1280
250114	A/Norway/07618/2024		5a.2a (C.1.9)	2024-10-09	MDCK1	2560	2560	2560	2560	1280
250106	A/Norway/08075/2024		5a.2a (C.1.9)	2024-10-24	MDCK1	2560	1280	2560	2560	640
250124	A/Norway/06974/2024		5a.2a (C.1.9.1)	2024-09-19	MDCK1	5120	2560	5120	5120	2560
250111	A/Norway/07650/2024		5a.2a (C.1.9.1)	2024-10-14	MDCK1	2560	5120	5120	2560	2560
250103	A/Norway/08128/2024		5a.2a (C.1.9.1)	2024-10-28	MDCK1	2560	1280	1280	2560	640
250117	A/Norway/07382/2024		5a.2a (C.1.9.3)	2024-10-07	MDCK1	2560	2560	2560	2560	1280
250116	A/Norway/07557/2024		5a.2a (C.1.9.3)	2024-10-07	MDCK1	5120	2560	2560	2560	1280
250113	A/Norway/07606/2024		5a.2a (C.1.9.3)	2024-10-12	MDCK2	2560	2560	2560	2560	1280
250110	A/Norway/07750/2024		5a.2a (C.1.9.3)	2024-10-16	MDCK1	5120	2560	5120	2560	1280
250109	A/Norway/07677/2024		5a.2a (C.1.9.3)	2024-10-18	MDCK1	2560	2560	2560	2560	1280
250101	A/Norway/08137/2024		5a.2a (C.1.9.3)	2024-10-31	MDCK1	2560	5120	2560	2560	1280
250071	A/Catalonia/NSVH102435379/2024		5a.2a (C.1.9.4)	2024-09-28	MDCK2	2560	2560	2560	1280	1280
250108	A/Norway/07739/2024		5a.2a (C.1.9.4)	2024-10-18	MDCK1	2560	2560	2560	2560	1280
250073	A/Catalonia/NSVH102418392/2024		5a.2a.1 (D)	2024-09-03	SIAT1/MDCK1	2560	1280	5120	2560	2560
250122	A/Norway/06906/2024		5a.2a.1 (D.5)	2024-09-23	MDCK2	1280	1280	2560	1280	1280
250112	A/Norway/07505/2024		5a.2a.1 (D.5)	2024-10-13	MDCK2	1280	1280	2560	2560	1280
						Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< relates to the lowest dilution of antiserum used

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-12-10)

	Viruses	Other information	Genetic group	Collection Date	Passage	A/Sydney 5/21 MDCK	A/Lisboa/ 188/2023 MDCK	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
						Haemagglutination inhibition titre				
						Post-infection ferret antiserum				
					Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23
					Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	)CK3/MDCK4	2560	2560	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	;IAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	;IAT2/MDCK3	2560	1280	5120	2560	2560
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	2560	2560	640
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	2560	2560
	TEST VIRUSES									
250169	A/Ibra/52425408/2024		5a.2a (C.1.9.1)	2024-09-08	MDCK1	1280	1280	2560	2560	640
250172	A/Salalah/52423942/2024		5a.2a (C.1.9.3)	2024-08-21	MDCK1	2560	2560	1280	2560	640
250171	A/Salalah/52423941/2024		5a.2a (C.1.9.3)	2024-08-24	MDCK1	5120	2560	2560	2560	1280
250167	A/Salalah/52426108/2024		5a.2a (C.1.9.3)	2024-09-13	MDCK1	2560	2560	2560	2560	1280
250166	A/AlMusanaah/7246354/2024		5a.2a (C.1.9.3)	2024-09-22	MDCK1	2560	2560	2560	2560	1280
						Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< relates to the lowest dilution of antiserum used

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-12-18)

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre: Post-infection ferret antiserum				
						A/Sydney 5/21	A/Lisboa/ 188/2023	A/Victoria/ 4897/2022	IVR-238 A/Vic/4897/22	A/Wisconsin/ 67/2022
					Passage	MDCK	MDCK	MDCK	Egg	MDCK
					Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23
					Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	CK3/MDCK4	2560	2560	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	IAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	IAT2/MDCK3	2560	2560	5120	2560	2560
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	2560	1280	2560	2560	640
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	2560	1280	1280
	TEST VIRUSES									
250181	A/Niedersachsen/137/2024		5a.2a (C.1.9.3)	2024-09-24	P2/MDCK1	5120	2560	2560	2560	1280
250182	A/Niedersachsen/136/2024		5a.2a (C.1.9.3)	2024-10-11	P2/MDCK1	2560	2560	2560	2560	1280
250183	A/Saarland/19/2024		5a.2a (C.1.9.3)	2024-10-21	P1/MDCK1	2560	2560	2560	2560	1280
250184	A/Schleswig-Holstein/50/2024		5a.2a (C.1.9.3)	2024-11-07	P1/MDCK1	5120	2560	5120	2560	1280
	< relates to the lowest dilution of antiserum used					Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2025-01-07)

	Viruses	Other information	Genetic group	Collection Date	Passage	Haemagglutination inhibition titre – Post-infection ferret antiserum: A/Sydney 5/21	A/Lisboa/ 188/2023	A/Victoria/ 4897/2022	IVR-238 A/Vic/4897/22	A/Wisconsin/ 67/2022
					Passage	MDCK	MDCK	MDCK	Egg	MDCK
					Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23
					Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	1280	2560	2560	640
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	5120	2560	1280
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	1280	2560	640
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	2560	1280	1280
	TEST VIRUSES									
250471	A/Mauritius/I-180/2024		5a.2a (C.1.9)	2024-10-23	MDCKx/MDCK1	640	640	1280	1280	640
250440	A/England/160/2024		5a.2a (C.1.9)	2024-10-24	MDCK1/MDCK1	2560	2560	2560	2560	1280
250409	A/HongKong/2853/2024		5a.2a (C.1.9)	2024-11-02	MDCK 1/MDCK1	1280	640	1280	1280	640
250407	A/HongKong/2860/2024		5a.2a (C.1.9)	2024-11-06	MDCK 1/MDCK1	1280	1280	1280	1280	640
250405	A/HongKong/2850/2024		5a.2a (C.1.9)	2024-11-07	MDCK 1/MDCK1	1280	1280	1280	1280	640
250401	A/HongKong/2854/2024		5a.2a (C.1.9)	2024-11-08	MDCK 1/MDCK1	2560	2560	2560	2560	640
250399	A/HongKong/2855/2024		5a.2a (C.1.9)	2024-11-10	MDCK 1/MDCK1	2560	2560	2560	2560	1280
250397	A/HongKong/2856/2024		5a.2a (C.1.9)	2024-11-10	MDCK 1/MDCK1	2560	1280	1280	2560	640
250395	A/HongKong/2851/2024		5a.2a (C.1.9)	2024-11-12	MDCK 1/MDCK1	2560	2560	2560	2560	1280
250391	A/HongKong/2874/2024		5a.2a (C.1.9)	2024-11-19	MDCK 1/MDCK1	1280	1280	1280	1280	640
250444	A/England/156/2024		5a.2a (C.1.9.3)	2024-09-26	SIAT1/MDCK1	2560	5120	5120	2560	1280
250288	A/Netherlands/01820/2024		5a.2a (C.1.9.3)	2024-10-09	hCK1/MDCK1	1280	640	1280	2560	320
250442	A/England/150/2024		5a.2a (C.1.9.3)	2024-10-11	SIAT1MDCK1	2560	2560	2560	2560	1280
250514	A/Latvia/10-076872/2024		5a.2a (C.1.9.3)	2024-10-20	P2/MDCK1	640	640	640	640	320
250513	A/Latvia/10-072784/2024		5a.2a (C.1.9.3)	2024-10-22	P2/MDCK1	320	320	640	640	320
250441	A/England/151/2024		5a.2a (C.1.9.3)	2024-10-23	SIAT1/MDCK1	2560	2560	2560	2560	1280
250512	A/Latvia/10-092659/2024		5a.2a (C.1.9.3)	2024-10-27	P2/MDCK1	1280	1280	1280	1280	640
250511	A/Latvia/10-092671/2024		5a.2a (C.1.9.3)	2024-10-28	P2/MDCK1	640	640	1280	1280	640
250433	A/Hungary/283/2024		5a.2a (C.1.9.3)	2024-10-29	MDCK1/MDCK1	1280	1280	1280	1280	640
250287	A/Netherlands/01851/2024		5a.2a (C.1.9.3)	2024-10-31	hCK1/MDCK1	320	640	640	640	320
250510	A/Latvia/11-007364/2024		5a.2a (C.1.9.3)	2024-11-03	P2/MDCK1	640	640	1280	1280	640
250432	A/Hungary/286/2024		5a.2a (C.1.9.3)	2024-11-05	MDCK1/MDCK1	640	640	640	640	320
250403	A/HongKong/2852/2024		5a.2a (C.1.9.3)	2024-11-08	MDCK 1/MDCK1	2560	2560	2560	2560	640
250393	A/HongKong/2873/2024		5a.2a (C.1.9.3)	2024-11-19	MDCK 1/MDCK1	1280	1280	2560	2560	1280
250509	A/Latvia/11-060592/2024		5a.2a (C.1.9.3)	2024-11-20	P2/MDCK1	1280	1280	2560	2560	640
250445	A/England/152/2024		5a.2a (C.1.9.4)	2024-09-14	SIAT1/MDCK1	2560	2560	5120	2560	1280
250434	A/Hungary/284/2024		5a.2a (C.1.9.4)	2024-10-22	MDCK1/MDCK1	2560	2560	2560	2560	1280
250447	A/England/147/2024		5a.2a.1 (D)	2024-09-11	MDCK1/MDCK1	1280	1280	5120	2560	1280
250446	A/England/159/2024		5a.2a.1 (D.1)	2024-09-12	MDCK1/MDCK1	1280	1280	2560	1280	1280
250435	A/Hungary/281/2024		5a.2a.1 (D.3)	2024-10-15	MDCK1/MDCK1	640	640	1280	640	640
250443	A/England/148/2024		5a.2a.1 (D.5)	2024-10-02	MDCK1/MDCK1	1280	1280	5120	2560	1280
250469	A/Mauritius/I-184/2024		pending	2024-10-30	MDCKx/MDCK1	2560	2560	2560	2560	1280
						Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< relates to the lowest dilution of antiserum used

Legend: < 4-fold; 4-fold; 8-fold; > 8-fold; < not recognised by the antiserum

Table H1-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2025-01-10)

	Viruses	Other information	Genetic group	Collection Date	Passage / Ferret number / Genetic group	A/Sydney 5/21 MDCK F46/22 5a.2a (C.1)	A/Lisboa/ 188/2023 MDCK F09/24 5a.2a (C.1.9)	A/Victoria/ 4897/2022 MDCK F05/23 5a.2a.1 (D)	IVR-238 A/Vic/4897/22 Egg F07/23 5a.2a.1 (D)	A/Wisconsin/ 67/2022 MDCK F17/23 5a.2a.1 (C.1.1)
						Haemagglutination inhibition titre – Post-infection ferret antiserum				
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	1280	2560	2560	640
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	5120	2560	1280
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	1280	2560	640
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	2560	1280	1280
	TEST VIRUSES									
250937	A/Segovia/167/2024		5a.2a (C.1.9)	2024-09-28	hCKx/MDCK1	2560	1280	5120	2560	2560
250305	A/Iceland/31058/2024		5a.2a (C.1.9)	2024-10-07	MDCK1	2560	2560	2560	2560	1280
250337	A/Ghana/6261/2024		5a.2a (C.1.9)	2024-10-09	MDCK1	2560	2560	2560	2560	1280
250598	A/France/PAC-IPP11827/2024		5a.2a (C.1.9)	2024-10-14	C1/MDCK1	2560	1280	2560	2560	1280
250336	A/Ghana/6427/2024		5a.2a (C.1.9)	2024-10-14	MDCK1	2560	2560	2560	2560	1280
250543	A/Lisboa/182/2024		5a.2a (C.1.9)	2024-10-31	SIAT1/SIAT1	2560	2560	5120	2560	2560
250589	A/France/CVL-IPP12623/2024		5a.2a (C.1.9)	2024-11-05	C1/MDCK1	2560	1280	1280	1280	640
250585	A/France/GES-RELAB-IPP12770/2024		5a.2a (C.1.9)	2024-11-08	C1/MDCK1	2560	1280	5120	2560	1280
250604	A/France/BRE-IPP10524/2024		5a.2a (C.1.9.1)	2024-09-14	C1/MDCK1	1280	1280	1280	1280	1280
250367	A/Ireland/00071989/2024		5a.2a (C.1.9.1)	2024-10-03	MDCK1	1280	2560	2560	2560	1280
250299	A/Iceland/34879/2024		5a.2a (C.1.9.1)	2024-11-11	MDCK1	1280	1280	1280	1280	640
250355	A/Ireland/00084429/2024		5a.2a (C.1.9.1)	2024-11-19	MDCK1	1280	640	1280	1280	640
250935	A/Segovia/203/2024		5a.2a (C.1.9.1)	2024-11-19	hCKx/MDCK1	2560	2560	2560	2560	1280
250731	A/Netherlands/10642/2024		5a.2a (C.1.9.3)	2024-09-25	MDCK-MIX2/MDCK1	2560	2560	2560	2560	1280
250729	A/Netherlands/10647/2024		5a.2a (C.1.9.3)	2024-09-27	MDCK-MIX2/MDCK1	640	1280	1280	1280	640
250600	A/France/CVL-IPP11758/2024		5a.2a (C.1.9.3)	2024-10-14	C1/MDCK1	1280	1280	2560	1280	1280
250515	A/Latvia/10-064394/2024		5a.2a (C.1.9.3)	2024-10-20	P2/MDCK1	640	640	1280	1280	640
250596	A/France/BFC-IPP11985/2024		5a.2a (C.1.9.3)	2024-10-21	C1/MDCK1	1280	1280	1280	1280	640
250303	A/Iceland/33152/2024		5a.2a (C.1.9.3)	2024-10-24	MDCK1	2560	1280	5120	2560	1280
250719	A/Denmark/2397/2024		5a.2a (C.1.9.3)	2024-10-30	MDCK3/MDCK1	1280	1280	2560	1280	1280
250591	A/France/IDF-RELAB-IPP12839/2024		5a.2a (C.1.9.3)	2024-10-30	C1/MDCK1	1280	1280	1280	1280	640
250362	A/Ireland/00080447/2024		5a.2a (C.1.9.3)	2024-11-04	MDCK1	2560	1280	2560	2560	1280
250300	A/Iceland/34614/2024		5a.2a (C.1.9.3)	2024-11-07	MDCK1	2560	2560	2560	2560	1280
250358	A/Ireland/00082541/2024		5a.2a (C.1.9.3)	2024-11-12	MDCK1	1280	640	1280	1280	1280
250357	A/Ireland/00082961/2024		5a.2a (C.1.9.3)	2024-11-13	MDCK1	2560	2560	2560	2560	1280
250551	A/Bulgaria/3480/2024		5a.2a (C.1.9.3)	2024-11-26	MDCK1	1280	1280	2560	1280	1280
250298	A/Iceland/37256/2024		5a.2a (C.1.9.3)	2024-11-29	MDCK1	1280	1280	1280	1280	640
250593	A/France/GES-RELAB-IPP12529/2024		5a.2a (C.1.9.4)	2024-10-29	C2/MDCK1	2560	1280	1280	1280	640
250720	A/Denmark/2345/2024		5a.2a.1 (D)	2024-10-09	MDCK2/MDCK1	2560	1280	2560	1280	640
250939	A/Salamanca/159/2024		5a.2a.1 (D.3)	2024-09-18	hCKx/MDCK1	2560	2560	5120	2560	2560
250353	A/Ireland/00084171/2024		5a.2a.1 (D.3)	2024-11-20	MDCK1	1280	1280	2560	2560	2560
250606	A/France/IDF-RELAB-IPP12450/2024		5a.2a.1 (D.4)	2024-09-05	C1/MDCK1	2560	2560	2560	2560	1280
250602	A/France/IDF-RELAB-IPP12458/2024		5a.2a.1 (D.5)	2024-09-19	C1/MDCK1	1280	1280	1280	1280	640
250301	A/Iceland/34025/2024		5a.2a.1 (D.5)	2024-11-01	MDCK1	1280	1280	2560	1280	1280
250587	A/France/GES-RELAB-IPP12883/2024		5a.2a.1 (D.5)	2024-11-07	C1/MDCK1	1280	640	1280	1280	640
						Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< relates to the lowest dilution of antiserum used

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2025-01-15)

	Viruses	Other information	Genetic group	Collection Date	Passage	A/Sydney 5/21	A/Lisboa/188/2023	A/Victoria/4897/2022	IVR-238 A/Vic/4897/22	A/Wisconsin/67/2022
						Haemagglutination inhibition titre				
						Post-infection ferret antiserum				
					Passage	MDCK	MDCK	MDCK	Egg	MDCK
					Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23
					Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	CK3/MDCK4	2560	1280	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	IAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	IAT2/MDCK3	1280	1280	2560	1280	1280
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	2560	2560	640
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	2560	1280	1280
	TEST VIRUSES									
250307	A/Iceland/27024/2024		5a.2a (C.1.9.3)	2024-09-02	MDCK1	1280	1280	1280	1280	640
250306	A/Iceland/27752/2024		5a.2a (C.1.9.3)	2024-09-06	MDCK1	1280	1280	2560	1280	640
250366	A/Ireland/00073671/2024		5a.2a (C.1.9.3)	2024-10-08	MDCK1	2560	2560	2560	2560	1280
250364	A/Ireland/00076673/2024		5a.2a (C.1.9.3)	2024-10-18	MDCK1	2560	1280	2560	2560	1280
250363	A/Ireland/00079631/2024		5a.2a (C.1.9.3)	2024-10-31	MDCK1	2560	2560	2560	2560	1280
250361	A/Ireland/00080860/2024		5a.2a (C.1.9.3)	2024-11-05	MDCK1	2560	2560	2560	2560	1280
250360	A/Ireland/00081991/2024		5a.2a (C.1.9.3)	2024-11-08	MDCK1	2560	1280	1280	2560	640
250356	A/Ireland/00083485/2024		5a.2a (C.1.9.3)	2024-11-15	MDCK1	1280	1280	1280	1280	640
250354	A/Ireland/00084524/2024		5a.2a (C.1.9.3)	2024-11-19	MDCK1	1280	1280	1280	2560	640
250199	A/Qatar/88547/2024		5a.2a.1 (D.3)	2024-11-08	MDCK1	1280	1280	2560	1280	1280
						Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< relates to the lowest dilution of antiserum used

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2025-01-21)

	Viruses	Other information	Genetic group	Collection Date	Passage / Ferret number / Genetic group	A/Sydney 5/21 MDCK F46/22 5a.2a (C.1)	A/Lisboa/ 188/2023 MDCK F09/24 5a.2a (C.1.9)	A/Victoria/ 4897/2022 MDCK F05/23 5a.2a.1 (D)	IVR-238 A/Vic/4897/22 Egg F07/23 5a.2a.1 (D)	A/Wisconsin/ 67/2022 MDCK F17/23 5a.2a.1 (C.1.1)
						Haemagglutination inhibition titre				
						Post-infection ferret antiserum				
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	DCK3/MDCK4	2560	2560	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	IAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	IAT2/MDCK3	1280	2560	5120	2560	2560
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	2560	2560	640
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	2560	1280	1280
	TEST VIRUSES									
250650	A/CoteD'Ivoire/3988/2024		5a.2a (C.1.9)	2024-10-01	MDCK1	2560	1280	2560	2560	1280
250237	A/Qatar/82042/2024		5a.2a (C.1.9)	2024-10-20	MDCK1	2560	1280	2560	2560	1280
250811	A/Belgium/8076/2024		5a.2a (C.1.9)	2024-11-14	MDCK1	1280	1280	2560	2560	640
250806	A/Belgium/8782/2024		5a.2a (C.1.9)	2024-11-26	MDCK1	2560	1280	2560	2560	1280
250793	A/Switzerland/3570/2024		5a.2a (C.1.9)	2024-11-27	MDCK1	5120	2560	5120	5120	2560
250698	A/Cardiff/5630/2024		5a.2a (C.1.9)	2024-12-09	MDCK1	2560	2560	2560	2560	1280
250790	A/Switzerland/7649/2024		5a.2a (C.1.9)	2024-12-09	MDCK1	5120	2560	5120	5120	1280
250789	A/Switzerland/8778/2024		5a.2a (C.1.9)	2024-12-10	MDCK1	5120	5120	5120	5120	2560
250226	A/Qatar/83328/2024		5a.2a (C.1.9.3)	2024-10-23	MDCK1	2560	2560	2560	2560	1280
250557	A/Bulgaria/3158/2024		5a.2a (C.1.9.3)	2024-10-28	MDCK2	5120	2560	5120	2560	1280
250556	A/Bulgaria/3179/2024		5a.2a (C.1.9.3)	2024-10-30	MDCK1	1280	1280	1280	1280	640
251037	A/Stockholm/SE24-54630/2024		5a.2a (C.1.9.3)	2024-11-01	IAT1/MDCK1	2560	2560	2560	2560	1280
251035	A/Linkoping/SE24-15525/2024		5a.2a (C.1.9.3)	2024-11-10	IAT1/MDCK1	2560	1280	1280	1280	640
250647	A/CoteD'Ivoire/4607/2024		5a.2a (C.1.9.3)	2024-11-14	MDCK1	2560	2560	2560	2560	1280
250646	A/CoteD'Ivoire/4611/2024		5a.2a (C.1.9.3)	2024-11-14	MDCK1	2560	1280	1280	1280	640
250808	A/Belgium/8777/2024		5a.2a (C.1.9.3)	2024-11-25	MDCK1	2560	1280	2560	1280	640
250807	A/Belgium/8780/2024		5a.2a (C.1.9.3)	2024-11-25	MDCK1	2560	2560	2560	2560	1280
250809	A/Belgium/913/2024		5a.2a (C.1.9.3)	2024-11-25	MDCK1	5120	2560	2560	2560	1280
250552	A/Bulgaria/3479/2024		5a.2a (C.1.9.3)	2024-11-26	MDCK2	2560	2560	2560	2560	1280
250705	A/Caerphilly/5142/2024		5a.2a (C.1.9.3)	2024-11-27	MDCK1	2560	1280	2560	2560	1280
250703	A/Cardiff/5162/2024		5a.2a (C.1.9.3)	2024-11-29	MDCK1	2560	2560	5120	2560	1280
250802	A/Belgium/8986/2024		5a.2a (C.1.9.3)	2024-12-04	MDCK1	2560	2560	2560	2560	2560
250792	A/Switzerland/9140/2024		5a.2a (C.1.9.3)	2024-12-04	MDCK1	5120	2560	5120	5120	2560
250791	A/Switzerland/0747/2024		5a.2a (C.1.9.3)	2024-12-06	MDCK1	5120	5120	5120	5120	2560
250594	A/France/IDF-RELAB-IPP12406/2024		5a.2a.1 (D.1)	2024-10-23	MDCK1	1280	1280	2560	1280	1280
250648	A/CoteD'Ivoire/4606/2024		5a.2a.1 (D.1)	2024-11-14	MDCK1	1280	1280	2560	1280	1280
250710	A/Mold/5864/2024		5a.2a.1 (D.1)	2024-11-19	MDCK1	1280	1280	2560	1280	640
250708	A/Newport/5865/2024		5a.2a.1 (D.1)	2024-11-21	MDCK1	2560	2560	5120	2560	2560
250707	A/Monmouth/5971/2024		5a.2a.1 (D.1)	2024-11-25	MDCK1	2560	2560	5120	2560	2560
250701	A/Coleford/5435/2024		5a.2a.1 (D.1)	2024-12-02	MDCK1	2560	2560	5120	2560	2560
250649	A/CoteD'Ivoire/4210/2024		5a.2a.1 (D.3)	2024-10-21	MDCK1	1280	1280	2560	1280	1280
250553	A/Bulgaria/3380/2024		5a.2a.1 (D.5)	2024-11-20	MDCK1	2560	2560	5120	2560	2560
250805	A/Belgium/8727/2024		5a.2a.1 (D.5)	2024-11-29	MDCK1	2560	2560	5120	2560	2560
	< relates to the lowest dilution of antiserum used					Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2025-01-24)

	Viruses	Other information	Genetic group	Collection Date	Passage / Ferret number / Genetic group	Haemagglutination inhibition titre – Post-infection ferret antiserum: A/Sydney 5/21 MDCK F46/22 5a.2a (C.1)	A/Lisboa/ 188/2023 MDCK F09/24 5a.2a (C.1.9)	A/Victoria/ 4897/2022 MDCK F05/23 5a.2a.1 (D)	IVR-238 A/Vic/4897/22 Egg F07/23 5a.2a.1 (D)	A/Wisconsin/ 67/2022 MDCK F17/23 5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	CK3/MDCK4	5120	2560	5120	2560	2560
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	IAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	IAT2/MDCK3	2560	2560	5120	2560	2560
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	2560	1280	2560	2560	1280
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	2560	1280	1280
	TEST VIRUSES									
251059	A/Boras/SE24-14131/2024		5a.2a (C.1.9)	2024-10-09	SIAT1/MDCK	2560	1280	2560	2560	640
251055	A/Goteborg/SE24-14317/2024		5a.2a (C.1.9)	2024-10-09	SIAT1/MDCK	5120	2560	5120	5120	2560
250241	A/Qatar/81611/2024		5a.2a (C.1.9)	2024-10-18	MDCK2	2560	2560	2560	2560	1280
251049	A/Eskilstuna/SE24-14606/2024		5a.2a (C.1.9)	2024-10-21	SIAT1/MDCK	2560	2560	2560	2560	1280
251122	A/Nordrhein-Westfalen/108/2024		5a.2a (C.1.9)	2024-11-28	P1/MDCK	2560	2560	2560	2560	1280
250699	A/Newport/5617/2024		5a.2a (C.1.9)	2024-12-06	MDCK2	2560	2560	5120	2560	1280
251169	A/Israel/R18419/2024		5a.2a (C.1.9)	2024-12-25	Px/MDCK	2560	2560	2560	2560	1280
251190	A/Israel/R14647/2024		5a.2a (C.1.9.1)	2024-10-09	Px/MDCK	2560	2560	1280	1280	2560
250858	A/Asturias/2285/2024		5a.2a (C.1.9.1)	2024-11-11	MDCK1	5120	2560	5120	5120	2560
251186	A/Israel/R17237/2024		5a.2a (C.1.9.1)	2024-12-08	Px/MDCK	2560	1280	2560	2560	1280
251115	A/Bayern/98/2024		5a.2a (C.1.9.1)	2024-12-16	P1/MDCK	2560	1280	2560	2560	640
251057	A/Gavle/SE24-14224/2024		5a.2a (C.1.9.3)	2024-10-09	SIAT1/MDCK	2560	2560	2560	2560	1280
251053	A/Stockholm/SE24-54383/2024		5a.2a (C.1.9.3)	2024-10-15	SIAT1/MDCK	2560	2560	2560	2560	1280
251051	A/Linkoping/SE24-14515/2024		5a.2a (C.1.9.3)	2024-10-16	SIAT1/MDCK	2560	2560	2560	2560	1280
250243	A/Qatar/81081/2024		5a.2a (C.1.9.3)	2024-10-16	MDCK2	2560	2560	5120	2560	1280
250882	A/ComunidadValenciana/2259/2024		5a.2a (C.1.9.3)	2024-10-23	SIAT1	2560	2560	2560	2560	1280
251043	A/Skovde/SE24-14886/2024		5a.2a (C.1.9.3)	2024-10-25	SIAT1/MDCK	2560	2560	2560	2560	1280
250813	A/Belgium/7696/2024		5a.2a (C.1.9.3)	2024-10-28	MDCK1	2560	2560	2560	2560	1280
250859	A/LaRioja/2275/2024		5a.2a (C.1.9.3)	2024-11-04	MDCK1	2560	2560	2560	2560	1280
251189	A/Israel/R15888/2024		5a.2a (C.1.9.3)	2024-11-10	Px/MDCK	2560	2560	2560	2560	1280
250812	A/Belgium/8058/2024		5a.2a (C.1.9.3)	2024-11-11	MDCK1	2560	2560	2560	2560	1280
250857	A/Asturias/2288/2024		5a.2a (C.1.9.3)	2024-11-15	MDCK1	2560	2560	2560	2560	1280
250810	A/Belgium/8297/2024		5a.2a (C.1.9.3)	2024-11-21	MDCK2	2560	2560	2560	2560	1280
251124	A/Berlin/40/2024		5a.2a (C.1.9.3)	2024-11-21	P1/MDCK	2560	2560	2560	2560	1280
250856	A/Navarra/2224/2024		5a.2a (C.1.9.3)	2024-11-21	MDCK1	2560	2560	2560	2560	1280
251123	A/Bayern/96/2024		5a.2a (C.1.9.3)	2024-11-25	P1/MDCK	2560	2560	2560	2560	1280
251103	A/Bratislava/9733/2024		5a.2a (C.1.9.3)	2024-11-25	DCK2/MDCK	1280	1280	1280	1280	5120
250702	A/Newport/5303/2024		5a.2a (C.1.9.3)	2024-11-29	MDCK2	5120	2560	5120	5120	2560
251121	A/Nordrhein-Westfalen/109/2024		5a.2a (C.1.9.3)	2024-12-04	P1/MDCK	2560	2560	2560	2560	1280
251120	A/Bremen/28/2024		5a.2a (C.1.9.3)	2024-12-06	P1/MDCK	2560	2560	2560	2560	1280
251187	A/Israel/R17318/2024		5a.2a (C.1.9.3)	2024-12-08	Px/MDCK	2560	2560	2560	2560	1280
251118	A/Rheinland-Pfalz/41/2024		5a.2a (C.1.9.3)	2024-12-09	P1/MDCK	5120	2560	2560	2560	1280
251117	A/Rheinland-Pfalz/43/2024		5a.2a (C.1.9.3)	2024-12-12	P1/MDCK	2560	2560	2560	2560	1280
250716	A/Blackwood/1760/2024		5a.2a (C.1.9.3)	2024-12-13	SIAT1	2560	2560	2560	2560	1280
251116	A/Thuringen/77/2024		5a.2a (C.1.9.3)	2024-12-16	P1/MDCK	5120	2560	2560	2560	1280
251114	A/Brandenburg/36/2024		5a.2a (C.1.9.3)	2024-12-17	P1/MDCK	5120	2560	5120	2560	2560
251113	A/Berlin/47/2024		5a.2a (C.1.9.3)	2024-12-18	P1/MDCK	5120	5120	5120	1280	2560
251112	A/Berlin/46/2024		5a.2a (C.1.9.3)	2024-12-19	P1/MDCK	5120	5120	5120	5120	2560
251173	A/Israel/R18226/2024		5a.2a (C.1.9.3)	2024-12-23	Px/MDCK	2560	1280	2560	2560	1280
251174	A/Israel/R18227/2024		5a.2a (C.1.9.3)	2024-12-23	Px/MDCK	1280	1280	2560	2560	1280
251175	A/Israel/R18254/2024		5a.2a (C.1.9.3)	2024-12-23	Px/MDCK	2560	2560	2560	2560	1280
251045	A/Goteborg/SE24-14949/2024		5a.2a (C.1.9.4)	2024-10-24	SIAT1/MDCK	2560	2560	5120	2560	1280
250253	A/Qatar/73573/2024		5a.2a.1 (D.3)	2024-09-22	MDCK2	2560	2560	5120	2560	2560
250207	A/Qatar/86159/2024		5a.2a.1 (D.3)	2024-10-31	MDCK2	2560	2560	5120	2560	2560
251177	A/Israel/R18120/2024		5a.2a.1 (D.3)	2024-12-22	Px/MDCK	1280	1280	2560	2560	1280
251168	A/Israel/R18370/2024		5a.2a.1 (D.3)	2024-12-25	Px/MDCK	2560	2560	5120	2560	2560
251041	A/Eskilstuna/SE24-15080/2024		5a.2a.1 (D.5)	2024-10-30	SIAT1/MDCK	1280	1280	5120	2560	1280
251039	A/Orebro/SE24-15290/2024		5a.2a.1 (D.5)	2024-10-31	SIAT1/MDCK	2560	2560	5120	2560	2560
						Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< relates to the lowest dilution of antiserum used

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2025-01-28)

	Viruses	Other information	Genetic group	Collection Date	Passage	A/Sydney 5/21 MDCK	A/Lisboa/ 188/2023 MDCK	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
						Haemagglutination inhibition titre				
						Post-infection ferret antiserum				
					Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23
					Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	5120	2560	1280
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q22:	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	1280	2560	640
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	2560	1280	1280
	TEST VIRUSES									
250996	A/Madagascar/02988/2024		5a.2a (C.1.8)	2024-09-30	MDCK1	1280	1280	1280	1280	640
250993	A/Madagascar/03221/2024		5a.2a (C.1.8)	2024-10-23	MDCK1	1280	1280	1280	1280	640
250995	A/Madagascar/03041/2024		5a.2a (C.1.9)	2024-10-07	MDCK1	2560	2560	2560	2560	1280
251166	A/Israel/R18372/2024		5a.2a (C.1.9)	2024-12-25	MDCK1	2560	2560	2560	2560	1280
251180	A/Israel/R17665/2024		5a.2a (C.1.9.1)	2024-12-15	MDCK1	1280	1280	1280	1280	640
251172	A/Israel/R18270/2024		5a.2a (C.1.9.1)	2024-12-23	MDCK1	1280	1280	1280	1280	640
250238	A/Qatar/82022/2024		5a.2a (C.1.9.3)	2024-10-20	MDCK3	2560	2560	2560	2560	640
251216	A/Lyon/CHU/0241842175/2024		5a.2a (C.1.9.3)	2024-11-26	MDCK2/MDCK1	640	640	1280	640	320
251188	A/Israel/R16808/2024		5a.2a (C.1.9.3)	2024-11-29	MDCK1	2560	2560	2560	2560	1280
251185	A/Israel/R17238/2024		5a.2a (C.1.9.3)	2024-12-08	MDCK1	1280	1280	2560	1280	640
251184	A/Israel/R17338/2024		5a.2a (C.1.9.3)	2024-12-09	MDCK1	1280	1280	1280	1280	640
251183	A/Israel/R17468/2024		5a.2a (C.1.9.3)	2024-12-10	MDCK1	2560	1280	2560	2560	1280
251212	A/Grenoble/0241960676/2024		5a.2a (C.1.9.3)	2024-12-11	MDCK2/MDCK1	1280	1280	1280	1280	640
251181	A/Israel/R17603/2024		5a.2a (C.1.9.3)	2024-12-13	MDCK1	1280	1280	2560	1280	1280
251179	A/Israel/R17886/2024		5a.2a (C.1.9.3)	2024-12-17	MDCK1	2560	2560	2560	2560	1280
251208	A/Lyon/0241972354/2024		5a.2a (C.1.9.3)	2024-12-17	MDCK2/MDCK1	2560	2560	2560	2560	640
251178	A/Israel/R17977/2024		5a.2a (C.1.9.3)	2024-12-19	MDCK1	1280	1280	1280	1280	640
251170	A/Israel/R18297/2024		5a.2a (C.1.9.3)	2024-12-24	MDCK1	2560	2560	2560	2560	1280
251171	A/Israel/R18311/2024		5a.2a (C.1.9.3)	2024-12-24	MDCK1	1280	1280	1280	1280	640
251205	A/Cannes/0242035716/2024		5a.2a (C.1.9.3)	2024-12-25	MDCK2/MDCK1	1280	1280	1280	1280	640
251167	A/Israel/R18387/2024		5a.2a (C.1.9.3)	2024-12-25	MDCK1	2560	2560	2560	2560	1280
251164	A/Israel/R18512/2024		5a.2a (C.1.9.3)	2024-12-26	MDCK1	1280	1280	2560	2560	640
251165	A/Israel/R18527/2024		5a.2a (C.1.9.3)	2024-12-26	MDCK1	5120	5120	5120	5120	2560
251204	A/Lyon/CHU/0242011127/2025		5a.2a.1 (D)	2024-12-25	MDCK2/MDCK1	2560	2560	5120	2560	2560
251209	A/Papeete/0242013579/2025		5a.2a.1 (D.5)	2024-12-16	MDCK2/MDCK1	1280	1280	2560	1280	1280
251206	A/StEtienne/0242013332/2025		5a.2a.1 (D.5)	2024-12-21	MDCK2/MDCK1	1280	1280	2560	1280	1280
	< relates to the lowest dilution of antiserum used					Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2025-01-31)

						Haemagglutination inhibition titre				
						Post-infection ferret antiserum				
	Viruses	Other information			Passage	A/Sydney 5/21 MDCK	A/Lisboa/ 188/2023 MDCK	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
					Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23
			Genetic group	Collection Date	Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK3	2560	2560	5120	2560	2560
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	1280	1280	5120	1280	1280
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	2560	1280	2560	2560	1280
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	2560	2560	2560
	TEST VIRUSES									
251378	A/Slovenia/NLZOH-8008-21-1936/2024		5a.2a (C.1.9)	2024-11-20	MDCKx/MDCK	2560	2560	2560	2560	1280
251276	A/Kazakhstan/2043/2024		5a.2a (C.1.9)	2024-12-09	MDCK1	2560	2560	2560	2560	1280
251213	A/Lyon/0241957113/2024		5a.2a (C.1.9)	2024-12-09	MDCK2/MDCK	1280	1280	2560	2560	1280
251275	A/Kazakhstan/1986/2024		5a.2a (C.1.9)	2024-12-10	MDCK1	2560	2560	2560	2560	1280
251274	A/Kazakhstan/1988/2024		5a.2a (C.1.9)	2024-12-11	MDCK1	2560	2560	2560	2560	1280
251273	A/Kazakhstan/1989/2024		5a.2a (C.1.9)	2024-12-11	MDCK1	2560	1280	1280	2560	1280
251271	A/Kazakhstan/1929/2024		5a.2a (C.1.9)	2024-12-12	MDCK1	2560	2560	2560	2560	1280
251269	A/Kazakhstan/2009/2024		5a.2a (C.1.9)	2024-12-13	MDCK1	2560	2560	2560	2560	1280
251211	A/Lyon/0241987424/2024		5a.2a (C.1.9)	2024-12-16	MDCK2/MDCK	2560	1280	2560	2560	2560
251210	A/Lyon/0241999528/2024		5a.2a (C.1.9)	2024-12-16	MDCK2/MDCK	2560	2560	2560	2560	1280
251207	A/Lyon/0241999610/2024		5a.2a (C.1.9)	2024-12-17	MDCK2/MDCK	2560	2560	2560	2560	1280
251222	A/Agen/0241461539/2024		5a.2a (C.1.9.3)	2024-09-09	MDCK2/MDCK	1280	1280	1280	1280	640
251221	A/ClermontFerrand/0241493056/2024		5a.2a (C.1.9.3)	2024-09-20	MDCK2/MDCK	2560	2560	2560	2560	5120
251219	A/Toulon/0241527008/2024		5a.2a (C.1.9.3)	2024-09-26	MDCK2/MDCK	1280	1280	1280	1280	640
251302	A/Kazakhstan/1964/2024		5a.2a (C.1.9.3)	2024-10-08	MDCK	2560	2560	2560	2560	1280
251218	A/Cannes/0241656802/2024		5a.2a (C.1.9.3)	2024-10-21	MDCK2/MDCK	2560	2560	2560	2560	1280
251380	A/Slovenia/NLZOH-8008-21-1895/2024		5a.2a (C.1.9.3)	2024-11-11	SIAT1/MDCK	2560	2560	5120	2560	1280
251298	A/Kazakhstan/64/2024		5a.2a (C.1.9.3)	2024-11-15	P1/MDCK	2560	2560	2560	2560	1280
251297	A/Kazakhstan/65/2024		5a.2a (C.1.9.3)	2024-11-15	P1/MDCK	2560	1280	1280	1280	640
251295	A/Kazakhstan/66/2024		5a.2a (C.1.9.3)	2024-11-18	P1/MDCK	2560	2560	2560	2560	1280
251294	A/Kazakhstan/67/2024		5a.2a (C.1.9.3)	2024-11-18	P1/MDCK	1280	1280	1280	1280	640
251293	A/Kazakhstan/68/2024		5a.2a (C.1.9.3)	2024-11-21	P1/MDCK	2560	2560	2560	2560	1280
251292	A/Kazakhstan/1910/2024		5a.2a (C.1.9.3)	2024-11-25	MDCK1	2560	1280	2560	2560	1280
251291	A/Kazakhstan/1911/2024		5a.2a (C.1.9.3)	2024-11-25	MDCK1	2560	2560	5120	2560	1280
251217	A/Lyon/0241873441/2024		5a.2a (C.1.9.3)	2024-11-25	MDCK2/MDCK	5120	2560	5120	2560	1280
251215	A/Toulon/0241882136/2024		5a.2a (C.1.9.3)	2024-11-28	MDCK2/MDCK	5120	2560	5120	2560	1280
251214	A/Lyon/CHU/0241865678/2024		5a.2a (C.1.9.3)	2024-11-29	MDCK2/MDCK	5120	2560	5120	5120	2560
251287	A/Kazakhstan/72/2024		5a.2a (C.1.9.3)	2024-11-30	P1/MDCK	1280	1280	2560	2560	1280
251284	A/Kazakhstan/1952/2024		5a.2a (C.1.9.3)	2024-12-05	MDCK1	5120	2560	2560	2560	1280
251283	A/Kazakhstan/1953/2024		5a.2a (C.1.9.3)	2024-12-05	MDCK1	2560	2560	1280	2560	640
251282	A/Kazakhstan/1956/2024		5a.2a (C.1.9.3)	2024-12-05	MDCK1	5120	2560	2560	2560	1280
251281	A/Kazakhstan/2041/2024		5a.2a (C.1.9.3)	2024-12-05	MDCK1	1280	1280	1280	1280	1280
251280	A/Kazakhstan/1954/2024		5a.2a (C.1.9.3)	2024-12-06	MDCK1	2560	2560	2560	2560	1280
251278	A/Kazakhstan/1957/2024		5a.2a (C.1.9.3)	2024-12-08	MDCK1	2560	2560	2560	2560	1280
251277	A/Kazakhstan/1958/2024		5a.2a (C.1.9.3)	2024-12-08	MDCK1	2560	2560	2560	2560	1280
251182	A/Israel/R17580/2024		5a.2a (C.1.9.3)	2024-12-12	MDCK2	2560	2560	2560	2560	1280
251272	A/Kazakhstan/1926/2024		5a.2a (C.1.9.3)	2024-12-12	MDCK1	1280	1280	1280	1280	640
251270	A/Kazakhstan/2004/2024		5a.2a (C.1.9.3)	2024-12-13	MDCK1	2560	2560	2560	2560	1280
251374	A/Slovenia/NLZOH-8008-21-17/2025		5a.2a (C.1.9.3)	2025-01-06	SIATx/MDCK	2560	2560	2560	2560	1280
251220	A/Toulouse/0241506797/2024		5a.2a.1 (D.3)	2024-09-23	MDCK2/MDCK	2560	2560	5120	5120	5120
	< relates to the lowest dilution of antiserum used					Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2025-02-04)

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre – Post-infection ferret antiserum				
						A/Sydney 5/21	A/Lisboa/ 188/2023	A/Victoria/ 4897/2022	IVR-238 A/Vic/4897/22	A/Wisconsin/ 67/2022
					Passage	MDCK	MDCK	MDCK	Egg	MDCK
					Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23
					Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	5120	5120	5120	5120	2560
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK3	2560	2560	5120	2560	2560
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	1280	1280	5120	2560	2560
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	2560	2560	2560	5120	1280
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	2560	1280	2560
	TEST VIRUSES									
251431	A/Palermo/109709/2024		5a.2a (C.1.9)	2024-11-25	MDCK3/MDCK	1280	640	1280	1280	640
251358	A/Switzerland/3570/2024		5a.2a (C.1.9)	2024-11-26	MDCK1	2560	2560	2560	2560	1280
251428	A/Milano/308/2024		5a.2a (C.1.9)	2024-12-02	MDCK3/MDCK	2560	1280	2560	2560	1280
251427	A/Parma/53/2024		5a.2a (C.1.9)	2024-12-02	MDCK3/MDCK	2560	2560	2560	2560	1280
251622	A/France/IDF-IPP13888/2024		5a.2a (C.1.9)	2024-12-04	MDCK1/MDCK	2560	2560	5120	5120	2560
251619	A/France/BRE-IPP13991/2024		5a.2a (C.1.9)	2024-12-05	MDCK1/MDCK	1280	1280	1280	2560	2560
251610	A/France/BRE-IPP14493/2024		5a.2a (C.1.9)	2024-12-15	MDCK2/MDCK	2560	2560	2560	2560	1280
251612	A/France/HDF-IPP14409/2024		5a.2a (C.1.9)	2024-12-15	MDCK1/MDCK	2560	2560	2560	2560	1280
251605	A/France/BFC-IPP14595/2024		5a.2a (C.1.9)	2024-12-16	MDCK1/MDCK	2560	1280	2560	2560	1280
251597	A/France/CVL-IPP14654/2024		5a.2a (C.1.9)	2024-12-19	MDCK1/MDCK	2560	2560	5120	2560	2560
251357	A/Switzerland/9136/2024		5a.2a (C.1.9)	2024-12-20	MDCK1	2560	2560	2560	2560	1280
251249	A/Luxembourg/2485332/2024		5a.2a (C.1.9.1)	2024-12-18	MDCK1	1280	1280	2560	2560	1280
251352	A/Switzerland/6056/2025		5a.2a (C.1.9.1)	2025-01-07	MDCK1	1280	1280	1280	1280	640
251300	A/Kazakhstan/63/2024		5a.2a (C.1.9.3)	2024-11-12	P1/MDCK1	5120	2560	5120	5120	2560
251432	A/Parma/51/2024		5a.2a (C.1.9.3)	2024-11-23	MDCK3/MDCK	2560	2560	2560	2560	1280
251290	A/Kazakhstan/69/2024		5a.2a (C.1.9.3)	2024-11-25	P1/MDCK1	1280	1280	1280	1280	640
251289	A/Kazakhstan/70/2024		5a.2a (C.1.9.3)	2024-11-28	P1/MDCK1	1280	1280	1280	1280	640
251430	A/Milano/305/2024		5a.2a (C.1.9.3)	2024-11-28	MDCK3/MDCK	2560	1280	1280	1280	640
251429	A/Milano/307/2024		5a.2a (C.1.9.3)	2024-11-29	MDCK3/MDCK	1280	1280	1280	1280	1280
251288	A/Kazakhstan/71/2024		5a.2a (C.1.9.3)	2024-11-30	P1/MDCK1	1280	1280	1280	1280	640
251620	A/France/NOR-RELAB-IPP14116/2024		5a.2a (C.1.9.3)	2024-12-04	MDCK1/MDCK	2560	2560	5120	2560	1280
251254	A/Luxembourg/5983073/2024		5a.2a (C.1.9.3)	2024-12-05	MDCK1	2560	2560	2560	2560	1280
251617	A/France/HDF-IPP13873/2024		5a.2a (C.1.9.3)	2024-12-06	MDCK1/MDCK	2560	2560	5120	5120	2560
251560	A/Denmark/2598/2024		5a.2a (C.1.9.3)	2024-12-07	MDCK2/MDCK	2560	1280	2560	2560	1280
251559	A/Denmark/2637/2024		5a.2a (C.1.9.3)	2024-12-12	MDCK2/MDCK	1280	1280	1280	1280	640
251613	A/France/CVL-IPP14252/2024		5a.2a (C.1.9.3)	2024-12-12	MDCK1/MDCK	5120	2560	5120	2560	2560
251252	A/Luxembourg/8888029/2024		5a.2a (C.1.9.3)	2024-12-13	MDCK1	2560	2560	2560	2560	1280
251609	A/France/BRE-IPP14494/2024		5a.2a (C.1.9.3)	2024-12-16	MDCK1/MDCK	2560	2560	5120	2560	2560
251607	A/France/NOR-IPP14552/2024		5a.2a (C.1.9.3)	2024-12-16	MDCK1/MDCK	1280	1280	1280	1280	640
251603	A/France/PDL-IPP14618/2024		5a.2a (C.1.9.3)	2024-12-16	MDCK1/MDCK	1280	1280	2560	2560	1280
251250	A/Luxembourg/3349245/2024		5a.2a (C.1.9.3)	2024-12-16	MDCK1	2560	1280	2560	2560	1280
251601	A/France/IDF-IPP14579/2024		5a.2a (C.1.9.3)	2024-12-17	MDCK1/MDCK	2560	1280	1280	1280	640
251356	A/Switzerland/9196/2024		5a.2a (C.1.9.3)	2024-12-20	MDCK1	1280	1280	1280	1280	640
251340	A/Algeria/16823/2024		5a.2a (C.1.9.3)	2024-12-22	MDCK1	1280	1280	1280	1280	640
251395	A/Ireland/00094983/2024		5a.2a (C.1.9.3)	2024-12-29	MDCK1	1280	1280	1280	1280	640
251392	A/Ireland/00094914/2024		5a.2a (C.1.9.3)	2024-12-30	MDCK1	2560	2560	2560	2560	1280
251393	A/Ireland/00095275/2024		5a.2a (C.1.9.3)	2024-12-30	MDCK1	2560	2560	2560	2560	1280
251391	A/Ireland/00095229/2024		5a.2a (C.1.9.3)	2024-12-31	MDCK1	2560	2560	2560	2560	1280
251389	A/Ireland/00001641/2025		5a.2a (C.1.9.3)	2025-01-04	MDCK1	2560	1280	5120	2560	2560
251353	A/Switzerland/3160/2025		5a.2a (C.1.9.3)	2025-01-06	MDCK1	1280	1280	1280	1280	640
251434	A/Messina/109463/2024		5a.2a (C.1.9.4)	2024-10-28	MDCK3/MDCK	2560	1280	2560	2560	1280
251433	A/Palermo/109340/2024		5a.2a (C.1.9.4)	2024-10-29	MDCK3/MDCK	2560	1280	1280	2560	1280
251599	A/France/BFC-IPP14622/2024		5a.2a.1 (D)	2024-12-18	MDCK1/MDCK	2560	2560	5120	2560	2560
251595	A/France/BFC-IPP14894/2024		5a.2a.1 (D)	2024-12-23	MDCK1/MDCK	2560	2560	5120	2560	2560
251355	A/Switzerland/7784/2024		5a.2a.1 (D)	2024-12-27	MDCK1	1280	640	2560	1280	1280
251593	A/France/BFC-IPP15052/2024		5a.2a.1 (D.3)	2024-12-23	MDCK1/MDCK	2560	2560	5120	2560	2560
251354	A/Switzerland/6849/2025		5a.2a.1 (D.3)	2025-01-02	MDCK1	1280	1280	2560	1280	1280
						Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< relates to the lowest dilution of antiserum used

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-13. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2025-02-07)

	Viruses	Other information	Genetic group	Collection Date	Passage	A/Sydney 5/21 MDCK	A/Lisboa/ 188/2023 MDCK	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
						Haemagglutination inhibition titre				
						Post-infection ferret antiserum				
					Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23
					Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK3	5120	2560	2560	2560	2560
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	5120	2560	2560
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	2560	1280	2560	5120	1280
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	5120	2560	2560
	TEST VIRUSES									
251756	A/Romania_BB/585570/2024		5a.2a (C.1.9)	2024-12-09	SIAT1/MDCK1	2560	2560	2560	2560	1280
251460	A/Montenegro/1015352502/2025		5a.2a (C.1.9)	2025-01-12	MDCK1	2560	2560	5120	5120	2560
251459	A/Montenegro/1015502502/2025		5a.2a (C.1.9)	2025-01-12	MDCK1	5120	2560	2560	2560	1280
251632	A/France/GES-RELAB-IPP12532/2024		5a.2a (C.1.9.3)	2024-10-29	MDCK2/MDCK1	1280	1280	2560	2560	1280
251773	A/Iran/4C28/2024		5a.2a (C.1.9.3)	2024-11-09	SIAT1/MDCK1	2560	2560	2560	2560	1280
251772	A/Iran/12V29/2024		5a.2a (C.1.9.3)	2024-11-12	SIAT1/MDCK1	1280	1280	1280	1280	1280
251771	A/Iran/13G28/2024		5a.2a (C.1.9.3)	2024-11-16	SIAT1/MDCK1	2560	2560	2560	2560	1280
251768	A/Iran/12P28/2024		5a.2a (C.1.9.3)	2024-11-24	SIAT1/MDCK1	2560	2560	5120	2560	2560
251626	A/France/GES-RELAB-IPP13676/2024		5a.2a (C.1.9.3)	2024-11-29	MDCK2/MDCK1	2560	1280	2560	2560	1280
251624	A/France/HDF-IPP13859/2024		5a.2a (C.1.9.3)	2024-12-04	MDCK1/MDCK1	1280	1280	1280	1280	1280
251755	A/Romania_CJ/585811/2024		5a.2a (C.1.9.3)	2024-12-12	SIAT1/MDCK1	2560	1280	2560	1280	640
251767	A/Iran/2X29/2024		5a.2a (C.1.9.3)	2024-12-15	SIAT1/MDCK1	2560	1280	1280	1280	640
251766	A/Iran/26U29/2024		5a.2a (C.1.9.3)	2024-12-16	SIAT1/MDCK1	1280	1280	2560	2560	1280
251376	A/Slovenia/NLZOH-8008-21-2168/2024		5a.2a (C.1.9.3)	2024-12-16	MDCKfx/MDCK1	2560	2560	5120	2560	2560
251764	A/Iran/12U29/2024		5a.2a (C.1.9.3)	2024-12-18	SIAT1/MDCK1	2560	2560	2560	2560	1280
251753	A/Romania_OT/586209/2024		5a.2a (C.1.9.3)	2024-12-18	SIAT1/MDCK1	2560	2560	2560	2560	1280
251752	A/Romania_OT/586210/2024		5a.2a (C.1.9.3)	2024-12-18	SIAT1/MDCK1	2560	1280	2560	2560	640
251751	A/Romania_BB/586825/2024		5a.2a (C.1.9.3)	2024-12-24	SIAT1/MDCK1	2560	2560	2560	2560	1280
251750	A/Romania_BB/586827/2024		5a.2a (C.1.9.3)	2024-12-24	SIAT1/MDCK1	1280	1280	2560	2560	1280
251749	A/Romania_AR/586446/2024		5a.2a (C.1.9.3)	2024-12-30	SIAT1/MDCK1	2560	1280	2560	2560	1280
251748	A/Romania_BB/586855/2024		5a.2a (C.1.9.3)	2024-12-30	SIAT1/MDCK1	2560	2560	2560	2560	1280
251747	A/Romania_BB/586857/2024		5a.2a (C.1.9.3)	2024-12-30	SIAT1/MDCK1	2560	1280	1280	1280	640
251390	A/Ireland/00001346/2025		5a.2a (C.1.9.3)	2025-01-02	MDCK2	2560	2560	2560	2560	1280
251745	A/Romania_BB/586905/2025		5a.2a (C.1.9.3)	2025-01-03	SIAT1/MDCK1	2560	2560	2560	2560	1280
251744	A/Romania_BB/586867/2025		5a.2a (C.1.9.3)	2025-01-04	SIAT1/MDCK1	2560	1280	2560	2560	640
251494	A/Poland/58/2025		5a.2a (C.1.9.3)	2025-01-06	MDCK1	1280	1280	1280	1280	2560
251492	A/Poland/11/2025		5a.2a (C.1.9.3)	2025-01-08	MCDK1	2560	2560	2560	2560	1280
251388	A/Ireland/00003707/2025		5a.2a (C.1.9.3)	2025-01-09	MDCK2	2560	1280	2560	2560	1280
251743	A/Romania_BH/586594/2025		5a.2a (C.1.9.3)	2025-01-10	SIAT1/MDCK1	2560	2560	2560	2560	1280
251742	A/Romania_BH/586601/2025		5a.2a (C.1.9.3)	2025-01-10	SIAT1/MDCK1	2560	2560	2560	2560	1280
251741	A/Romania_BH/586977/2025		5a.2a (C.1.9.3)	2025-01-13	SIAT1/MDCK1	1280	1280	1280	1280	640
251740	A/Romania_BH/586981/2025		5a.2a (C.1.9.3)	2025-01-13	SIAT1/MDCK1	1280	1280	1280	1280	640
251739	A/Romania_BH/587157/2025		5a.2a (C.1.9.3)	2025-01-15	SIAT1/MDCK1	2560	1280	2560	2560	1280
251628	A/France/HDF-IPP13305/2024		5a.2a.1 (D)	2024-11-23	MDCK1/MDCK1	1280	1280	2560	1280	1280
251630	A/France/HDF-IPP13225/2024		5a.2a.1 (D.3)	2024-11-22	MDCK1/MDCK1	1280	1280	2560	1280	1280
	< relates to the lowest dilution of antiserum used					Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H1-14. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2025-02-11)

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre – Post-infection ferret antiserum: A/Sydney/5/21	A/Lisboa/188/2023	A/Victoria/4897/2022	IVR-238 A/Vic/4897/22	A/Wisconsin/67/2022
					Passage	MDCK	MDCK	MDCK	Egg	MDCK
					Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23
					Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
	REFERENCE VIRUSES									
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK3	2560	2560	2560	5120	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	1280	1289	2560	1280	1280
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	1280	2560	640
	A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	640	2560	2560	2560
	TEST VIRUSES									
251672	A/Cameroon/11661/2024		5a.2a (C.1.9)	2024-11-06	MDCK1	1280	640	640	640	320
251670	A/Cameroon/11991/2024		5a.2a (C.1.9)	2024-11-12	MDCK1	2560	2560	2560	2560	1280
251668	A/Cameroon/12020/2024		5a.2a (C.1.9)	2024-11-15	MDCK1	2560	2560	2560	2560	1280
251667	A/Cameroon/12199/2024		5a.2a (C.1.9)	2024-11-22	MDCK1	1280	1280	1280	1280	640
251769	A/Iran/15P28/2024		5a.2a (C.1.9.1)	2024-11-23	SIAT1/MDCK1/MDCK2	160	160	80	320	40
251467	A/Montenegro/1009752502/2025		5a.2a (C.1.9.3)	2025-01-09	MDCK1	2560	2560	2560	2560	1280
251466	A/Montenegro/1009792502/2025		5a.2a (C.1.9.3)	2025-01-09	MDCK1	2560	2560	5120	2560	2560
	< relates to the lowest dilution of antiserum used					Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025	

Legend: < 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Fold-reduction table: A/H1N1

Reference Virus	Clade	<4-fold difference		4-fold difference		>4-fold difference		
		Number	Percentage	Number	Percentage	Number	Percentage	Total
A/Sydney/5/2021 Cell	5a.2a (C.1)	365	91.7	30	7.5	3	0.8	398
A/Lisboa/188/2023 Cell	5a.2a (C.1.9)	377	94.7	19	4.8	2	0.5	398
A/Victoria/4897/2022 Cell	5a.2a.1 (D)	311	78.1	82	20.6	5	1.3	398
IVR-238 (A/Victoria/4897/2022) Egg	5a.2a.1 (D)	362	91.0	35	8.8	1	0.3	398
A/Wisconsin/67/2022 Cell	5a.2a.1 (C.1.1)	344	86.4	52	13.1	2	0.5	398

A/H1N1: Egg isolates

Virus	Status	Genetic group (subclade) [Additional subst.]	Comment
A/Norway/31694/2022	NIB-133 (2-way pass)	5a.2a.1 (C.1.1)	Egg adaptation: (Q223R)
A/Catalonia/NSVH161512065/2022	NIB-134 (2-way pass)	5a.2a.1 (C.1.1)	Egg adaptation: (Q223R)
A/Switzerland/74135/74135/2024	available to send out	C.1.8	Egg adaptation: (Q223R)
A/Austria/1678861/2024	available to send out	C.1.8	Egg adaptation: (Q223R)
A/Switzerland/6509/2024	available to send out	C.1.9	Egg adaptation: (Q223R)
A/Austria/1679300/2024	available to send out	C.1.9.2	Egg adaptation: (Q223R)

A/H1N1: HI reagents and references

Virus	Genetic group	Virus passage	Ferret ID
A/Sydney/5/2021	5a.2a (C.1)	MDCK3/MDCK4	F46/22
A/Victoria/4897/2022	5a.2a.1 (D)	SIAT2/MDCK3	F05/23
IVR-238 (A/Victoria/4897/2022)	5a.2a.1 (D)	E3/D6/E1	F07/23
A/Wisconsin/67/2022	5a.2a.1 (C.1.1)	MDCK2	F17/23
A/Lisboa/188/2023	5a.2a.1 (C.1.9)	SIAT1/MDCK2	F09/24

Summary: A/H1N1

Genetic analyses

6B.1A.**5a.2a** and 6B.1A.**5a.2a.1** clade viruses both continued to circulate with changing relative proportions throughout this period.

The SH 2024 past influenza season was characterised by predominance of clade 5a.2a subclades C.1.9 and C.1.8 with minor circulation of clade 5a.2a.1 subclade D. Since September 2024, subclade C.1.9 continues to predominate, with minor cocirculation of subclades derived from 5a.2a.1 subclade D, whereas subclade C.1.8 has been detected in very low numbers so far.

The majority (80%) of A/H1N1 viruses sequenced globally belong to subclade C.1.9, characterised by substitution K169Q. Subclade C.1.9 has further split into four subclades: C.1.9.1 with P137S, C.1.9.2 with N38D and K153R, C.1.9.3 with S83P and I510T and C.1.9.4 with Q54K, D86N, N125D and I149V. Of these, C.1.9.3 represent 60% of all the A/H1N1 viruses sequenced so far and predominates in all Europe, North and Central America, the Caribbean, the Middle East, Western Africa, South East Asia, Russian Federation, Kazakhstan, Australia and Japan. Basal C.1.9 subclade cocirculates as a minority predominating in China, South East Asia, Austria and Slovakia. Subclade C.1.9.1 circulates in low frequency mostly in Europe, US and Australia, whereas C.1.9.2 and C.1.9.4 are very sporadically detected.

Within the 5a.2a.1 viruses, major subclade D with T216A represented by A/Victoria/4897/2022 has split into 5 subclades, of which subclade D.3 is the most frequently detected with increasing frequency in the US, Europe and Middle East, though still representing a minority of A/H1N1 sequences detected so far this season (8%). The recently emerged subclade D.5 characterised by R45K was the most frequently detected at the beginning of the season but has since decreased frequency, although it still predominates in Brazil. Subclade D.5 and other 5a.2a.1 subclades continue to be detected in significant proportions only in South America and the US.

Antigenic analyses

HI titres show that NH 2023-24, 2024-25 and SH 2024 strains, cell-based A/Wisconsin/67/2022 and egg-based A/Victoria/4897/2022, generally recognised both 5a.2a and 5a.2a.1 clade viruses well. Some four-fold drops were observed with the cell- based A/Wisconsin/67/2022 strain that were not seen with the egg-based A/Victoria/4897/2022.

Influenza A/H3N2

Genetic analyses

Global distribution of A/H3N2 subclades

Global geographical distribution of influenza A/H3N2 genetic clades (subclades) viruses with collection dates from 1st November 2024 to 7th February 2025, obtained with full HA sequences as deposited in GISAID, downloaded on 14th February 2025 and classified using Nextclade. Map prepared with Microreact. Geographic markers scaled to detection proportions.

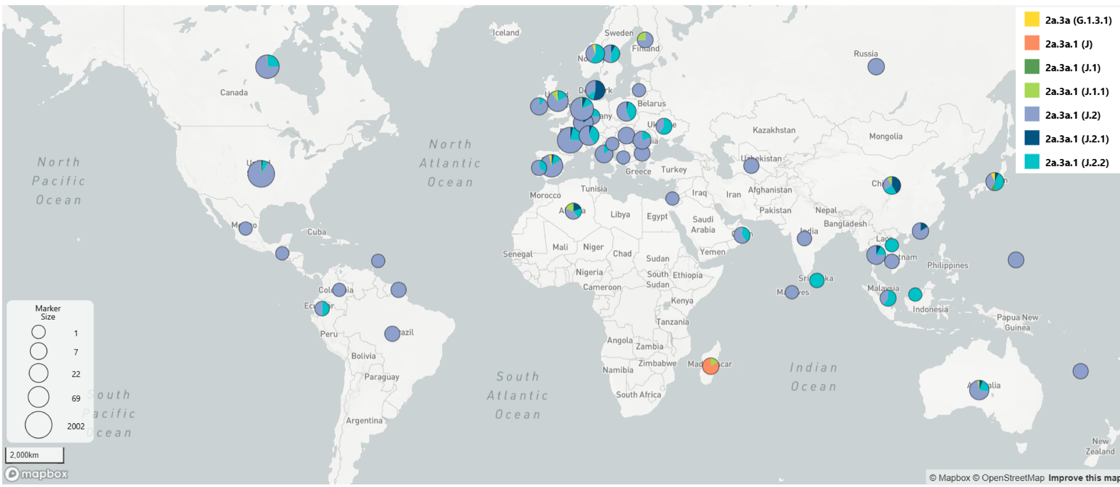


Figure 10: Global distribution of A/H3N2 genetic subclades

Global time-dependent variation of A/H3N2 subclades

Global time-dependent variation in frequencies of genetic clades-subclades of A/H3N2 viruses collected from 1st September 2024 to 7th February 2025.

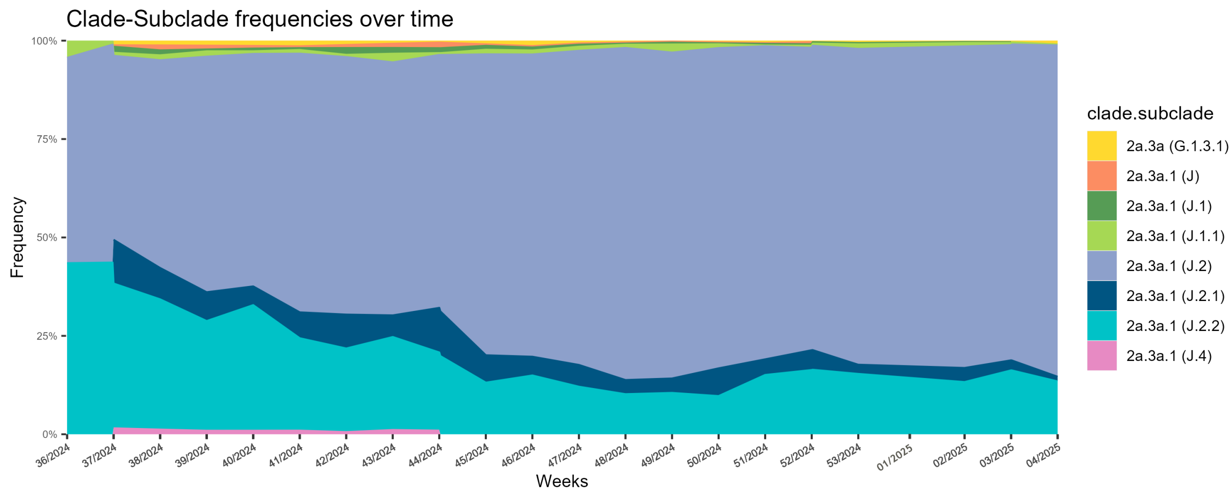


Figure 11: A/H3N2 clade-subclade frequencies over time (weeks 36/2024 to 4/2025)

Phylogenetic analysis of A/H3N2 sequences

Maximum likelihood phylogenetic tree inferred using IQTree2 with HA sequence data obtained from GISAID, collection dates from 1st September 2024 to 7th February 2025, manually curated and then downsampled using Treemmer to retain a representative tree topology of 200 sequences, with emphasis on sequences generated at the WIC. Amino acid signatures annotated with Treetime. References and CVVs are marked as Cell or Egg.

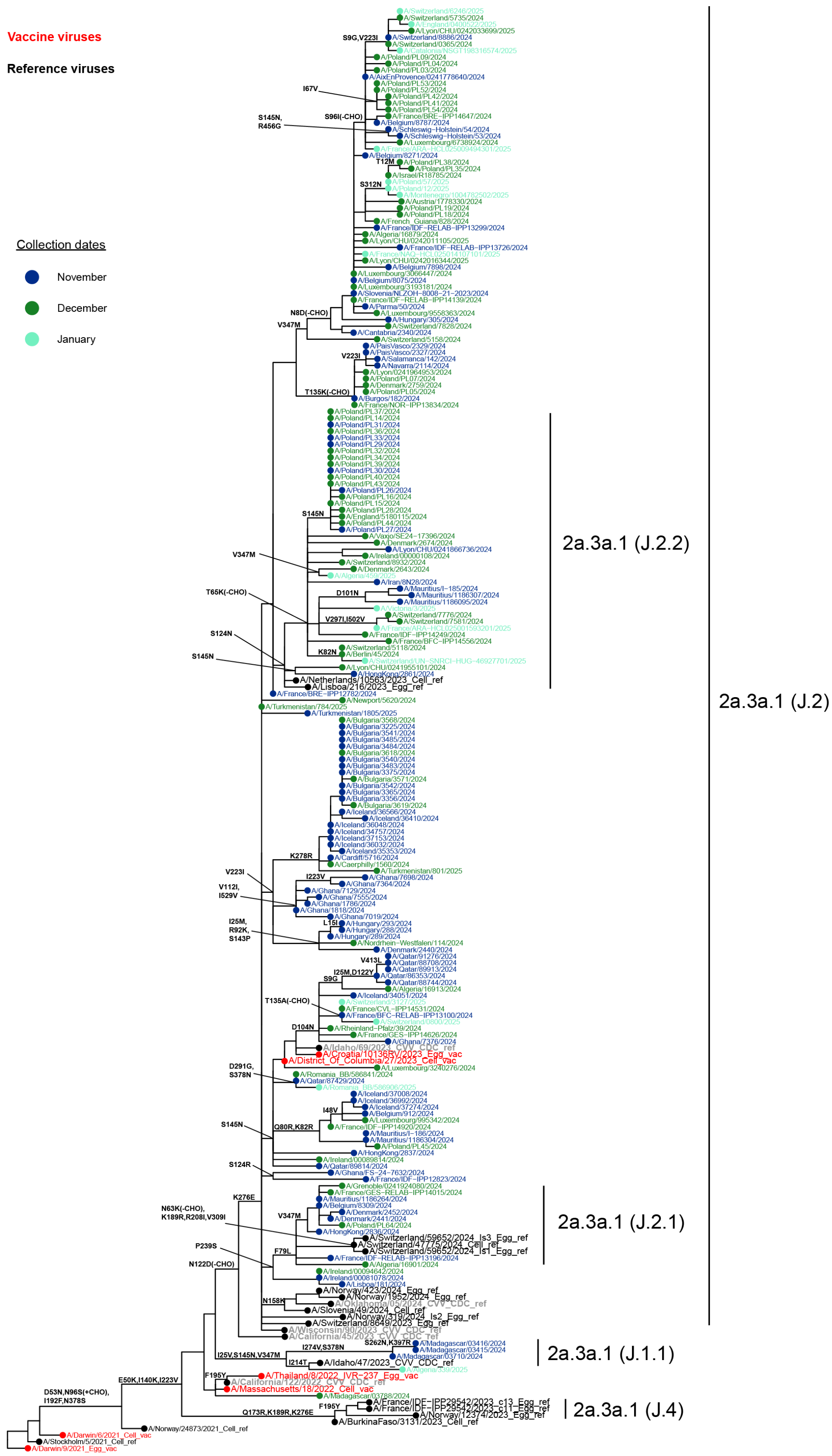


Figure 12: A/H3N2 Maximum likelihood phylogenetic tree, HA segment

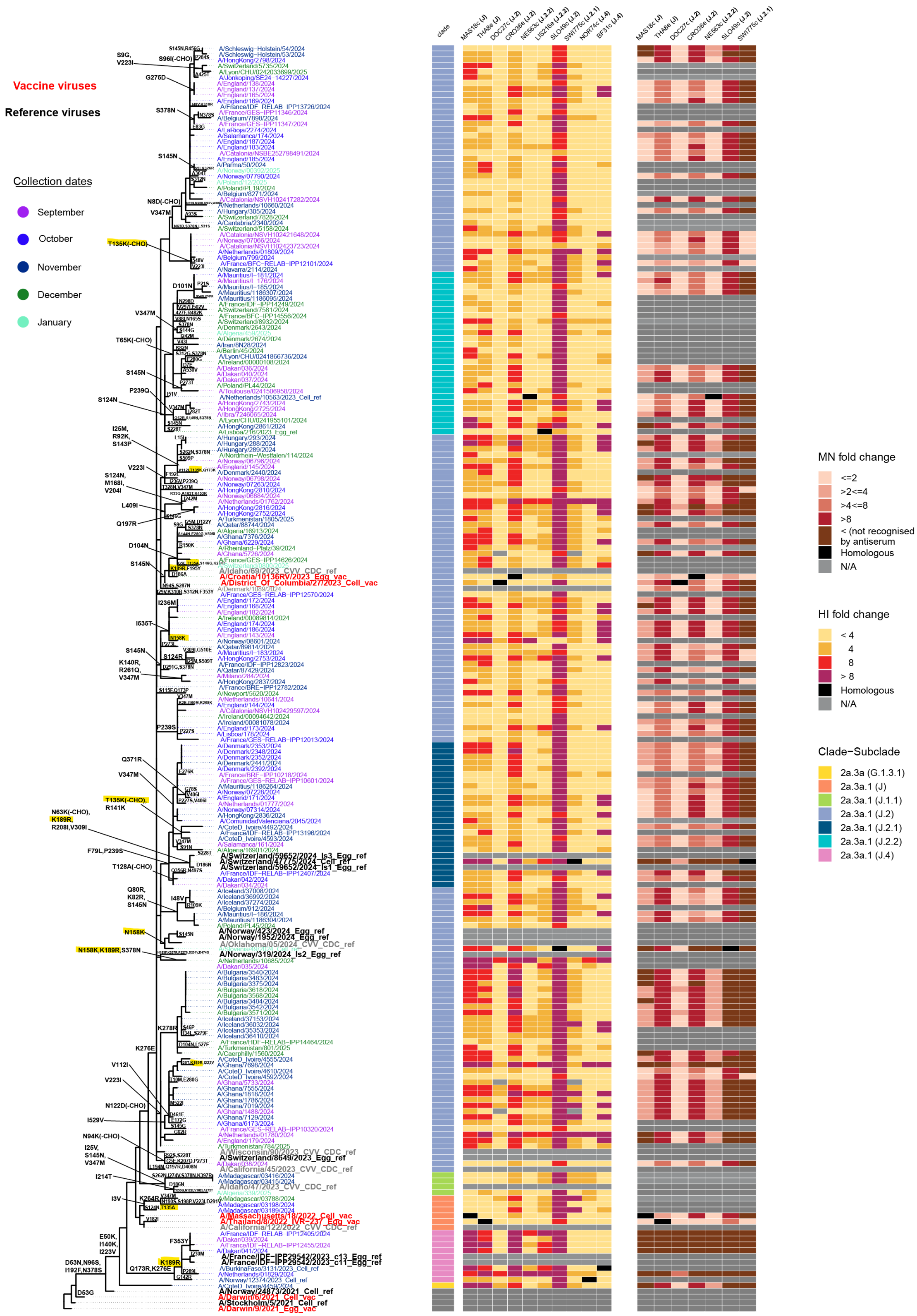


Figure 13: ML phylogenetic tree with combined heatmap representing antigenic reactivities by HI and MN

Antigenic analyses: A/H3N2

Haemagglutination inhibition tables: A/H3N2

Table H3-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-10-15

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre: Post-infection ferret antiserum										
					Passage	A/Albania /289813/2022 SIAT	assachusetts /18/2022 SIAT	A/Thailand /08/2022 Egg	A/Sydney /856/2023 SIAT	A/Croatia/ 0136RV/2023 SIAT	A/Croatia/ 0136RV/2023 Egg	/Netherlands /10563/2023 SIAT	A/Lisboa/ 216/2023 Egg	A/Slovenia /49/2024 SIAT	A/Norway /12374/2023 SIAT	/BurkinaFaso /3131/2023 SIAT
					Ferret number	F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F15/24	F11/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES															
	A/Albania/289813/2022	N122D	2a.3a.1 (J)	2022-12-13	IDCK1/SIAT2	2560	1280	2560	2560	640	640	640	640	160	320	160
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	1280	640	1280	1280	320	320	640	320	160	160	80
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	1280	320	1280	640	640	640	640	640	320	320	160
	A/Sydney/856/2023		2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT2	2560	640	1280	2560	320	640	640	640	320	160	160
	A/Croatia/10136RV/2023		2a.3a.1 (J.2)	2023-12-04	SIAT3	320	160	320	640	320	320	640	320	320	160	80
	A/Croatia/10136RV/2023		2a.3a.1 (J.2)		E3/E1	1280	320	640	1280	640	640	640	640	640	320	80
	A/Netherlands/10563/2023		2a.3a.1 (J.2.2)	2023-11-10	(-MIX2/SIAT3	1280	320	640	1280	640	640	640	640	640	320	160
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	640	320	640	1280	320	320	640	640	320	320	160
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	IDCKx/SIAT3	160	160	320	80	160	160	160	320	2560	160	160
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	320	160	320	320	160	160	320	320	160	160	160
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	40	40	160	<40	80	80	160	80	40	40	160
	TEST VIRUSES															
250026	A/Ghana/1031/2024		2a.3a.1 (J.2)	2024-07-03	SIAT1	1280	640	1280	640	640	640	640	640	640	320	160
250025	A/Ghana/1081/2024		2a.3a.1 (J.2)	2024-07-15	SIAT1	640	160	640	640	320	320	320	320	320	160	160
250024	A/Ghana/1293/2024		2a.3a.1 (J.2)	2024-08-13	SIAT1	320	160	320	640	320	640	640	320	320	160	80
250023	A/Ghana/5160/2024		2a.3a.1 (J.2)	2024-08-23	SIAT1	640	320	640	640	320	320	640	640	320	320	160
250022	A/Ghana/1406/2024		2a.3a.1 (J.2)	2024-08-27	SIAT1	640	320	640	640	320	320	640	640	320	320	160
250021	A/Ghana/5733/2024		2a.3a.1 (J.2)	2024-09-16	SIAT1	320	160	320	320	160	160	320	320	160	160	80
250019	A/Ghana/5726/2024		2a.3a.1 (J.2)	2024-09-17	SIAT1	320	160	320	640	320	320	640	320	320	160	80
250020	A/Ghana/1488/2024		2a.3a.1 (J.2)	2024-09-17	SIAT1	640	160	640	640	320	320	640	640	160	160	80
							Vaccine SH 2024 NH 2024-25			Vaccine SH 2025						

< relates to the lowest dilution of antiserum used

[1] hyperimmune sheep serum: ND = Not Done

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H3-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-11-22

	Viruses	Other information	Genetic group	Collection Date	Passage / Ferret number / Genetic group	A/Massachusetts/18/2022 SIAT F36/23 2a.3a.1 (J)	A/Thailand/08/2022 Egg F34/23 2a.3a.1 (J)	A/DistrictOfColumbia/27/2023 SIAT F38/24 2a.3a.1 (J.2)	A/Croatia/10136RV/2023 Egg F16/24 2a.3a.1 (J.2)	A/Netherlands/10563/2023 SIAT F08/24 2a.3a.1 (J.2.2)	A/Lisboa/216/2023 Egg F15/24 2a.3a.1 (J.2.2)	A/Slovenia/49/2024 SIAT F11/24 2a.3a.1 (J.2)	A/Switzerland/47775/2024 SIAT F43/24 2a.3a.1 (J.2.1)	A/Norway/12374/2023 SIAT F21/24 2a.3a.1 (J.4)	A/BurkinaFaso/3131/2023 SIAT F32/24 2a.3a.1 (J.4)	A/Norway/423/2023 Egg F40/24 2a.3a.1 (J.2)	A/Norway/1952/2024 Egg F44/24 2a.3a.1 (J.2)	A/DistrictOfColumbia/27/2023 SIAT F41/24 2a.3a.1 (J.2)
						Haemagglutination inhibition titre – Post-infection ferret antiserum												
	REFERENCE VIRUSES																	
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	320	640	160	160	320	320	160	80	160	80	80	80	320
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	640	640	320	640	640	320	160	320	80	320	320	640
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	160	320	640	640	640	320	320	160	320	160	160	160	640
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	1280	640	640	640	640	160	320	160	320	320	1280
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	:-MIX2/SIAT3	320	640	640	320	640	640	320	160	160	80	320	320	640
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	640	640	320	640	640	320	160	320	80	320	320	640
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	IDCKx/SIAT3	40	80	80	40	80	160	1280	40	40	<40	640	1280	160
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	IDCK1/SIAT2	<40	80	320	80	160	80	40	160	40	80	<40	40	160
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	80	320	320	160	320	320	160	80	160	80	80	160	320
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	<40	80	160	80	160	80	40	160	40	160	<40	40	160
243662	A/Norway/423/2023	N158K	2a.3a.1 (J.2)	2024-01-12	E6 (Am2Al4)	160	640	640	160	160	640	2560	80	160	<40	2560	2560	160
243658	A/Norway/1952/2024	S145N, N158K	2a.3a.1 (J.2)	2024-02-23	E6 (Am2Al4)	40	160	160	80	40	160	1280	40	80	<40	640	2560	160
	TEST VIRUSES																	
250058	A/SaintPetersburg/RII-39/2024		2a.3a.1 (J.2)	2024-02-16	SIAT2/SIAT1	80	160	160	80	160	160	80	40	80	40	40	80	160
250059	A/SaintPetersburg/RII-38/2024		2a.3a.1 (J.2)	2024-02-16	SIAT2/SIAT1	160	160	80	80	80	40	40	<40	<40	<40	<40	<40	80
250057	A/SaintPetersburg/RII-46/2024		2a.3a.1 (J.2)	2024-03-01	IDCK2/SIAT1	160	320	320	160	320	160	160	80	80	<40	80	80	320
250056	A/Krasnodar/RII-02/2024		2a.3a.1 (J.2)	2024-03-03	IDCK2/SIAT1	80	160	160	80	160	160	80	40	40	<40	40	80	160
250055	A/Krasnodar/RII-05/2024		2a.3a.1 (J.2)	2024-03-04	SIAT1/SIAT1	160	320	640	320	320	320	160	160	160	320	160	160	640
250053	A/Krasnodar/RII-06/2024		2a.3a.1 (J.2)	2024-03-05	SIAT1/SIAT1	80	160	320	320	320	160	160	80	160	80	80	160	320
250054	A/Krasnodar/RII-04/2024		2a.3a.1 (J.2)	2024-03-05	SIAT2/SIAT1	160	320	320	160	320	160	160	80	80	80	80	160	320
250051	A/Kemerovo/RII-02/2024		2a.3a.1 (J.2)	2024-03-06	IDCK1/SIAT1	320	640	640	320	1280	640	320	160	160	80	160	160	640
250052	A/Kemerovo/RII-01/2024		2a.3a.1 (J.2)	2024-03-06	IDCK1/SIAT1	160	320	320	160	320	320	160	80	160	40	80	160	320
250050	A/Krasnodar/RII-03/2024		2a.3a.1 (J.2)	2024-03-07	SIAT1/SIAT1	40	160	160	160	160	160	80	80	80	40	40	80	320
250049	A/Perm/RII-03/2024		2a.3a.1 (J.2)	2024-03-09	SIAT1/SIAT1	160	320	320	320	320	320	320	80	160	80	80	160	320
250048	A/Perm/RII-04/2024		2a.3a.1 (J.2)	2024-03-14	IDCK1/SIAT1	320	320	640	320	640	640	320	320	160	160	160	320	640
250047	A/Saint-Petersburg/RII-48/2024		2a.3a.1 (J.2)	2024-04-04	IDCK1/SIAT1	160	320	320	160	320	320	320	160	160	80	160	160	640
250046	A/SaintPetersburg/RII-47/2024		2a.3a.1 (J.2)	2024-04-15	IDCK1/SIAT1	80	160	160	80	160	160	80	40	80	40	40	80	320
250091	A/Catalonia/NSVH102417282/2024		2a.3a.1 (J.2)	2024-09-01	SIAT1/SIAT1	160	320	320	160	320	320	160	80	160	80	80	160	320
250087	A/Catalonia/NSVH102421648/2024		2a.3a.1 (J.2)	2024-09-07	SIAT1/SIAT1	80	160	160	80	160	160	160	40	80	<40	80	160	320
250084	A/Catalonia/NSVH102422395/2024		2a.3a.1 (J.2)	2024-09-09	SIAT1/SIAT1	160	320	320	160	320	320	320	80	160	40	160	160	320
250086	A/Catalonia/NSBE252798491/2024		2a.3a.1 (J.2)	2024-09-09	SIAT2	160	640	640	320	640	320	320	160	160	160	160	160	640
250082	A/Catalonia/NSVH102423723/2024		2a.3a.1 (J.2)	2024-09-11	SIAT1/SIAT1	160	320	640	160	320	320	320	80	160	40	160	160	320
250080	A/Catalonia/NSVH102426635/2024		2a.3a.1 (J.2)	2024-09-15	SIAT1/SIAT1	320	640	320	320	640	320	320	160	160	160	160	160	640
250078	A/Catalonia/NSVH102429597/2024		2a.3a.1 (J.2)	2024-09-20	SIAT1/SIAT1	160	320	320	160	320	320	160	80	80	80	160	160	640
						Vaccine SH 2024 NH 2024-25		Vaccine SH 2025										

< relates to the lowest dilution of antiserum used

[1] hyperimmune sheep serum; ND =

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H3-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-11-27

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre – Post-infection ferret antiserum: A/Massachusetts/18/2022	A/Thailand/08/2022	A/DistrictOfColumbia/27/2023	A/Croatia/10136RV/2023	A/Netherlands/10563/2023	A/Lisboa/216/2023	A/Slovenia/49/2024	A/Switzerland/47775/2024	A/Norway/12374/2023	A/BurkinaFaso/3131/2023
					Passage	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
					Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	320	640	160	160	160	320	80	80	80	80
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	640	640	320	320	640	320	160	160	160
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	160	320	640	320	320	320	320	160	160	80
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	640	640	640	640	640	320	320	160
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	(-MIX2/SIAT3	160	320	320	320	320	320	320	80	160	80
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	160	320	320	160	320	320	160	80	160	80
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	IDCKx/SIAT3	40	160	160	40	80	160	2560	80	40	<40
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	IDCK1/SIAT2	<40	80	160	40	80	40	<40	160	40	80
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	80	160	320	80	160	160	80	80	80	80
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	<40	80	160	40	80	40	<40	80	40	80
	TEST VIRUSES														
250137	A/Norway/06798/2024		2a.3a.1 (J.2)	2024-09-16	SIAT1	80	160	320	80	160	160	80	80	80	80
250135	A/Norway/06884/2024		2a.3a.1 (J.2)	2024-09-21	SIAT1	160	320	320	160	320	320	160	80	160	80
250136	A/Norway/06796/2024		2a.3a.1 (J.2)	2024-09-21	SIAT1	80	160	160	80	160	160	80	40	40	40
250134	A/Norway/07066/2024		2a.3a.1 (J.2)	2024-09-23	SIAT1	160	320	320	160	320	320	320	80	160	40
250131	A/Norway/07263/2024		2a.3a.1 (J.2)	2024-10-07	SIAT1	80	160	160	80	160	160	80	40	80	40
250129	A/Norway/07790/2024		2a.3a.1 (J.2)	2024-10-15	SIAT1	80	160	320	160	320	320	160	80	80	40
250132	A/Norway/07314/2024		2a.3a.1 (J.2.1)	2024-10-04	SIAT1	160	320	320	160	320	320	160	80	160	80
250133	A/Norway/07228/2024		2a.3a.1 (J.2.1)	2024-10-04	SIAT1	160	320	320	160	320	320	160	80	80	40
						Vaccine SH 2024 NH 2024-25		Vaccine SH 2025							

< relates to the lowest dilution of antiserum used
[1] hyperimmune sheep serum: ND =

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H3-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-12-10

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre									
						Post-infection ferret antiserum									
						A/Massachusetts /18/2022	A/Thailand /08/2022	A/DistrictOf Columbia/27/2023	A/Croatia/ 10136RV/2023	A/Netherlands /10563/2023	A/Lisboa/ 216/2023	A/Slovenia /49/2024	A/Switzerland /47775/2024	A/Norway /12374/2023	A/BurkinaFaso /3131/2023
					Passage	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
					Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	640	1280	160	160	320	320	160	80	160	80
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	1280	640	320	640	640	320	160	320	160
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	160	320	640	320	640	320	320	80	160	80
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	1280	1280	1280	1280	1280	640	320	320	160
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	(-MIX2/SIAT3	160	320	320	320	640	640	320	80	160	80
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	640	640	320	640	640	320	80	160	80
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	IDCKx/SIAT3	40	160	160	40	80	320	2560	40	80	<40
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	IDCK1/SIAT2	<40	80	160	80	80	80	<40	160	80	80
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	80	320	320	160	320	320	160	80	160	80
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	<40	80	160	40	80	80	<40	160	80	80
	TEST VIRUSES														
250178	A/Ibra/52422154/2024		2a.3a.1 (J.2)	2024-08-04	SIAT1	640	1280	320	320	320	320	160	80	160	80
250177	A/Firq/7245406/2024		2a.3a.1 (J.2)	2024-08-20	SIAT1	320	640	640	320	640	640	320	160	160	80
250176	A/Salalah/52423943/2024		2a.3a.1 (J.2)	2024-08-22	SIAT1	80	160	160	80	320	160	160	80	80	40
250185	A/Schleswig-Holstein/54/2024		2a.3a.1 (J.2)	2024-11-06	P2/SIAT1	320	640	640	640	640	320	320	160	160	160
250186	A/Schleswig-Holstein/53/2024		2a.3a.1 (J.2)	2024-11-06	P1/SIAT1	160	320	640	320	640	320	320	160	160	80
250173	A/Ibra/7246065/2024		2a.3a.1 (J.2.2)	2024-09-10	SIAT2	640	1280	1280	640	1280	1280	640	160	320	160
	< relates to the lowest dilution of antiserum used					Vaccine SH 2024 NH 2024-25		Vaccine SH 2025							

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H3-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-01-07

	Viruses	Other information	Genetic group	Collection Date	Passage / Ferret number / Genetic group	Haemagglutination inhibition titre – Post-infection ferret antiserum: A/Massachusetts/18/2022 SIAT F36/23 2a.3a.1 (J)	A/Thailand/08/2022 Egg F34/23 2a.3a.1 (J)	A/DistrictOfColumbia/27/2023 SIAT F38/24 2a.3a.1 (J.2)	A/Croatia/10136RV/2023 Egg F16/24 2a.3a.1 (J.2)	A/Netherlands/10563/2023 SIAT F08/24 2a.3a.1 (J.2.2)	A/Lisboa/216/2023 Egg F15/24 2a.3a.1 (J.2.2)	A/Slovenia/49/2024 SIAT F11/24 2a.3a.1 (J.2)	A/Switzerland/47775/2024 SIAT F43/24 2a.3a.1 (J.2.1)	A/Norway/12374/2023 SIAT F21/24 2a.3a.1 (J.4)	A/BurkinaFaso/3131/2023 SIAT F32/24 2a.3a.1 (J.4)	A/Croatia/10136RV/2023 Egg F45/24 2a.3a.1 (J.2)
	REFERENCE VIRUSES															
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	320	640	160	160	320	160	80	40	80	40	160
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	640	640	320	320	640	320	160	160	80	640
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	160	320	320	160	320	320	320	160	80	40	320
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	640	640	640	640	320	160	320	80	1280
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	MDCK-MIX2/SIAT3	160	320	320	320	640	640	320	160	160	80	320
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	640	640	320	640	640	320	80	160	80	640
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT3	<40	80	80	40	80	160	1280	80	40	<40	80
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	MDCK1/SIAT2	<40	40	160	40	80	40	<40	160	40	40	80
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	80	160	320	80	320	160	160	80	80	40	160
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	<40	80	160	40	80	80	<40	80	80	80	80
	TEST VIRUSES															
250293	A/Netherlands/01762/2024		2a.3a.1 (J.2)	2024-09-03	hCK1/SIAT1	40	80	80	40	80	80	40	<40	<40	<40	80
250292	A/Netherlands/01780/2024		2a.3a.1 (J.2)	2024-09-03	hCK1/SIAT1	40	40	80	40	80	80	40	<40	<40	<40	80
250465	A/England/165/2024		2a.3a.1 (J.2)	2024-09-10	SIAT1/SIAT1	80	160	160	80	160	160	80	40	80	<40	160
250464	A/England/145/2024		2a.3a.1 (J.2)	2024-09-13	SIAT1/SIAT1	40	80	160	80	160	160	80	40	80	40	160
250463	A/England/179/2024		2a.3a.1 (J.2)	2024-09-15	SIAT2/SIAT1	40	80	160	80	160	160	80	40	80	<40	160
250462	A/England/137/2024		2a.3a.1 (J.2)	2024-09-20	SIAT1/SIAT1	80	160	160	80	160	160	80	40	80	<40	160
250461	A/England/138/2024		2a.3a.1 (J.2)	2024-09-23	SIAT1/SIAT1	160	320	320	160	320	320	160	80	80	40	320
250460	A/England/182/2024		2a.3a.1 (J.2)	2024-09-27	SIAT1/SIAT1	80	160	160	160	320	160	160	40	80	<40	320
250459	A/England/143/2024		2a.3a.1 (J.2)	2024-09-30	SIAT1/SIAT1	40	80	160	80	160	160	80	40	40	<40	160
250425	A/HongKong/2753/2024		2a.3a.1 (J.2)	2024-10-01	MDCK1/SIAT1	40	80	160	80	160	80	80	40	40	<40	160
250458	A/England/168/2024		2a.3a.1 (J.2)	2024-10-02	SIAT1/SIAT1	40	80	160	80	160	160	80	40	40	<40	160
250423	A/HongKong/2752/2024		2a.3a.1 (J.2)	2024-10-02	MDCK1/SIAT1	80	160	160	80	160	160	80	40	80	<40	160
250457	A/England/169/2024		2a.3a.1 (J.2)	2024-10-03	SIAT1/SIAT1	80	160	320	160	320	320	160	40	80	40	320
250290	A/Netherlands/01809/2024		2a.3a.1 (J.2)	2024-10-03	hCK1/SIAT1	40	80	160	80	160	160	80	<40	80	<40	160
250421	A/HongKong/2798/2024		2a.3a.1 (J.2)	2024-10-06	MDCK1/SIAT1	80	160	160	160	320	320	160	80	80	40	160
250455	A/England/144/2024		2a.3a.1 (J.2)	2024-10-10	SIAT1/SIAT1	80	160	160	80	320	320	160	40	80	40	320
250454	A/England/172/2024		2a.3a.1 (J.2)	2024-10-10	SIAT1/SIAT1	40	160	320	160	160	160	160	40	80	40	160
250453	A/England/173/2024		2a.3a.1 (J.2)	2024-10-15	SIAT1/SIAT1	80	160	160	80	160	80	80	40	40	<40	160
250452	A/England/183/2024		2a.3a.1 (J.2)	2024-10-15	SIAT1/SIAT1	80	160	160	160	160	160	160	80	80	40	160
250451	A/England/186/2024		2a.3a.1 (J.2)	2024-10-15	SIAT1/SIAT1	80	160	320	160	320	160	160	40	80	<40	160
250450	A/England/174/2024		2a.3a.1 (J.2)	2024-10-17	SIAT1/SIAT1	80	160	320	160	320	160	160	40	80	<40	320
250449	A/England/185/2024		2a.3a.1 (J.2)	2024-10-18	SIAT1/SIAT1	80	320	320	160	320	320	160	80	80	40	320
250448	A/England/187/2024		2a.3a.1 (J.2)	2024-10-18	SIAT1/SIAT1	80	160	160	160	320	320	160	80	80	40	320
250419	A/HongKong/2810/2024		2a.3a.1 (J.2)	2024-10-21	MDCK1/SIAT1	80	160	320	160	320	320	160	40	80	40	320
250417	A/HongKong/2816/2024		2a.3a.1 (J.2)	2024-10-25	MDCK1/SIAT1	40	80	160	80	160	160	80	40	80	<40	160
250492	A/Mauritius/I-183/2024		2a.3a.1 (J.2)	2024-10-30	MDCKx/SIAT1	80	160	160	160	160	160	160	80	80	40	160
250413	A/HongKong/2837/2024		2a.3a.1 (J.2)	2024-11-04	MDCK1/SIAT1	40	160	160	160	320	160	160	80	80	40	320
250486	A/Mauritius/I-186/2024		2a.3a.1 (J.2)	2024-11-05	MDCKx/SIAT1	40	80	160	80	160	160	80	40	80	<40	160
250439	A/Hungary/289/2024		2a.3a.1 (J.2)	2024-11-15	MDCK1/SIAT1	80	160	160	80	160	160	80	40	80	<40	160
250438	A/Hungary/288/2024		2a.3a.1 (J.2)	2024-11-18	MDCK1/SIAT1	<40	80	80	40	160	160	80	40	80	<40	160
250437	A/Hungary/293/2024		2a.3a.1 (J.2)	2024-11-25	MDCK1/SIAT1	40	80	80	80	160	160	80	40	40	<40	160
250436	A/Hungary/305/2024		2a.3a.1 (J.2)	2024-11-25	MDCK1/SIAT1	80	160	320	160	320	320	160	40	80	40	160
250291	A/Netherlands/01777/2024		2a.3a.1 (J.2.1)	2024-09-06	hCK1/SIAT1	80	160	160	80	160	160	80	40	80	<40	160
250456	A/England/171/2024		2a.3a.1 (J.2.1)	2024-10-06	SIAT1/SIAT1	80	160	160	80	160	160	80	40	80	<40	320
250415	A/HongKong/2836/2024		2a.3a.1 (J.2.1)	2024-11-01	MDCK1/SIAT1	80	160	160	80	320	320	160	80	80	<40	160
250295	A/Netherlands/01697/2024		2a.3a.1 (J.2.2)	2024-07-16	hCK1/SIAT1	<40	40	80	80	80	80	40	<40	40	<40	80
250505	A/Mauritius/I-176/2024		2a.3a.1 (J.2.2)	2024-09-11	MDCKx/SIAT1	80	160	160	80	160	160	80	40	80	40	160
250429	A/HongKong/2725/2024		2a.3a.1 (J.2.2)	2024-09-28	MDCK1/SIAT1	80	160	160	80	160	320	160	40	80	<40	160
250427	A/HongKong/2743/2024		2a.3a.1 (J.2.2)	2024-09-30	MDCK1/SIAT1	80	160	160	80	160	160	80	40	80	<40	160
250497	A/Mauritius/I-181/2024		2a.3a.1 (J.2.2)	2024-10-24	MDCKx/SIAT1	80	160	160	80	160	160	80	40	80	<40	160
250488	A/Mauritius/I-185/2024		2a.3a.1 (J.2.2)	2024-11-05	MDCKx/SIAT1	160	160	320	160	320	320	160	80	160	40	320
250411	A/HongKong/2861/2024		2a.3a.1 (J.2.2)	2024-11-11	MDCK1/SIAT1	<40	80	160	80	160	80	80	40	40	<40	160
250297	A/Netherlands/01563/2024		no sequence	2024-04-29	hCK1/SIAT1	<40	40	80	80	160	80	40	<40	40	<40	80
						Vaccine SH 2024 NH 2024-25		Vaccine SH 2025								

< relates to the lowest dilution of antiserum used

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H3-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-01-10

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre – Post-infection ferret antiserum									
						A/Massachusetts/18/2022	A/Thailand/08/2022	A/DistrictOfColumbia/27/2023	A/Croatia/10136RV/2023	A/Netherlands/10563/2023	A/Lisboa/216/2023	A/Slovenia/49/2024	A/Switzerland/47775/2024	A/Norway/12374/2023	A/BurkinaFaso/3131/2023
					Passage	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
					Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	320	640	160	160	160	160	80	80	160	40
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	640	640	320	640	640	320	160	160	80
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	80	320	320	320	640	320	160	80	160	40
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	1280	640	640	640	640	160	320	160
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	MDCK-MIX2/SIAT3	320	320	640	320	640	640	320	160	320	80
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	640	640	320	640	640	320	80	80	80
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT3	40	160	160	40	80	320	2560	80	40	<40
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	MDCK1/SIAT2	<40	80	160	80	80	80	<40	160	80	40
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	80	160	160	160	320	160	160	80	80	40
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	<40	80	160	40	80	80	<40	80	80	80
	TEST VIRUSES														
250630	A/France/IDF-RELAB-IPP12462/2024		2a.3a.1 (J.2)	2024-09-19	C1/SIAT1	160	640	320	160	320	320	160	80	160	80
250349	A/Ghana/6173/2024		2a.3a.1 (J.2)	2024-10-01	SIAT1	80	160	160	80	160	160	80	40	80	40
250348	A/Ghana/6229/2024		2a.3a.1 (J.2)	2024-10-01	SIAT1	40	80	160	80	160	160	80	40	80	<40
250544	A/Lisboa/178/2024		2a.3a.1 (J.2)	2024-10-07	SIAT1	160	320	320	160	320	160	160	80	80	40
250624	A/France/BFC-RELAB-IPP12101/2024		2a.3a.1 (J.2)	2024-10-15	C1/SIAT1	160	160	320	160	160	320	160	40	80	<40
250622	A/France/IDF-RELAB-IPP12167/2024		2a.3a.1 (J.2)	2024-10-18	C1/SIAT1	80	160	320	160	160	160	160	80	80	40
250620	A/France/IDF-RELAB-IPP12198/2024		2a.3a.1 (J.2)	2024-10-18	C1/SIAT1	80	160	160	160	160	160	160	80	80	40
250614	A/France/IDF-RELAB-IPP12629/2024		2a.3a.1 (J.2)	2024-10-30	C1/SIAT1	160	320	320	160	320	320	160	80	80	40
250612	A/France/GES-RELAB-IPP12570/2024		2a.3a.1 (J.2)	2024-10-31	C1/SIAT1	160	320	320	160	320	320	160	80	160	40
250610	A/France/GES-RELAB-IPP12571/2024		2a.3a.1 (J.2)	2024-10-31	C1/SIAT1	80	160	160	80	160	320	160	40	80	<40
250346	A/Ghana/7019/2024		2a.3a.1 (J.2)	2024-11-01	SIAT1	40	80	80	80	160	160	40	<40	40	<40
250259	A/Qatar/87429/2024		2a.3a.1 (J.2)	2024-11-04	SIAT1	80	160	320	160	320	320	160	40	80	40
250344	A/Ghana/7129/2024		2a.3a.1 (J.2)	2024-11-06	SIAT1	40	80	160	80	160	160	80	<40	40	<40
250368	A/Ireland/00081078/2024		2a.3a.1 (J.2)	2024-11-06	SIAT1	160	320	320	160	320	320	160	80	160	40
250257	A/Qatar/88744/2024		2a.3a.1 (J.2)	2024-11-09	SIAT1	80	160	320	160	320	160	160	40	80	40
250608	A/France/BRE-IPP12782/2024		2a.3a.1 (J.2)	2024-11-12	C1/SIAT1	320	320	320	160	320	320	160	80	160	40
250255	A/Qatar/89814/2024		2a.3a.1 (J.2)	2024-11-12	SIAT1	80	160	320	160	320	320	160	80	80	40
250341	A/Ghana/1786/2024		2a.3a.1 (J.2)	2024-11-13	SIAT1	80	160	160	80	160	160	80	40	80	40
250316	A/Iceland/35353/2024		2a.3a.1 (J.2)	2024-11-13	SIAT1	80	160	160	80	160	160	80	40	80	<40
250340	A/Ghana/7555/2024		2a.3a.1 (J.2)	2024-11-15	SIAT1	40	80	80	80	160	160	80	40	80	40
250339	A/Ghana/1818/2024		2a.3a.1 (J.2)	2024-11-19	SIAT1	40	80	80	40	80	80	40	<40	40	<40
250315	A/Iceland/36032/2024		2a.3a.1 (J.2)	2024-11-20	SIAT1	80	160	160	80	160	160	80	<40	40	<40
250314	A/Iceland/36048/2024		2a.3a.1 (J.2)	2024-11-20	SIAT1	40	160	80	80	160	160	80	<40	40	<40
250313	A/Iceland/36410/2024		2a.3a.1 (J.2)	2024-11-23	SIAT1	80	160	160	160	160	160	80	40	80	40
250312	A/Iceland/36566/2024		2a.3a.1 (J.2)	2024-11-25	SIAT1	40	160	160	80	160	160	80	40	40	<40
250476	A/Mauritius/1186304/2024		2a.3a.1 (J.2)	2024-11-26	SIAT1	80	160	320	160	320	160	160	80	80	40
250338	A/Ghana/7698/2024		2a.3a.1 (J.2)	2024-11-27	SIAT1	<40	40	80	40	80	<40	<40	40	<40	<40
250311	A/Iceland/36992/2024		2a.3a.1 (J.2)	2024-11-27	SIAT1	40	80	160	80	160	160	80	40	40	<40
250310	A/Iceland/37008/2024		2a.3a.1 (J.2)	2024-11-28	SIAT1	80	160	320	160	320	160	160	40	80	40
250309	A/Iceland/37153/2024		2a.3a.1 (J.2)	2024-11-28	SIAT1	80	160	160	80	160	160	80	40	80	40
250308	A/Iceland/37274/2024		2a.3a.1 (J.2)	2024-11-29	SIAT1	80	160	160	80	160	160	80	40	80	40
250616	A/France/IDF-RELAB-IPP12407/2024		2a.3a.1 (J.2.1)	2024-10-23	C1/SIAT1	80	160	160	80	160	160	80	40	80	<40
250479	A/Mauritius/1186264/2024		2a.3a.1 (J.2.1)	2024-11-24	SIAT1	80	320	160	160	160	320	160	80	80	40
250475	A/Mauritius/1186307/2024		2a.3a.1 (J.2.2)	2024-11-26	SIAT1	160	320	320	160	320	320	160	40	80	40
250289	A/Netherlands/01829/2024	Starting 1 in 10 dilution	2a.3a.1 (J.4)	2024-10-17	hCK1/SIAT1	40	20	40	20	40	20	20	40	20	40
250618	A/France/IDF-RELAB-IPP12405/2024		2a.3a.1 (J.4)	2024-10-22	C1/SIAT1	<40	80	160	40	80	40	<40	80	40	40
	< relates to the lowest dilution of antiserum used					Vaccine SH 2024 NH 2024-25			Vaccine SH 2025						

< 4-fold   4-fold   8-fold   > 8-fold   < not recognised by the antiserum

Table H3-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-01-14

	Viruses	Other information	Genetic group	Collection Date	Passage	A/Massachusetts/18/2022 SIAT	A/Thailand/08/2022 Egg	A/DistrictOfColumbia/27/2023 SIAT	A/Croatia/10136RV/2023 Egg	A/Netherlands/10563/2023 SIAT	A/Lisboa/216/2023 Egg	A/Slovenia/49/2024 SIAT	A/Switzerland/47775/2024 SIAT	A/Norway/12374/2023 SIAT	A/BurkinaFaso/3131/2023 SIAT
					Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
						Haemagglutination inhibition titre — Post-infection ferret antiserum									
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	640	640	320	160	320	320	160	80	320	80
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	640	1280	1280	640	640	640	640	320	320	160
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	80	320	320	320	320	320	160	80	160	80
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	640	640	1280	1280	1280	1280	640	320	320	160
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	K-MIX2/SIAT3	320	320	640	320	320	640	320	160	320	160
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	640	640	320	640	640	640	160	160	160
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	IDCKx/SIAT3	40	160	160	40	80	160	2560	80	80	40
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	IDCK1/SIAT2	<40	80	160	80	80	80	<40	160	40	80
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	160	320	320	160	320	320	160	80	160	80
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	<40	80	160	40	80	80	<40	80	80	80
	TEST VIRUSES														
250728	A/Denmark/1089/2024		2a.3a.1 (J.2)	2024-02-05	SIAT2/SIAT1	160	320	320	160	320	320	160	80	80	40
251022	A/Dakar/038/2024		2a.3a.1 (J.2)	2024-09-14	IDCK1/SIAT1	320	640	640	320	640	640	320	80	160	80
250628	A/France/GES-IPP11346/2024		2a.3a.1 (J.2)	2024-09-20	C2/SIAT2	320	320	320	160	320	320	320	80	160	80
250626	A/France/GES-IPP11347/2024		2a.3a.1 (J.2)	2024-09-20	C2/SIAT2	160	320	320	160	320	320	160	80	160	80
250734	A/Netherlands/10641/2024		2a.3a.1 (J.2)	2024-09-25	K-MIX2/SIAT1	320	640	640	320	640	640	320	80	160	40
250945	A/Salamanca/174/2024		2a.3a.1 (J.2)	2024-10-07	hCKx/SIAT1	320	320	320	320	640	640	320	160	320	80
250723	A/Denmark/2440/2024		2a.3a.1 (J.2)	2024-11-11	SIAT3/SIAT1	160	160	320	160	320	320	160	80	160	80
250342	A/Ghana/7376/2024		2a.3a.1 (J.2)	2024-11-11	SIAT2	80	160	320	160	320	160	160	40	80	40
250573	A/Bulgaria/3356/2024		2a.3a.1 (J.2)	2024-11-12	SIAT1	80	320	320	160	320	320	160	40	80	40
250256	A/Qatar/89913/2024		2a.3a.1 (J.2)	2024-11-12	SIAT2	80	320	320	320	320	320	160	80	160	80
250572	A/Bulgaria/3365/2024		2a.3a.1 (J.2)	2024-11-15	SIAT1	80	160	160	80	320	160	160	40	80	40
250571	A/Bulgaria/3375/2024		2a.3a.1 (J.2)	2024-11-18	SIAT1	80	320	320	160	320	320	160	40	80	40
250569	A/Bulgaria/3483/2024		2a.3a.1 (J.2)	2024-11-25	SIAT1	80	160	160	80	160	160	80	40	80	40
250568	A/Bulgaria/3484/2024		2a.3a.1 (J.2)	2024-11-25	SIAT1	80	160	320	160	320	160	160	40	80	40
250566	A/Bulgaria/3540/2024		2a.3a.1 (J.2)	2024-11-28	SIAT1	80	160	160	80	160	160	80	40	80	40
250564	A/Bulgaria/3542/2024		2a.3a.1 (J.2)	2024-11-28	SIAT1	160	320	320	160	320	160	160	80	160	80
250560	A/Bulgaria/3571/2024		2a.3a.1 (J.2)	2024-12-02	SIAT1	80	160	160	80	160	160	80	40	80	40
250562	A/Bulgaria/3568/2024		2a.3a.1 (J.2)	2024-12-04	SIAT1	80	320	160	80	320	320	80	40	160	40
250559	A/Bulgaria/3618/2024		2a.3a.1 (J.2)	2024-12-04	SIAT1	80	160	160	80	160	160	80	40	80	40
250947	A/Salamanca/161/2024		2a.3a.1 (J.2.1)	2024-09-06	hCKx/SIAT1	160	320	320	160	320	320	160	80	80	40
250636	A/France/BRE-IPP10218/2024		2a.3a.1 (J.2.1)	2024-09-08	C1/SIAT1	320	640	160	160	320	160	80	80	80	40
250634	A/France/GES-RELAB-IPP10601/2024		2a.3a.1 (J.2.1)	2024-09-13	C1/SIAT1	320	640	320	320	320	640	160	80	160	80
250732	A/Netherlands/10645/2024		2a.3a.1 (J.2.1)	2024-09-26	K-MIX2/SIAT1	320	640	640	320	640	640	320	80	160	40
251018	A/Dakar/042/2024		2a.3a.1 (J.2.1)	2024-10-07	IDCK1/SIAT1	160	320	320	160	640	320	320	80	160	40
250727	A/Denmark/2348/2024		2a.3a.1 (J.2.1)	2024-10-10	SIAT2/SIAT1	80	160	160	80	160	160	160	80	80	40
250726	A/Denmark/2352/2024		2a.3a.1 (J.2.1)	2024-10-14	SIAT2/SIAT1	160	320	320	160	320	320	160	80	160	80
250725	A/Denmark/2353/2024		2a.3a.1 (J.2.1)	2024-10-15	SIAT2/SIAT1	80	160	320	160	320	320	160	80	80	40
250724	A/Denmark/2392/2024		2a.3a.1 (J.2.1)	2024-10-28	SIAT2/SIAT1	160	320	320	160	320	160	160	80	160	40
250722	A/Denmark/2441/2024		2a.3a.1 (J.2.1)	2024-11-11	SIAT3/SIAT1	160	320	320	160	320	320	160	80	160	80
250721	A/Denmark/2452/2024		2a.3a.1 (J.2.1)	2024-11-14	SIAT2/SIAT1	80	160	160	80	160	160	160	80	80	40
251024	A/Dakar/036/2024		2a.3a.1 (J.2.2)	2024-09-11	IDCK1/SIAT1	160	320	320	160	320	320	160	80	160	40
251023	A/Dakar/037/2024		2a.3a.1 (J.2.2)	2024-09-13	IDCK1/SIAT1	160	640	640	160	640	640	160	80	160	40
251020	A/Dakar/040/2024		2a.3a.1 (J.2.2)	2024-09-20	IDCK1/SIAT1	160	320	320	160	320	320	160	80	160	40
250632	A/France/IDF-RELAB-IPP12455/2024		2a.3a.1 (J.4)	2024-09-17	C1/SIAT2	<40	80	160	80	80	80	<40	160	80	80
251021	A/Dakar/039/2024		2a.3a.1 (J.4)	2024-09-19	IDCK1/SIAT1	<40	160	320	80	160	160	40	160	80	160
251019	A/Dakar/041/2024		2a.3a.1 (J.4)	2024-10-02	IDCK1/SIAT1	<40	160	320	80	160	80	40	160	80	160
250943	A/Salamanca/204/2024		no sequence	2024-11-21	hCKx/SIAT1	320	640	640	320	640	640	320	160	160	80

< relates to the lowest dilution of antiserum used

Vaccine SH 2024 NH 2024-25 (A/Massachusetts/18/2022, A/Thailand/08/2022)

Vaccine SH 2025 (A/DistrictOfColumbia/27/2023, A/Croatia/10136RV/2023)

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H3-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-01-17

	Viruses	Other information	Genetic group	Collection Date	Passage / Ferret number / Genetic group	Haemagglutination inhibition titre: Post-infection ferret antiserum									
					Passage	A/Massachusetts/18/2022 SIAT	A/Thailand/08/2022 Egg	A/DistrictOfColumbia/27/2023 SIAT	A/Croatia/10136RV/2023 Egg	A/Netherlands/10563/2023 SIAT	A/Lisboa/216/2023 Egg	A/Slovenia/49/2024 SIAT	A/Switzerland/47775/2024 SIAT	A/Norway/12374/2023 SIAT	A/BurkinaFaso/3131/2023 SIAT
					Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	640	1280	320	320	640	320	160	80	320	160
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	1280	640	640	640	640	320	320	320	160
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	80	160	320	160	320	320	160	80	160	80
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	1280	640	1280	640	640	320	320	160
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	MDCK-MIX2/SIAT3	160	320	320	160	320	640	320	80	160	80
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	640	1280	1280	640	640	640	640	160	320	160
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT3	40	320	320	80	160	320	2560	80	80	40
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	MDCK1/SIAT2	<40	40	160	40	80	40	<40	160	80	80
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	160	320	320	160	320	320	160	80	160	80
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	<40	80	160	40	80	40	<40	80	80	80
	TEST VIRUSES														
250661	A/CoteD'Ivoire/4459/2024		2a.3a (G.1.3.1)	2024-11-04	SIAT1	<40	80	80	80	160	160	80	40	80	40
251000	A/Madagascar/03189/2024		2a.3a.1 (J)	2024-10-21	SIAT1	320	640	160	80	160	160	80	<40	80	20
250999	A/Madagascar/03198/2024		2a.3a.1 (J)	2024-10-23	SIAT1	320	640	320	160	160	160	80	80	80	80
251001	A/Madagascar/03788/2024		2a.3a.1 (J)	2024-12-04	SIAT1	320	640	80	80	160	160	80	<40	40	40
250998	A/Madagascar/03415/2024		2a.3a.1 (J.1.1)	2024-11-11	SIAT1	320	640	160	160	160	160	80	40	40	40
250997	A/Madagascar/03416/2024		2a.3a.1 (J.1.1)	2024-11-11	SIAT1	320	640	160	160	160	160	80	40	40	80
251025	A/Dakar/035/2024		2a.3a.1 (J.2)	2024-09-11	MDCK1/SIAT1	<40	80	160	80	160	80	<40	80	40	80
250820	A/Belgium/7898/2024		2a.3a.1 (J.2)	2024-11-04	SIAT1	80	160	160	160	160	160	80	40	40	40
250662	A/CoteD'Ivoire/4452/2024		2a.3a.1 (J.2)	2024-11-04	SIAT1	160	320	320	320	640	320	320	80	160	80
250656	A/CoteD'Ivoire/4555/2024		2a.3a.1 (J.2)	2024-11-11	SIAT1	80	160	320	160	320	320	160	80	160	40
250819	A/Belgium/8075/2024		2a.3a.1 (J.2)	2024-11-14	SIAT1	160	320	320	160	320	320	160	80	160	80
250654	A/CoteD'Ivoire/4592/2024		2a.3a.1 (J.2)	2024-11-14	SIAT1	320	640	640	640	640	640	320	160	320	160
250652	A/CoteD'Ivoire/4610/2024		2a.3a.1 (J.2)	2024-11-14	SIAT1	160	320	320	160	320	320	160	80	80	40
250800	A/Switzerland/8886/2024		2a.3a.1 (J.2)	2024-11-21	SIAT1	40	160	160	80	160	160	80	40	40	40
250817	A/Belgium/8271/2024		2a.3a.1 (J.2)	2024-11-24	SIAT1	160	320	320	160	320	320	160	80	80	80
250816	A/Belgium/912/2024		2a.3a.1 (J.2)	2024-11-25	SIAT1	80	80	160	160	320	160	80	40	40	40
250567	A/Bulgaria/3485/2024		2a.3a.1 (J.2)	2024-11-25	SIAT2	40	160	160	80	160	160	80	40	40	40
250815	A/Belgium/8787/2024		2a.3a.1 (J.2)	2024-11-28	SIAT1	160	320	320	320	320	320	160	80	80	80
250565	A/Bulgaria/3541/2024		2a.3a.1 (J.2)	2024-11-28	SIAT1	80	160	160	80	160	160	80	<40	40	40
250798	A/Switzerland/0365/2024		2a.3a.1 (J.2)	2024-12-04	SIAT1	80	160	160	160	160	160	80	40	80	40
250713	A/Newport/5620/2024		2a.3a.1 (J.2)	2024-12-06	SIAT1	80	160	160	80	160	160	160	40	80	40
250797	A/Switzerland/5735/2024		2a.3a.1 (J.2)	2024-12-09	SIAT1	80	160	160	160	160	160	160	40	80	40
250795	A/Switzerland/5158/2024		2a.3a.1 (J.2)	2024-12-10	SIAT1	80	320	160	160	320	160	80	40	80	40
250712	A/Caerphilly/1560/2024		2a.3a.1 (J.2)	2024-12-13	SIAT1	80	160	160	80	160	160	80	40	80	40
251030	A/Dakar/030/2024		2a.3a.1 (J.2.1)	2024-08-02	MDCK2/SIAT1	160	320	320	160	320	320	160	80	80	40
251028	A/Dakar/032/2024		2a.3a.1 (J.2.1)	2024-08-15	MDCK1/SIAT1	160	320	320	160	320	320	160	80	80	80
251026	A/Dakar/034/2024		2a.3a.1 (J.2.1)	2024-09-10	MDCK1/SIAT1	160	320	320	160	320	320	160	80	80	40
250664	A/CoteD'Ivoire/4448/2024		2a.3a.1 (J.2.1)	2024-10-29	SIAT1	80	160	160	160	160	160	160	40	160	40
250663	A/CoteD'Ivoire/4450/2024		2a.3a.1 (J.2.1)	2024-10-30	SIAT1	40	80	80	80	160	160	80	<40	80	10
250660	A/CoteD'Ivoire/4460/2024		2a.3a.1 (J.2.1)	2024-11-04	SIAT1	320	640	640	320	320	640	640	80	320	40
250657	A/CoteD'Ivoire/4492/2024		2a.3a.1 (J.2.1)	2024-11-04	SIAT1	160	320	640	160	320	320	160	80	320	40
250653	A/CoteD'Ivoire/4593/2024		2a.3a.1 (J.2.1)	2024-11-14	SIAT1	160	320	320	160	320	320	320	80	160	40
250818	A/Belgium/8309/2024		2a.3a.1 (J.2.1)	2024-11-23	SIAT1	80	160	160	160	320	160	80	40	80	40
251029	A/Dakar/031/2024		2a.3a.1 (J.2.2)	2024-08-13	MDCK2/SIAT1	160	320	160	160	320	320	160	40	80	40
251027	A/Dakar/033/2024		2a.3a.1 (J.2.2)	2024-08-16	MDCK1/SIAT1	80	160	160	160	160	160	80	40	80	40
250799	A/Switzerland/8932/2024		2a.3a.1 (J.2.2)	2024-12-04	SIAT1	80	160	160	160	160	160	80	40	40	20
250796	A/Switzerland/5118/2024		2a.3a.1 (J.2.2)	2024-12-10	SIAT1	160	320	160	320	320	320	160	80	80	40
251031	A/Dakar/029/2024		2a.3a.1 (J.4)	2024-06-21	MDCK1/SIAT1	<40	80	160	80	80	40	<40	80	40	80
						Vaccine SH 2024 NH 2024-25		Vaccine SH 2025							

< relates to the lowest dilution of antiserum used

< 4-fold; 4-fold; 8-fold; > 8-fold; < not recognised by the antiserum

Table H3-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-01-21

	Viruses	Other information	Genetic group	Collection Date	Passage / Ferret number / Genetic group	Haemagglutination inhibition titre – Post-infection ferret antiserum									
						A/Massachusetts/18/2022	A/Thailand/08/2022	A/DistrictOfColumbia/27/2023	A/Croatia/10136RV/2023	A/Netherlands/10563/2023	A/Lisboa/216/2023	A/Slovenia/49/2024	A/Switzerland/47775/2024	A/Norway/12374/2023	A/BurkinaFaso/3131/2023
					Passage	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
					Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	640	1280	320	160	320	320	160	80	160	160
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	640	640	320	320	320	320	160	160	80
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	80	160	320	320	320	160	160	80	80	80
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	1280	640	640	640	320	160	320	160
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	(-MIX2/SIAT3	160	320	320	320	320	320	160	80	160	80
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	640	1280	1280	640	640	640	640	160	320	160
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	IDCKx/SIAT4	40	160	160	80	80	160	2560	40	40	<40
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	IDCK1/SIAT2	<40	80	160	80	80	40	<40	160	40	80
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	80	320	320	160	320	160	160	80	80	80
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	<40	80	160	40	80	40	<40	80	40	80
	TEST VIRUSES														
250637	A/France/GES-RELAB-IPP10320/2024		2a.3a.1 (J.2)	2024-09-04	C1/SIAT1	80	320	320	320	320	320	160	80	80	80
250876	A/LaRioja/2278/2024		2a.3a.1 (J.2)	2024-10-01	SIAT1	160	320	320	320	320	320	320	80	160	40
250822	A/Belgium/799/2024		2a.3a.1 (J.2)	2024-10-02	SIAT2	80	320	160	160	160	320	160	40	80	20
250875	A/ComunidadValenciana/1925/2024		2a.3a.1 (J.2)	2024-10-02	SIAT1	160	320	320	160	320	320	80	80	160	80
250873	A/LaRioja/2274/2024		2a.3a.1 (J.2)	2024-10-12	SIAT1	160	320	320	320	320	320	160	80	80	80
251063	A/Goteborg/SE24-14319/2024		2a.3a.1 (J.2)	2024-10-13	SIAT1/SIAT1	80	160	160	160	160	160	80	40	40	40
250869	A/LaRioja/2276/2024		2a.3a.1 (J.2)	2024-10-15	SIAT1	160	320	320	320	320	320	160	80	160	80
251061	A/Jonkoping/SE24-14227/2024		2a.3a.1 (J.2)	2024-10-15	SIAT1/SIAT1	80	160	160	160	320	160	160	40	80	40
250867	A/Cantabria/2340/2024		2a.3a.1 (J.2)	2024-11-01	SIAT1	160	320	320	160	320	320	160	80	160	80
250866	A/Navarra/2114/2024		2a.3a.1 (J.2)	2024-11-09	SIAT1	80	160	160	160	320	160	160	40	80	20
250941	A/Salamanca/142/2024		2a.3a.1 (J.2)	2024-11-26	SIAT1	80	160	160	160	320	160	160	40	80	10
250940	A/Burgos/182/2024		2a.3a.1 (J.2)	2024-11-27	SIAT1	320	640	320	320	320	320	320	80	160	40
250864	A/PaisVasco/2327/2024		2a.3a.1 (J.2)	2024-11-30	SIAT1	40	80	80	80	80	80	40	<40	20	<10
250863	A/PaisVasco/2329/2024		2a.3a.1 (J.2)	2024-11-30	SIAT1	80	160	320	160	320	320	160	40	80	20
251127	A/Rheinland-Pfalz/39/2024		2a.3a.1 (J.2)	2024-12-05	P1/SIAT1	80	160	160	160	160	160	80	40	40	40
251125	A/Nordrhein-Westfalen/114/2024		2a.3a.1 (J.2)	2024-12-19	P1/SIAT1	80	320	320	160	160	160	80	80	80	40
250951	A/Salamanca/2656/2024		2a.3a.1 (J.2.1)	2024-08-18	SIAT1	80	320	160	320	320	320	160	80	80	40
250950	A/Salamanca/2657/2024		2a.3a.1 (J.2.1)	2024-08-26	SIAT1	80	160	320	160	320	160	160	80	80	40
250949	A/Salamanca/2662/2024		2a.3a.1 (J.2.1)	2024-08-26	SIAT1	80	160	160	160	320	160	160	80	80	40
250948	A/Salamanca/2663/2024		2a.3a.1 (J.2.1)	2024-08-29	SIAT1	160	320	320	160	320	320	160	80	80	80
250883	A/CastillaLaMancha/2087/2024	Flu A	2a.3a.1 (J.2.1)	2024-10-23	SIAT1	160	320	320	160	320	320	160	80	160	80
250881	A/ComunidadValenciana/2045/2024	Flu A	2a.3a.1 (J.2.1)	2024-10-30	SIAT1	80	160	160	160	160	160	80	40	80	40
250651	A/CoteD'Ivoire/4612/2024		2a.3a.1 (J.2.1)	2024-11-14	SIAT2	80	160	320	160	160	160	160	40	160	10
250484	A/Mauritius/1186095/2024		2a.3a.1 (J.2.2)	2024-11-06	SIAT1	160	320	320	320	320	320	160	80	160	80
251126	A/Berlin/45/2024		2a.3a.1 (J.2.2)	2024-12-09	P2/SIAT1	80	320	320	320	320	320	160	80	80	40
250865	A/Navarra/2163/2024		no sequence	2024-11-13	SIAT1	80	160	160	160	160	160	160	40	80	10
	< relates to the lowest dilution of antiserum used					Vaccine SH 2024 NH 2024-25		Vaccine SH 2025							

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H3-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-01-29

	Viruses	Other information	Genetic group	Collection Date	Passage	A/Massachusetts/18/2022	A/Thailand/08/2022	A/DistrictOfColumbia/27/2023	A/Croatia/10136RV/2023	A/Netherlands/10563/2023	A/Lisboa/216/2023	A/Slovenia/49/2024	A/Switzerland/47775/2024	A/Norway/12374/2023	A/BurkinaFaso/3131/2023
						Haemagglutination inhibition titre									
						Post-infection ferret antiserum									
					Passage	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
					Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	640	1280	320	160	320	320	80	80	80	80
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	640	640	320	320	640	320	160	160	160
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	80	320	320	320	320	160	160	80	80	80
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	1280	1280	640	640	320	160	160	160
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	MDCK-MIX2/SIAT3	160	320	320	320	320	320	160	80	160	40
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	1280	640	320	640	640	320	160	160	80
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT4	<40	80	80	160	80	160	1280	40	20	<40
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	MDCK1/SIAT2	<40	40	160	80	80	40	<40	160	40	40
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	80	160	320	160	160	160	160	80	80	80
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	<40	80	160	80	80	40	<40	80	40	160
	TEST VIRUSES														
251233	A/Toulon/0241495940/2024		2a.3a.1 (J.2)	2024-09-18	MDCK3/SIAT1	160	320	320	320	320	320	160	80	80	80
251232	A/Limoges/0241559093/2024		2a.3a.1 (J.2)	2024-09-22	MDCK2/SIAT2	160	320	320	320	320	320	160	80	80	40
251231	A/StEtienne/0241649490/2024		2a.3a.1 (J.2)	2024-10-07	MDCK2/SIAT1	160	320	320	320	320	320	160	80	80	80
251230	A/AixEnProvence/0241778640/2024		2a.3a.1 (J.2)	2024-11-03	MDCK2/SIAT1	80	320	320	320	320	320	160	80	80	80
251128	A/Turkmenistan/1805/2025		2a.3a.1 (J.2)	2024-11-22	SIAT1	80	160	160	160	160	160	80	40	80	40
251129	A/Turkmenistan/784/2025		2a.3a.1 (J.2)	2024-12-05	SIAT1	80	160	160	160	160	160	80	40	80	40
251130	A/Turkmenistan/801/2025		2a.3a.1 (J.2)	2024-12-07	SIAT1	80	320	160	160	160	160	80	40	40	40
251227	A/Lyon/0241964953/2024		2a.3a.1 (J.2)	2024-12-11	MDCK2/SIAT2	80	320	320	160	320	320	160	40	80	20
251225	A/Lyon/CHU/0242011105/2025		2a.3a.1 (J.2)	2024-12-25	MDCK2/SIAT2	160	320	320	320	320	320	160	80	160	80
251224	A/Lyon/CHU/0242016344/2025		2a.3a.1 (J.2)	2024-12-26	MDCK2/SIAT2	160	320	320	320	320	320	160	80	160	80
251223	A/Lyon/CHU/0242033699/2025		2a.3a.1 (J.2)	2024-12-31	MDCK2/SIAT2	80	320	320	320	320	160	160	80	80	80
251228	A/Grenoble/0241924080/2024		2a.3a.1 (J.2.1)	2024-12-05	MDCK2/SIAT2	80	320	320	320	320	320	160	80	80	80
251234	A/Toulouse/0241506958/2024		2a.3a.1 (J.2.2)	2024-09-11	MDCK2/SIAT1	80	160	320	160	320	160	80	80	80	40
251229	A/Lyon/CHU/0241866736/2024		2a.3a.1 (J.2.2)	2024-11-30	MDCK2/SIAT1	80	160	160	160	160	160	80	80	80	40
251226	A/Lyon/CHU/0241955101/2024		2a.3a.1 (J.2.2)	2024-12-16	MDCK2/SIAT2	40	160	320	160	160	160	160	80	80	80
	< relates to the lowest dilution of antiserum used					Vaccine SH 2024 NH 2024-25		Vaccine SH 2025							

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H3-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-02-05

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre / Post-infection ferret antiserum									
						A/Massachusetts/18/2022	A/Thailand/08/2022	A/DistrictOfColumbia/27/2023	A/Croatia/10136RV/2023	A/Netherlands/10563/2023	A/Lisboa/216/2023	A/Slovenia/49/2024	A/Switzerland/47775/2024	A/Norway/12374/2023	A/BurkinaFaso/3131/2023
					Passage	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
					Ferret number	F36/23	F34/23	F38/24	F45/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	320	640	320	160	320	160	80	40	80	80
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	1280	640	640	640	640	320	160	160	160
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	80	320	320	320	320	320	160	80	160	80
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	1280	1280	1280	640	640	320	320	160
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	DCK-MIX2/SIAT3	160	320	320	320	320	320	160	80	160	80
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	640	320	640	320	320	320	80	160	80
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT4	40	160	160	160	80	160	2560	40	40	<10
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	MDCK1/SIAT2	<40	40	160	80	80	40	<40	160	40	40
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	160	320	320	320	320	320	160	80	80	80
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT3	<40	160	160	160	80	40	<40	160	40	160
	TEST VIRUSES														
251664	A/France/GES-RELAB-IPP10320/2024		2a.3a.1 (J.2)	2024-09-04	MDCK1/SIAT1	160	640	640	640	640	640	320	160	160	160
251384	A/Slovenia/NLZOH-8008-21-1540/2024		2a.3a.1 (J.2)	2024-09-22	MDCKfx/SIAT1	160	320	320	320	320	320	320	40	160	40
251661	A/France/GES-RELAB-IPP12013/2024		2a.3a.1 (J.2)	2024-10-18	MDCK1/SIAT1	160	320	320	320	320	320	160	80	80	80
251435	A/Parma/50/2024		2a.3a.1 (J.2)	2024-11-10	SIAT3/SIAT1	80	160	320	320	320	160	160	80	80	40
251658	A/France/IDF-IPP12823/2024		2a.3a.1 (J.2)	2024-11-13	MDCK1/SIAT1	80	160	320	320	320	160	160	80	80	80
251656	A/France/BFC-RELAB-IPP13100/2024		2a.3a.1 (J.2)	2024-11-16	MDCK1/SIAT1	80	160	160	320	320	160	160	<40	80	10
251654	A/France/IDF-RELAB-IPP13299/2024		2a.3a.1 (J.2)	2024-11-22	MDCK1/SIAT1	320	640	640	640	640	640	320	160	160	160
251382	A/Slovenia/NLZOH-8008-21-2023/2024		2a.3a.1 (J.2)	2024-11-26	SIAT1/SIAT1	80	160	160	160	160	160	80	40	80	40
251652	A/France/IDF-RELAB-IPP13726/2024		2a.3a.1 (J.2)	2024-11-27	MDCK2/SIAT1	160	320	320	320	320	320	160	80	80	80
251650	A/France/NOR-IPP13834/2024		2a.3a.1 (J.2)	2024-12-02	MDCK1/SIAT1	80	160	80	160	160	160	80	<40	40	10
251648	A/France/IDF-RELAB-IPP14139/2024		2a.3a.1 (J.2)	2024-12-03	MDCK1/SIAT1	80	320	320	320	320	320	160	80	80	80
251561	A/Denmark/2759/2024		2a.3a.1 (J.2)	2024-12-05	SIAT2/SIAT1	80	160	160	160	160	320	160	40	80	20
251425	A/Ireland/00089814/2024		2a.3a.1 (J.2)	2024-12-11	SIAT1	80	160	320	320	320	160	160	80	80	40
251639	A/France/CVL-IPP14531/2024		2a.3a.1 (J.2)	2024-12-16	MDCK1/SIAT1	80	160	320	320	320	160	160	40	80	10
251260	A/Luxembourg/3193181/2024		2a.3a.1 (J.2)	2024-12-16	SIAT1	80	320	320	320	320	160	160	40	80	40
251635	A/France/BRE-IPP14647/2024		2a.3a.1 (J.2)	2024-12-18	MDCK1/SIAT1	160	320	320	320	320	320	160	80	80	80
251637	A/France/GES-IPP14626/2024		2a.3a.1 (J.2)	2024-12-18	MDCK1/SIAT1	80	160	320	320	320	160	160	40	80	40
251257	A/Luxembourg/3066447/2024		2a.3a.1 (J.2)	2024-12-20	SIAT1	320	640	640	640	640	640	320	160	160	80
251633	A/France/IDF-IPP14920/2024		2a.3a.1 (J.2)	2024-12-21	MDCK1/SIAT1	160	320	640	640	640	320	160	80	80	80
251344	A/Algeria/16913/2024		2a.3a.1 (J.2)	2024-12-23	SIAT1	80	160	320	320	160	160	80	40	40	40
251362	A/Switzerland/7828/2024		2a.3a.1 (J.2)	2024-12-23	SIAT1	320	640	640	640	640	640	320	160	320	160
251191	A/Israel/R18785/2024		2a.3a.1 (J.2)	2024-12-30	SIAT1	160	320	320	320	320	320	160	80	80	80
251360	A/Switzerland/0800/2025		2a.3a.1 (J.2)	2025-01-07	SIAT1	80	160	320	320	320	160	160	40	80	20
251361	A/Switzerland/3127/2025		2a.3a.1 (J.2)	2025-01-07	SIAT1	160	320	320	640	320	320	320	40	80	20
251359	A/Switzerland/6246/2025		2a.3a.1 (J.2)	2025-01-08	SIAT1	80	160	320	160	320	160	160	80	80	80
251386	A/Slovenia/NLZOH-8008-21-1393/2024		2a.3a.1 (J.2.1)	2024-08-11	MDCKfx/SIAT1	160	320	320	320	320	160	160	80	80	40
251083	A/SouthSudan/983/2024		2a.3a.1 (J.2.1)	2024-08-16	SIAT1	160	320	320	320	320	320	160	80	80	80
251084	A/SouthSudan/988/2024		2a.3a.1 (J.2.1)	2024-08-16	SIAT1	160	320	320	320	320	320	160	80	160	80
251080	A/SouthSudan/987/2024		2a.3a.1 (J.2.1)	2024-08-20	SIAT1	80	320	320	320	320	320	160	80	80	40
251666	A/France/IDF-RELAB-IPP10386/2024		2a.3a.1 (J.2.1)	2024-09-03	MDCK1/SIAT1	160	320	320	320	320	320	160	80	160	80
251663	A/France/BRE-IPP10218/2024		2a.3a.1 (J.2.1)	2024-09-08	MDCK1/SIAT1	320	640	160	160	160	160	40	40	80	40
251660	A/France/IDF-RELAB-IPP13196/2024		2a.3a.1 (J.2.1)	2024-11-12	MDCK2/SIAT1	160	160	160	320	160	160	160	40	80	10
251646	A/France/GES-RELAB-IPP14015/2024		2a.3a.1 (J.2.1)	2024-12-04	MDCK1/SIAT1	80	320	320	160	320	160	160	80	80	40
251343	A/Algeria/16901/2024		2a.3a.1 (J.2.1)	2024-12-24	SIAT1	160	320	320	320	320	160	160	80	80	40
251563	A/Denmark/2643/2024		2a.3a.1 (J.2.2)	2024-12-08	SIAT2/SIAT1	160	320	320	320	320	320	160	80	160	80
251365	A/Switzerland/5118/2024		2a.3a.1 (J.2.2)	2024-12-09	SIAT1	160	320	160	320	320	320	160	80	80	40
251562	A/Denmark/2674/2024		2a.3a.1 (J.2.2)	2024-12-10	SIAT2/SIAT1	160	320	320	320	320	320	160	80	80	80
251644	A/France/IDF-IPP14249/2024		2a.3a.1 (J.2.2)	2024-12-11	MDCK1/SIAT1	80	320	320	320	320	320	160	80	80	40
251641	A/France/BFC-IPP14556/2024		2a.3a.1 (J.2.2)	2024-12-14	MDCK1/SIAT1	320	640	640	640	640	640	320	160	160	80
251364	A/Switzerland/7776/2024		2a.3a.1 (J.2.2)	2024-12-17	SIAT1	160	320	320	320	320	320	160	80	80	80
251363	A/Switzerland/7581/2024		2a.3a.1 (J.2.2)	2024-12-18	SIAT1	160	320	320	320	320	320	160	80	160	80
251424	A/Ireland/00000108/2024		2a.3a.1 (J.2.2)	2024-12-23	SIAT1	160	320	320	320	320	320	160	80	80	40
251342	A/Algeria/459/2025		2a.3a.1 (J.2.2)	2025-01-05	SIAT1	80	160	160	160	160	160	80	40	80	40
	< relates to the lowest dilution of antiserum used					Vaccine SH 2024 NH 2024-25		Vaccine SH 2025							

< 4-fold   4-fold   8-fold   > 8-fold   < not recognised by the antiserum

Table H3-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-02-11

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre / Post-infection ferret antiserum: A/Massachusetts/18/2022	A/Thailand/08/2022	A/DistrictOfColumbia/27/2023	A/Croatia/10136RV/2023	A/Netherlands/10563/2023	A/Lisboa/216/2023	A/Slovenia/49/2024	A/Switzerland/47775/2024	A/Norway/12374/2023	A/BurkinaFaso/3131/2023
					Passage	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
					Ferret number	F36/23	F34/23	F38/24	F45/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	320	320	320	320	320	320	80	80	80	160
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	640	640	640	640	320	160	320	160	160
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	160	320	320	640	320	640	80	160	80	160
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	640	1280	640	640	320	320	160	160
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	MDCK-MIX2/SIAT3	320	320	320	640	320	640	320	160	160	160
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	640	640	640	320	320	160	160	160	80
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT4	80	160	160	320	80	320	2560	80	40	20
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	MDCK1/SIAT2	40	80	320	160	160	80	<40	160	80	80
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	160	320	320	640	320	320	160	160	160	160
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT4	40	160	160	160	160	80	40	160	80	160
	TEST VIRUSES														
251341	A/Algeria/339/2025		2a.3a.1 (J.1.1)	2025-01-07	SIAT2	160	160	160	320	160	160	80	80	40	80
251436	A/Milano/284/2024		2a.3a.1 (J.2)	2024-09-24	SIAT3/SIAT2	160	160	320	320	160	320	80	80	80	80
251509	A/Poland/PL51/2024		2a.3a.1 (J.2)	2024-10-09	SIAT1	160	160	320	320	320	320	80	160	80	80
252082	A/Netherlands/10660/2024		2a.3a.1 (J.2)	2024-11-29	MDCK-MIX2/SIAT1	160	320	320	640	320	640	80	160	160	80
252080	A/Netherlands/10662/2024		2a.3a.1 (J.2)	2024-12-02	MDCK-MIX2/SIAT1	160	320	320	640	320	640	80	160	80	80
252078	A/Netherlands/10665/2024		2a.3a.1 (J.2)	2024-12-05	MDCK-MIX2/SIAT1	160	640	320	640	320	320	40	80	80	80
252076	A/Netherlands/10667/2024		2a.3a.1 (J.2)	2024-12-06	MDCK-MIX2/SIAT1	160	320	320	640	320	320	40	160	80	80
252072	A/Netherlands/10685/2024		2a.3a.1 (J.2)	2024-12-12	MDCK-MIX2/SIAT1	<40	40	40	40	<40	80	40	40	20	10
251642	A/France/HDF-RELAB-IPP14464/2024		2a.3a.1 (J.2)	2024-12-13	SIAT1	160	160	320	320	320	320	80	80	80	80
251521	A/Poland/PL19/2024		2a.3a.1 (J.2)	2024-12-18	SIAT1	160	320	320	640	320	640	160	160	80	80
251515	A/Poland/PL54/2024		2a.3a.1 (J.2)	2024-12-19	SIAT2	320	640	320	640	320	640	320	160	80	160
251423	A/Ireland/00094642/2024		2a.3a.1 (J.2)	2024-12-27	SIAT2	160	320	320	640	320	640	40	160	160	80
251758	A/Romania_BB/586841/2024		2a.3a.1 (J.2)	2024-12-27	SIAT1/SIAT1	160	160	320	640	320	320	160	160	80	80
251512	A/Poland/57/2025		2a.3a.1 (J.2)	2025-01-03	SIAT1	160	320	320	640	320	320	80	160	80	80
251757	A/Romania_BB/586906/2025		2a.3a.1 (J.2)	2025-01-03	SIAT1/SIAT1	160	320	320	640	320	40	40	80	80	80
251480	A/Montenegro/1004782502/2025		2a.3a.1 (J.2)	2025-01-08	SIAT1	160	160	320	320	160	320	160	160	80	80
251511	A/Poland/12/2025		2a.3a.1 (J.2)	2025-01-09	SIAT1	320	640	320	640	320	640	40	160	160	160
251514	A/Poland/PL64/2024		2a.3a.1 (J.2.1)	2024-12-20	SIAT1	320	160	320	640	320	640	320	160	80	160
252070	A/Netherlands/10722/2024		2a.3a.1 (J.2.1)	2024-12-24	MDCK-MIX2/SIAT1	160	320	320	640	320	320	80	160	80	80
251778	A/Iran/8N28/2024		2a.3a.1 (J.2.2)	2024-11-24	SIAT1/SIAT1	160	160	320	320	320	320	160	160	80	80
251517	A/Poland/PL40/2024		2a.3a.1 (J.2.2)	2024-12-19	SIAT1	160	160	320	320	320	320	160	80	80	80
	< relates to the lowest dilution of antiserum used					Vaccine SH 2024 NH 2024-25		Vaccine SH 2025							

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum

Table H3-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-02-14

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre									
						Post-infection ferret antiserum									
						A/Massachusetts/18/2022	A/Thailand/08/2022	A/DistrictOfColumbia/27/2023	A/Croatia/10136RV/2023	A/Netherlands/10563/2023	A/Lisboa/216/2023	A/Slovenia/49/2024	A/Switzerland/47775/2024	A/Norway/12374/2023	A/BurkinaFaso/3131/2023
					Passage	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
					Ferret number	F36/23	F34/23	F38/24	F45/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	320	640	160	320	320	320	80	40	80	80
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	1280	320	640	640	640	320	160	160	160
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	160	320	320	640	320	320	160	160	80	80
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	640	1280	640	640	640	160	160	160
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	(-MIX2/SIAT3	320	640	640	640	640	640	320	160	320	160
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	640	640	640	320	320	320	80	160	80
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	IDCKx/SIAT3	40	160	160	160	80	160	1280	40	20	<10
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	IDCK1/SIAT2	<40	80	160	80	80	80	<40	160	40	80
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	320	640	320	320	320	640	320	160	160	160
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT4	<40	160	160	160	80	80	<40	160	40	160
	TEST VIRUSES														
251882	A/Norway/08009/2024		2a.3a.1 (J.2)	2024-10-24	MDCK1	160	160	320	320	320	320	160	80	80	80
251880	A/Norway/08601/2024		2a.3a.1 (J.2)	2024-11-05	SIAT1	<40	80	80	160	40	160	2560	40	20	<10
251518	A/Poland/PL45/2024		2a.3a.1 (J.2)	2024-12-18	SIAT2	160	320	320	320	320	320	160	80	80	80
251516	A/Poland/PL53/2024		2a.3a.1 (J.2)	2024-12-19	SIAT2	80	320	320	320	320	160	160	80	80	80
251875	A/Norway/00392/2025		2a.3a.1 (J.2)	2025-01-08	SIAT1	80	160	320	320	320	160	160	80	80	80
251519	A/Poland/PL44/2024		2a.3a.1 (J.2.2)	2024-12-18	SIAT2	80	160	160	320	320	160	80	40	80	40
	< relates to the lowest dilution of antiserum used					Vaccine SH 2024 NH 2024-25		Vaccine SH 2025							

< 4-fold   4-fold   8-fold   > 8-fold   < not recognised by the antiserum

Table H3-14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-02-18

	Viruses	Other information	Genetic group	Collection Date	Passage	A/Massachusetts /18/2022 SIAT	A/Thailand /08/2022 Egg	A/DistrictOf Columbia/27/2023 SIAT	A/Croatia/ 10136RV/2023 Egg	A/Netherlands /10563/2023 SIAT	A/Lisboa/ 216/2023 Egg	A/Slovenia /49/2024 SIAT	A/Switzerland /47775/2024 SIAT	A/Norway /12374/2023 SIAT	A/BurkinaFaso /3131/2023 SIAT
						Haemagglutination inhibition titre									
						Post-infection ferret antiserum									
					Ferret number	F36/23	F34/23	F38/24	F45/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	320	640	160	160	160	160	80	40	40	80
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	640	320	640	320	640	320	160	160	160
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	160	320	320	320	320	320	160	80	80	80
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	640	640	640	640	320	160	160	80
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	(-MIX2/SIAT3	320	640	640	640	640	640	320	160	320	160
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	640	640	640	640	320	320	160	160	80
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	IDCKx/SIAT3	40	80	80	160	40	160	1280	40	20	<10
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	IDCK1/SIAT2	<40	40	160	80	80	40	<40	160	40	40
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	160	320	320	320	320	320	160	80	80	80
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT4	<40	160	160	160	160	80	<40	160	80	160
	TEST VIRUSES														
251876	A/Norway/00396/2025		2a.3a (G.1.3.1)	2025-01-08	SIAT2	40	160	160	160	160	160	80	80	80	40
252189	A/Netherlands/01577/2024	N158K	2a.3a.1 (J.2)	2024-05-06	hCK1/SIAT1	<40	40	40	80	40	80	1280	<40	10	<10
252190	A/Netherlands/01285/2024	K189R	2a.3a.1 (J.2.1)	2024-03-09	hCK1/SIAT1	<40	<40	80	40	40	<40	<40	80	20	40
251881	A/Norway/07962/2024		2a.3a.1 (J.2.2)	2024-10-28	MDCK2	160	320	320	320	320	320	160	80	80	80
252188	A/Netherlands/01829/2024	K189R	2a.3a.1 (J.4)	2024-10-17	hCK1/SIAT1	<40	40	80	40	40	40	<40	80	40	80
	< relates to the lowest dilution of antiserum used					Vaccine SH 2024 NH 2024-25		Vaccine SH 2025							

Legend	
< 4-fold	
4-fold	
8-fold	
> 8-fold	
<	not recognised by the antiserum

Haemagglutination inhibition from J.2 viruses with double substitution N158K + K189R

Table 1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-02-18

	Viruses	Other information	Genetic group	Collection Date	Passage	Haemagglutination inhibition titre – Post-infection ferret antiserum									
						A/Oklahoma/05/2024	A/Idaho/69/2023	A/Norway/423/2024	A/Norway/1952/2024	A/Slovenia/49/2024	A/Switzerland/47775/2024	A/Netherlands/01285/2024	A/Netherlands/01285/2024	A/Croatia/10136RV/2023	A/DistrictOfColumbia/27/2023
					Passage	EGG	EGG	EGG	EGG	SIAT	SIAT	CELL	CELL	Egg	SIAT
					Ferret number	F24-044	F24-053	F40/24	F44/24	F11/24	F43/24	FS24027 (K189R)	FS24028 (K189R)	F45/24	F38/24
					Genetic group	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)
	REFERENCE VIRUSES														
243827	A/Oklahoma/05/2024	N158K	2a.3a.1 (J.2)		E1/E1	2560	40	640	2560	2560	40	40	80	320	160
244236	A/Idaho/69/2023	K189R, S145N	2a.3a.1 (J.2)		E1/E1	<40	160	<40	80	<40	160	160	160	160	320
243850	A/Switzerland/59652/2024	K189R (EGG S228T)	2a.3a.1 (J.2)	2024-03-05	E3(Am1Al2)	<40	80	<40	<40	40	160	160	160	80	160
243850	A/Switzerland/59652/2024	K189R (EGG D186N)	2a.3a.1 (J.2)	2024-03-05	E3(Am1Al2)	<40	40	<40	<40	<40	160	160	160	80	80
243662	A/Norway/423/2024	N158K	2a.3a.1 (J.2)	2024-01-12	E6/E1	2560	40	640	2560	1280	<40	40	80	320	80
243658	A/Norway/1952/2024	N158K, S145N	2a.3a.1 (J.2)	2024-02-23	E6/E1	2560	80	640	5120	2560	40	80	80	320	160
252072	A/Netherlands/10685/2024	K189R, N158K	2a.3a.1 (J.2)	2024-12-12	‹-MIX2/SIAT1	160	40	160	320	320	<40	<40	80	<40	<40
241754	A/Slovenia/49/204	N158K	2a.3a.1 (J.2)	2024-01-08	ıDCKx/SIAT3	1280	40	1280	2560	2560	40	40	80	160	80
243851	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	ıDCK1/SIAT2	<40	40	<40	<40	<40	160	320	320	80	160
240929	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	40	160	160	160	320	160	320	320	640	640
243805	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	40	80	40	80	160	80	80	80	320	320

< relates to the lowest dilution of antiserum used
ND = Not Done

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table H3-15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2025-02-20

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre – Post-infection ferret antiserum									
						A/Massachusetts /18/2022	A/Thailand /08/2022	A/DistrictOf olumbia/27/2023	A/Croatia/ 10136RV/2023	A/Netherlands /10563/2023	A/Lisboa/ 216/2023	A/Slovenia /49/2024	A/Switzerland /47775/2024	A/Norway /12374/2023	A/BurkinaFaso /3131/2023
					Passage	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
					Ferret number	F36/23	F34/23	F38/24	F45/24	F08/24	F15/24	F11/24	F43/24	F21/24	F32/24
					Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2.1)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
	REFERENCE VIRUSES														
	A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT2	320	640	160	160	160	160	80	40	40	80
	A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	320	640	320	640	320	640	320	160	160	160
	A/DistrictOfColumbia/27/2023	S145N	2a.3a.1 (J.2)	2023-12-09	SIAT2/SIAT1	160	320	320	320	320	320	160	80	80	80
	A/Croatia/10136RV/2023	S145N	2a.3a.1 (J.2)	2023-12-04	E3/E1	320	640	640	640	640	640	320	160	160	80
	A/Netherlands/10563/2023	S124N	2a.3a.1 (J.2.2)	2023-11-10	(-MIX2/SIAT3	320	640	640	640	640	640	320	160	320	160
	A/Lisboa/216/2023	S124N	2a.3a.1 (J.2.2)	2023-12-15	E3 (Am1Al2)	320	640	640	640	640	320	320	160	160	80
	A/Slovenia/49/2024	N158K	2a.3a.1 (J.2)	2024-01-08	IDCKx/SIAT3	40	80	80	160	40	160	1280	40	20	<10
	A/Switzerland/47775/2024	K189R	2a.3a.1 (J.2.1)	2024-03-04	IDCK1/SIAT2	<40	40	160	80	80	40	<40	160	40	40
	A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	160	320	320	320	320	320	160	80	80	80
	A/BurkinaFaso/3131/2023	R142G, K189R	2a.3a.1 (J.4)	2023-11-20	SIAT4	<40	160	160	160	160	80	<40	160	80	160
	TEST VIRUSES														
252183	A/Netherlands/00024/2025	N158K + K189R	2a.3a.1 (J.2)	2025-01-01	SIAT1	<40	<40	<40	<40	<40	<40	160	<40	<10	<10
252181	A/Netherlands/00077/2025	N158K + K189R	2a.3a.1 (J.2)	2025-01-06	SIAT1	<40	<40	<40	<40	<40	40	320	40	10	<10
252182	A/Netherlands/00156/2025	N158K + K189R	2a.3a.1 (J.2)	2025-01-03	SIAT1	<40	<40	<40	<40	<40	80	320	40	10	<10
252179	A/Netherlands/00328/2025	N158K + K189R	2a.3a.1 (J.2)	2025-01-27	SIAT1	<40	<40	<40	<40	<40	40	320	<40	10	<10
252178	A/Netherlands/00401/2025	N158K + K189R	2a.3a.1 (J.2)	2025-01-27	SIAT1	<40	<40	<40	<40	<40	40	320	<40	<10	<10
252177	A/Netherlands/00520/2025	N158K + K189R	2a.3a.1 (J.2)	2025-01-27	SIAT1	<40	<40	<40	<40	<40	40	320	<40	<10	<10
252187	A/Netherlands/02093/2024	N158K + K189R	2a.3a.1 (J.2)	2024-12-17	SIAT1	<40	<40	<40	<40	<40	40	320	<40	<10	<10
252184	A/Netherlands/02175/2024	N158K + K189R	2a.3a.1 (J.2)	2024-12-29	SIAT1	<40	<40	<40	<40	<40	80	320	<40	10	<10
252185	A/Netherlands/02182/2024	N158K + K189R	2a.3a.1 (J.2)	2024-12-25	SIAT1	<40	<40	<40	<40	<40	80	320	40	10	<10
						Vaccine SH 2024 NH 2024-25		Vaccine SH 2025							

< relates to the lowest dilution of antiserum used

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Microneutralisation tables: A/H3N2

Table MN-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-12-02

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 18/2022	A/Thailand 8/2022	A/D.Columbia 27/2023	A/Croatia 10136/RV/2023	A/Netherlands 10563/2023	A/Slovenia 49/2024	A/Switzerland 47775/2024
			Passage history	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT
			Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F11/24	F43/24
Viruses	Genetic group	Collection Date	Passage History	J	J	J.2 + S145N	J.2 + S145N	J.2.2	J.2 + N158K	J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	1294	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/Ghana/5726/2024	J.2	2024-09-17	SIAT1	<	59	158	140	208	55	<
A/Ghana/1488/2024	J.2	2024-09-17	SIAT1	43	60	104	70	133	<	<
A/Ghana/5733/2024	J.2	2024-09-16	SIAT1	44	56	119	69	150	<	<
A/Catalonia/NSVH102429597/2024	J.2	2024-09-20	SIAT1/SIAT1	101	143	257	138	262	109	<
A/Catalonia/NSVH102423723/2024	J.2	2024-09-11	SIAT1/SIAT1	202	299	438	250	317	203	56
A/Catalonia/NSVH102421648/2024	J.2	2024-09-07	SIAT1/SIAT1	181	271	403	189	273	141	50
A/Catalonia/NSVH102417282/2024	J.2	2024-09-01	SIAT1/SIAT1	84	124	221	127	261	53	<

<=2-fold >2<4-fold >4<8-fold >8-fold not recognised by the antiserum

Table MN-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-12-09

				Neutralisation Titre						
				Post-infection ferret antisera						
			Passage history	A/Massachusetts 18/2022 SIAT	A/Thailand 8/2022 Egg	A/D.Columbia 27/2023 SIAT	A/Croatia 10136/RV/2023 Egg	A/Netherlands 10563/2023 SIAT	A/Slovenia 49/2024 SIAT	A/Switzerland 47775/2024 SIAT
			Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F11/24	F43/24
Viruses	Genetic group	Collection Date	Passage History	J	J	J.2 + S145N	J.2 + S145N	J.2.2	J.2 + N158K	J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	1294	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/Catalonia/NSBE252798491/2024	J.2	2024-09-09	SIAT2	137	220	284	171	263	88	<
A/Norway/07790/2024	J.2	2024-10-15	SIAT1	135	238	285	185	292	67	<
A/Norway/07263/2024	J.2	2024-10-07	SIAT1	51	74	111	70	118	<	<
A/Norway/07066/2024	J.2	2024-09-23	SIAT1	119	212	262	153	245	124	40
A/Norway/06884/2024	J.2	2024-09-21	SIAT1	149	279	229	244	278	114	<
A/Norway/06796/2024	J.2	2024-09-21	SIAT1	41	61	96	65	111	<	<
A/Norway/06798/2024	J.2	2024-09-16	SIAT1	<	60	147	59	46	<	<
A/Norway/07314/2024	J.2.1	2024-10-04	SIAT1	128	174	238	148	257	78	<
A/Norway/07228/2024	J.2.1	2024-10-04	SIAT1	127	189	243	176	252	66	<

<=2-fold   >2<4-fold   >4<8-fold   >8-fold   not recognised by the antiserum

Table MN-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2025-01-09

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts	A/Thailand	A/D.Columbia	A/Croatia	A/Netherlands	A/Slovenia	A/Switzerland
				18/2022	8/2022	27/2023	10136/RV/2023	10563/2023	49/2024	47775/2024
			Passage history	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT
			Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F11/24	F43/24
Viruses	Genetic group	Collection Date	Passage History	J	J	J.2 + S145N	J.2 + S145N	J.2.2	J.2 + N158K	J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	1294	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/Schleswig-Holstein/54/2024	J.2	2024-11-06	P2/SIAT1	<	40	129	86	140	42	<
A/Schleswig-Holstein/53/2024	J.2	2024-11-06	P1/SIAT1	<	40	138	83	133	<	<
A/Ibra/7246065/2024	J.2.2	2024-09-10	SIAT2	64	110	201	74	240	53	<

<=2-fold   >2<4-fold   >4<8-fold   >8-fold   not recognised by the antiserum

Table MN-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2025-01-13

Viruses	Genetic group	Collection Date	Passage History	Neutralisation Titre Post-infection ferret antisera						
			Passage history Ferret number	A/Massachusetts 18/2022 SIAT F36/23 J	A/Thailand 8/2022 Egg F34/23 J	A/D.Columbia 27/2023 SIAT F38/24 J.2 + S145N	A/Croatia 10136/RV/2023 Egg F16/24 J.2 + S145N	A/Netherlands 10563/2023 SIAT F08/24 J.2.2	A/Slovenia 49/2024 SIAT F11/24 J.2 + N158K	A/Switzerland 47775/2024 SIAT F43/24 J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	1294	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/Netherlands/01809/2024	J.2	2024-10-03	hCK1/SIAT1	205	337	662	244	400	175	79
A/Netherlands/01780/2024	J.2	2024-09-03	hCK1/SIAT1	<	51	147	53	135	<	<
A/Netherlands/01762/2024	J.2	2024-09-03	hCK1/SIAT1	67	97	196	101	199	46	<
A/HongKong/2837/2024	J.2	2024-11-04	MDCK 1/SIAT1	75	110	462	195	283	117	68
A/HongKong/2816/2024	J.2	2024-10-25	MDCK 1/SIAT1	<	40	70	<	64	<	<
A/HongKong/2810/2024	J.2	2024-10-21	MDCK 1/SIAT1	192	249	462	173	456	119	49
A/HongKong/2798/2024	J.2	2024-10-06	MDCK 1/SIAT1	112	134	277	127	254	66	<
A/HongKong/2752/2024	J.2	2024-10-02	MDCK 1/SIAT1	<	52	97	42	91	<	<
A/HongKong/2753/2024	J.2	2024-10-01	MDCK 1/SIAT1	66	78	381	182	294	83	50
A/Netherlands/01777/2024	J.2.1	2024-09-06	hCK1/SIAT1	106	160	476	146	386	67	<
A/HongKong/2836/2024	J.2.1	2024-11-01	MDCK 1/SIAT1	135	202	382	151	407	80	<
A/HongKong/2861/2024	J.2.2	2024-11-11	MDCK 1/SIAT1	<	52	147	73	121	49	<
A/HongKong/2743/2024	J.2.2	2024-09-30	MDCK 1/SIAT1	115	200	366	153	296	96	<
A/HongKong/2725/2024	J.2.2	2024-09-28	MDCK 1/SIAT1	125	205	315	145	402	80	<
A/Netherlands/01829/2024	J.4	2024-10-17	hCK1/SIAT1	<	<	44	<	<	<	<

<=2-fold   >2<4-fold   >4<8-fold   >8-fold   not recognised by the antiserum

Table MN-5. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2025-01-20

				Neutralisation Titre						
				Post-infection ferret antisera						
			Passage history	A/Massachusetts 18/2022 SIAT	A/Thailand 8/2022 Egg	A/D.Columbia 27/2023 SIAT	A/Croatia 10136/RV/2023 Egg	A/Netherlands 10563/2023 SIAT	A/Slovenia 49/2024 SIAT	A/Switzerland 47775/2024 SIAT
			Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F11/24	F43/24
Viruses	Genetic group	Collection Date	Passage History	J	J	J.2 + S145N	J.2 + S145N	J.2.2	J.2 + N158K	J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	1294	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/England/185/2024	J.2	2024-10-18	SIAT1/SIAT1	113	180	276	144	305	64	<
A/England/174/2024	J.2	2024-10-17	SIAT1/SIAT1	57	74	274	143	293	57	<
A/England/186/2024	J.2	2024-10-15	SIAT1/SIAT1	48	67	238	139	270	52	<
A/England/183/2024	J.2	2024-10-15	SIAT1/SIAT1	124	199	305	154	310	65	<
A/England/173/2024	J.2	2024-10-15	SIAT1/SIAT1	131	194	309	158	382	86	<
A/England/172/2024	J.2	2024-10-10	SIAT1/SIAT1	47	64	240	125	231	68	<
A/England/144/2024	J.2	2024-10-10	SIAT1/SIAT1	124	199	294	138	432	103	<
A/England/169/2024	J.2	2024-10-03	SIAT1/SIAT1	113	169	274	148	306	60	<
A/England/168/2024	J.2	2024-10-02	SIAT1/SIAT1	<	62	182	122	194	49	<
A/England/143/2024	J.2	2024-09-30	SIAT1/SIAT1	50	73	242	144	263	46	<
A/England/182/2024	J.2	2024-09-27	SIAT1/SIAT1	51	69	250	142	247	50	<
A/England/138/2024	J.2	2024-09-23	SIAT1/SIAT1	141	236	311	185	305	71	<
A/England/137/2024	J.2	2024-09-20	SIAT1/SIAT1	120	231	404	192	497	68	<
A/England/179/2024	J.2	2024-09-15	SIAT2/SIAT1	<	54	114	55	127	<	<
A/England/171/2024	J.2.1	2024-10-06	SIAT1/SIAT1	106	147	238	145	302	50	<

<=2-fold   >2<4-fold   >4<8-fold   >8-fold   not recognised by the antiserum

Table MN-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2025-01-23

				Neutralisation Titre						
				Post-infection ferret antisera						
			Passage history	A/Massachusetts 18/2022 SIAT	A/Thailand 8/2022 Egg	A/D.Columbia 27/2023 SIAT	A/Croatia 10136/RV/2023 Egg	A/Netherlands 10563/2023 SIAT	A/Slovenia 49/2024 SIAT	A/Switzerland 47775/2024 SIAT
			Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F11/24	F43/24
Viruses	Genetic group	Collection Date	Passage History	J	J	J.2 + S145N	J.2 + S145N	J.2.2	J.2 + N158K	J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	624	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/Hungary/305/2024	J.2	2024-11-25	MDCK 1/SIAT1	113	187	279	150	310	68	<
A/Hungary/293/2024	J.2	2024-11-25	MDCK 1/SIAT1	40	66	154	70	184	44	<
A/Hungary/288/2024	J.2	2024-11-18	MDCK 1/SIAT1	<	59	136	57	169	<	<
A/Hungary/289/2024	J.2	2024-11-15	MDCK 1/SIAT1	45	64	130	63	149	<	<
A/England/187/2024	J.2	2024-10-18	SIAT1/SIAT1	133	226	343	155	306	72	<
A/England/145/2024	J.2	2024-09-13	SIAT1/SIAT1	61	79	174	114	211	47	<
A/England/165/2024	J.2	2024-09-10	SIAT1/SIAT1	157	240	427	206	371	108	<
A/ Mauritius /I-186/ 2024	J.2	2024-11-05	MDCKx/SIAT1	66	113	330	194	302	69	<
A/ Mauritius /I-183/ 2024	J.2	2024-10-30	MDCKx/SIAT1	60	68	260	128	160	73	43
A/ Mauritius /I-181/ 2024	J.2.2	2024-10-24	MDCKx/SIAT1	107	158	257	125	244	63	<
A/ Mauritius /I-176/ 2024	J.2.2	2024-09-11	MDCKx/SIAT1	73	141	224	94	206	59	<
A/ Mauritius /I-185/ 2024	J.2.2	2024-11-05	MDCKx/SIAT1	117	170	320	140	338	62	<

<=2-fold >2<4-fold >4<8-fold >8-fold not recognised by the antiserum

Table MN-7. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2025-01-30

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 18/2022	A/Thailand 8/2022	A/D.Columbia 27/2023	A/Croatia 10136/RV/2023	A/Netherlands 10563/2023	A/Slovenia 49/2024	A/Switzerland 47775/2024
			Passage history	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT
			Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F11/24	F43/24
Viruses	Genetic group	Collection Date	Passage History	J	J	J.2 + S145N	J.2 + S145N	J.2.2	J.2 + N158K	J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	624	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/Qatar/89814/2024	J.2	2024-11-12	SIAT1	73	141	230	233	254	77	<
A/Qatar/88744/2024	J.2	2024-11-09	SIAT1	59	96	264	149	276	63	<
A/Qatar/87429/2024	J.2	2024-11-04	SIAT1	74	127	230	195	244	74	<
A/Iceland/37274/2024	J.2	2024-11-29	SIAT1	69	115	191	156	198	76	<
A/Iceland/37153/2024	J.2	2024-11-28	SIAT1	55	77	100	61	121	<	<
A/Iceland/37008/2024	J.2	2024-11-28	SIAT1	58	93	158	135	197	75	<
A/Iceland/36992/2024	J.2	2024-11-27	SIAT1	70	119	292	160	303	83	<
A/Iceland/36032/2024	J.2	2024-11-20	SIAT1	54	80	175	73	207	<	<
A/Ghana/7698/2024	J.2	2024-11-27	SIAT1	<	<	68	<	<	<	<
A/Ghana/1818/2024	J.2	2024-11-19	SIAT1	53	69	117	57	132	<	<
A/Ghana/7555/2024	J.2	2024-11-15	SIAT1	61	85	150	65	132	<	<
A/Ghana/1786/2024	J.2	2024-11-13	SIAT1	51	74	116	62	145	<	<
A/Ghana/7129/2024	J.2	2024-11-06	SIAT1	47	73	134	67	157	<	<
A/Ghana/7019/2024	J.2	2024-11-01	SIAT1	59	77	111	63	133	<	<
A/Ghana/6229/2024	J.2	2024-10-01	SIAT1	51	88	229	126	177	63	<

<=2-fold   >2<4-fold   >4<8-fold   >8-fold   not recognised by the antiserum

Table MN-8. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2025-02-03

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts	A/Thailand	A/D.Columbia	A/Croatia	A/Netherlands	A/Slovenia	A/Switzerland
				18/2022	8/2022	27/2023	10136/RV/2023	10563/2023	49/2024	47775/2024
			Passage history	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT
			Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F11/24	F43/24
Viruses	Genetic group	Collection Date	Passage History	J	J	J.2 + S145N	J.2 + S145N	J.2.2	J.2 + N158K	J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	624	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/France/BFC-RELAB-IPP12101/2024	J.2	2024-10-15	C1/SIAT1	144	492	723	261	475	217	66
A/France/GES-IPP11347/2024	J.2	2024-09-20	C2/SIAT2	112	285	216	130	282	74	<
A/Denmark/1089/2024	J.2	2024-02-05	SIAT2/SIAT1	86	268	221	131	263	70	<
A/Salamanca/174/2024	J.2	2024-10-07	hCKx/SIAT1	64	191	236	119	274	59	<
A/France/IDF-RELAB-IPP12407/2024	J.2.1	2024-10-23	C1/SIAT1	44	118	133	67	197	<	<
A/France/GES-RELAB-IPP10601/2024	J.2.1	2024-09-13	C1/SIAT1	67	150	182	107	258	57	<
A/Denmark/2441/2024	J.2.1	2024-11-11	SIAT3/SIAT1	73	207	144	97	164	68	<
A/Denmark/2392/2024	J.2.1	2024-10-28	SIAT2/SIAT1	86	226	160	112	231	57	<
A/Denmark/2353/2024	J.2.1	2024-10-15	SIAT2/SIAT1	75	223	145	113	174	60	<
A/Denmark/2352/2024	J.2.1	2024-10-14	SIAT2/SIAT1	71	181	131	87	160	67	<
A/Denmark/2348/2024	J.2.1	2024-10-10	SIAT2/SIAT1	65	158	171	92	236	53	<
A/Salamanca/161/2024	J.2.1	2024-09-06	hCKx/SIAT1	57	151	160	91	170	48	<
A/France/IDF-RELAB-IPP12405/2024	J.4	2024-10-22	C1/SIAT1	<	<	<	<	<	<	<
A/France/IDF-RELAB-IPP12455/2024	J.4	2024-09-17	C1/SIAT2	<	<	<	<	<	<	<
A/Salamanca/204/2024	no seq	2024-11-21	hCKx/SIAT1	47	118	160	87	176	<	<

<=2-fold   >2<4-fold   >4<8-fold   >8-fold   not recognised by the antiserum

Table MN-9. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2025-02-06

				Neutralisation Titre						
				Post-infection ferret antisera						
			Passage history	A/Massachusetts 18/2022	A/Thailand 8/2022	A/D.Columbia 27/2023	A/Croatia 10136/RV/2023	A/Netherlands 10563/2023	A/Slovenia 49/2024	A/Switzerland 47775/2024
				SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT
			Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F11/24	F43/24
Viruses	Genetic group	Collection Date	Passage History	J	J	J.2 + S145N	J.2 + S145N	J.2.2	J.2 + N158K	J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	624	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/Ireland/00081078/2024	J.2	2024-11-06	SIAT1	76	240	132	95	163	69	<
A/ Mauritius /1186304/ 2024	J.2	2024-11-26	SIAT1	52	101	137	111	158	61	<
A/Lisboa/178/2024	J.2	2024-10-07	SIAT1	69	172	136	85	158	57	<
A/Bulgaria/3618/2024	J.2	2024-12-04	SIAT1	41	86	87	52	116	<	<
A/Bulgaria/3571/2024	J.2	2024-12-02	SIAT1	44	76	78	52	110	<	<
A/Bulgaria/3568/2024	J.2	2024-12-04	SIAT1	50	97	111	58	140	<	<
A/Bulgaria/3542/2024	J.2	2024-11-28	SIAT1	40	75	112	50	147	<	<
A/Bulgaria/3540/2024	J.2	2024-11-28	SIAT1	<	72	92	51	130	<	<
A/Bulgaria/3484/2024	J.2	2024-11-25	SIAT1	<	76	113	58	164	<	<
A/Bulgaria/3483/2024	J.2	2024-11-25	SIAT1	<	60	90	40	137	<	<
A/Bulgaria/3375/2024	J.2	2024-11-18	SIAT1	<	62	103	<	130	<	<
A/ Mauritius /1186264/ 2024	J.2.1	2024-11-24	SIAT1	68	153	126	77	164	65	<
A/ Mauritius /1186307/ 2024	J.2.2	2024-11-26	SIAT1	132	331	254	135	298	109	<

<=2-fold   >2<4-fold   >4<8-fold   >8-fold   not recognised by the antiserum

Table MN-10. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2025-02-10

				Neutralisation Titre						
				Post-infection ferret antisera						
			Passage history	A/Massachusetts 18/2022	A/Thailand 8/2022	A/D.Columbia 27/2023	A/Croatia 10136/RV/2023	A/Netherlands 10563/2023	A/Slovenia 49/2024	A/Switzerland 47775/2024
				SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT
			Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F11/24	F43/24
Viruses	Genetic group	Collection Date	Passage History	J	J	J.2 + S145N	J.2 + S145N	J.2.2	J.2 + N158K	J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	624	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/CoteD'Ivoire/4459/2024	G.1.3.1	2024-11-04	SIAT1	<	67	93	44	109	<	<
A/Dakar/038/2024	J.2	2024-09-14	MDCK1/SIAT1	105	234	178	128	226	89	<
A/CoteD'Ivoire/4610/2024	J.2	2024-11-14	SIAT1	95	234	176	132	234	71	<
A/CoteD'Ivoire/4592/2024	J.2	2024-11-14	SIAT1	78	135	273	276	235	208	78
A/CoteD'Ivoire/4555/2024	J.2	2024-11-11	SIAT1	53	92	114	73	160	<	<
A/Caerphilly/1560/2024	J.2	2024-12-13	SIAT1	<	65	116	60	160	44	<
A/Newport/5620/2024	J.2	2024-12-06	SIAT1	70	167	141	109	201	57	<
A/Dakar/042/2024	J.2.1	2024-10-07	MDCK1/SIAT1	71	154	133	104	216	51	<
A/CoteD'Ivoire/4593/2024	J.2.1	2024-11-14	SIAT1	74	170	148	120	177	111	<
A/CoteD'Ivoire/4492/2024	J.2.1	2024-11-04	SIAT1	61	149	128	105	144	94	<
A/Dakar/037/2024	J.2.2	2024-09-13	MDCK1/SIAT1	63	125	158	99	260	52	<
A/Dakar/036/2024	J.2.2	2024-09-11	MDCK1/SIAT1	76	156	127	97	154	56	<
A/Dakar/040/2024	J.2.2	2024-09-20	MDCK1/SIAT1	67	133	142	104	249	47	<
A/Dakar/039/2024	J.4	2024-09-19	MDCK1/SIAT1	<	<	<	<	<	<	<
A/Dakar/041/2024	J.4	2024-10-02	MDCK1/SIAT1	<	<	<	<	<	<	<

<=2-fold >2<4-fold >4<8-fold >8-fold not recognised by the antiserum

Table MN-11. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2025-02-13

				Neutralisation Titre						
				Post-infection ferret antisera						
			Passage history	A/Massachusetts 18/2022	A/Thailand 8/2022	A/D.Columbia 27/2023	A/Croatia 10136/RV/2023	A/Netherlands 10563/2023	A/Slovenia 49/2024	A/Switzerland 47775/2024
				SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT
			Ferret number	F36/23	F34/23	F38/24	F16/24	F08/24	F11/24	F43/24
Viruses	Genetic group	Collection Date	Passage History	J	J	J.2 + S145N	J.2 + S145N	J.2.2	J.2 + N158K	J.2.1 + K189R
REFERENCE VIRUSES										
A/Massachusetts/18/2022	J	2022-06-04	SIAT3/SIAT1 10^{-3}	161	229	259	160	305	63	<
A/Thailand/8/2022	J	-	E3/E1 10^{-5}	437	1144	1166	783	961	428	133
A/District Of Columbia/27/2023	J.2 + S145N	2023-12-09	SIAT2/SIAT1 10^{-3}	57	92	184	172	188	67	<
A/Croatia/10136/RV/2023	J.2 + S145N	2023-12-04	E3 (Am1Al2) 10^{-3}	177	396	1161	624	1498	932	153
A/Netherlands/10563/2023	J.2.2	2023-11-10	MDCK-MIX2/SIAT3 10^{-3}	135	210	348	207	377	100	<
A/Slovenia/49/2024	J.2 + N158K	2024-01-08	MDCKx/SIAT3 10^{-3}	<	128	113	<	<	3351	<
A/Switzerland/47775/2024	J.2.1 + K189R	2024-03-04	MDCK1/SIAT2 10^{-3}	<	<	68	<	<	<	64
TEST VIRUSES										
A/Madagascar/03788/2024	J	2024-12-04	SIAT1	104	270	149	89	204	56	<
A/Madagascar/03416/2024	J.1.1	2024-11-11	SIAT1	68	134	197	107	223	73	<
A/Madagascar/03415/2024	J.1.1	2024-11-11	SIAT1	64	136	190	117	189	68	<
A/Switzerland/5158/2024	J.2	2024-12-10	SIAT1	79	227	169	128	240	77	<
A/Switzerland/5735/2024	J.2	2024-12-09	SIAT1	<	67	130	64	182	<	<
A/Belgium/8271/2024	J.2	2024-11-24	SIAT1	110	272	315	200	489	95	<
A/Belgium/799/2024	J.2	2024-10-02	SIAT2	84	267	258	137	246	140	45
A/Navarra/2114/2024	J.2	2024-11-09	SIAT1	<	75	77	56	96	55	<
A/Cantabria/2340/2024	J.2	2024-11-01	SIAT1	103	265	253	135	287	79	<
A/Jonkoping/SE24-14227/2024	J.2	2024-10-15	SIAT1/SIAT1	76	206	274	126	320	70	<
A/Nordrhein-Westfalen/114/2024	J.2	2024-12-19	P1/SIAT1	<	60	70	40	96	<	<
A/Rheinland-Pfalz/39/2024	J.2	2024-12-05	P1/SIAT1	<	56	122	73	158	40	<
A/ComunidadValenciana/2045/2024	J.2.1	2024-10-30	SIAT1	40	114	76	61	100	<	<
A/Switzerland/8932/2024	J.2.2	2024-12-04	SIAT1	78	187	181	128	311	74	<

<=2-fold >2<4-fold >4<8-fold >8-fold not recognised by the antiserum

Fold-reduction table: A/H3N2 (HI only)

		<4-fold difference		4-fold difference		>4-fold difference		
Reference Virus	Clade	Number	Percentage	Number	Percentage	Number	Percentage	Total
A/Massachusetts/18/2022 Cell	2a.3a.1 (J)	105	34.2	115	37.5	87	28.3	307
A/Thailand/08/2022 Egg	2a.3a.1 (J)	90	29.3	137	44.6	80	26.1	307
A/DistrictOfColumbia/27/2023 Cell	2a.3a.1 (J.2)	282	92.8	19	6.2	3	1.0	304
A/Croatia/10136RV/2023 Egg	2a.3a.1 (J.2)	66	21.5	131	42.7	110	35.8	307
A/Netherlands/10563/2023 Cell	2a.3a.1 (J.2.2)	245	79.8	53	17.3	9	2.9	307
A/Lisboa/216/2023 Egg	2a.3a.1 (J.2.2)	194	63.2	98	31.9	15	4.9	307
A/Slovenia/49/2024 Cell	2a.3a.1 (J.2)	2	0.7	6	2.0	299	97.4	307
A/Switzerland/47775/2024 Cell	2a.3a.1 (J.2.1)	180	59.2	109	35.9	15	4.9	304
A/Norway/12374/2023 Cell	2a.3a.1 (J.4)	287	93.5	14	4.6	6	2.0	307
A/BurkinaFaso/3131/2023 Cell	2a.3a.1 (J.4)	222	72.3	27	8.8	58	18.9	307

A/H3N2: Egg isolates

Virus	Status	Genetic group (subclade) [Additional subst.]	Comment
A/Switzerland/8649/2023	SAN-032 (2-way pass), X-421 (2-way pass)	2a.3a.1 (J.2) [N122D (-CHO), K276E]	Egg adaptation: (X92S, S228T)
A/Croatia/10136/RV/2023	NYMC X-425 (2-way pass), NYMC X-425A (2-way fail), NIB-146C (2-way pass)	2a.3a.1 (J.2) [S145N]	Egg adaptation: (D186A)
A/Lisboa/216/2023	NYMC X-423 (2-way pass), NIB-145 (2-way pass), IVR-255 (2-way pass)	2a.3a.1 (J.2) [S124N]	Egg adaptation: (S228T)
A/Norway/423/2024	NIB-148 (1-way pass)	2a.3a.1 (J.2) [N158K]	Egg adaptation: (V166M, F195Y)
A/Norway/1952/2024	NIB-147 (1-way pass)	2a.3a.1 (J.2) [S145N, N158K]	Egg adaptation: (F195Y)
A/Switz/59652/2024	sent out	2a.3a.1 (J.2) [K189R]	Egg adaptation: (Is1: D186N), (Is3: S228T)
A/Norway/319/2024	available to send out	2a.3a.1 (J.2) [K207R, D291V, D475G]	Egg adaptation: (Is2: H183F, P227S)
A/Norway/12374/2023	available to send out	2a.3a.1 (J.4) [G142R]	Egg adaptation: (F195Y)
A/France/IDF-IPP29542/2023	available to send out	2a.3a.1 (J.4) [K189R]	Egg adaptation: (clone 11: D186N, F195Y), (clone 13: F195Y, I230M)

A/H3N2: HI reagents and references

Virus	Genetic group	Virus passage	Ferret ID
A/Massachusetts/18/2022	2a.3a.1 (J)	SIAT3/SIAT1	F36/23
A/Thailand/08/2022	2a.3a.1 (J)	E3/E1	F34/23
A/DistrictOfColumbia/27/2023	2a.3a.1 (J.2)	SIAT2/SIAT1	F38/24
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	E3/E1	F16/24
A/Netherlands/10563/2023	2a.3a.1 (J.2.2)	MDCK-MIX2/SIAT3	F08/24
A/Lisboa/216/2023	2a.3a.1 (J.2.2)	E3 (Am1Al2)	F15/24
A/Slovenia/49/2024	2a.3a.1 (J.2)	MDCKx/SIAT3	F11/24
A/Switzerland/47775/2024	2a.3a.1 (J.2.1)	MDCK1/SIAT2	F43/24
A/Norway/12374/2023	2a.3a.1 (J.4)	SIAT4	F21/24
A/BurkinaFaso/3131/2023	2a.3a.1 (J.4)[K189R]	SIAT3	F32/24

A/H3N2: MN reagents and references

Virus	Genetic group	Virus passage	Ferret ID
A/Massachusetts/18/2022	2a.3a.1 (J)	SIAT3/SIAT1	F36/23
A/Thailand/08/2022	2a.3a.1 (J)	E3/E1	F34/23
A/DistrictOfColumbia/27/2023	2a.3a.1 (J.2) + S145N	SIAT2/SIAT1	F38/24
A/Croatia/10136/RV/2023	2a.3a.1 (J.2) + S145N	E3 (Am1Al2)	F16/24
A/Netherlands/10563/2023	2a.3a.1 (J.2.2)	MDCK-MIX2/SIAT3	F08/24
A/Slovenia/49/2024	2a.3a.1 (J.2) + N158K	MDCKx/SIAT3	F11/24
A/Switzerland/47775/2024	2a.3a.1 (J.2.1) + K189R	MDCK1/SIAT2	F43/24

Summary: A/H3N2

Genetic analyses

The SH 2024 winter season was characterised by predominance of subclade 2a.3a.1 (J.2). Since September 2024, subclade J.2 continues to predominate globally.

During this reporting period, the great majority of H3 viruses detected belong to clade 2a.3a.1 subclade J, vaccine reference A/Thailand/8/2022 (substitutions E50K, I140K and I223V). Subclade J split into 4 further subclades: of these, subclade J.2 (vaccine reference A/Croatia/10136RV/2023) characterised by N122D (-CHO) and K276E became the dominant subclade, predominating in the majority of continents with >90% frequency. Two subclades were recently designated within J.2: J.2.1 with substitution P239S (reference A/West Virginia/51/2024) and J.2.2 with substitution S124N (reference A/Lisboa/216/2023). Of these, basal subclade J.2 continues to predominate globally, with J.2.2 as second most frequent subclade predominating in Australia, some countries in South East Asia, Mauritius, Senegal, Oman, Poland, Ukraine and Norway, and circulating in low proportions elsewhere, whereas subclade J.2.1 predominates only in China, Denmark and Ivory Coast.

Other non-J.2 subclades were detected in very low proportions during this period: J.1.1 (substitution S145N) in Bhutan, Madagascar, Maldives and Algeria, basal J in Madagascar, J.4 in Western Africa, France and Netherlands, and G.1.3.1 in Western Africa and Spain.

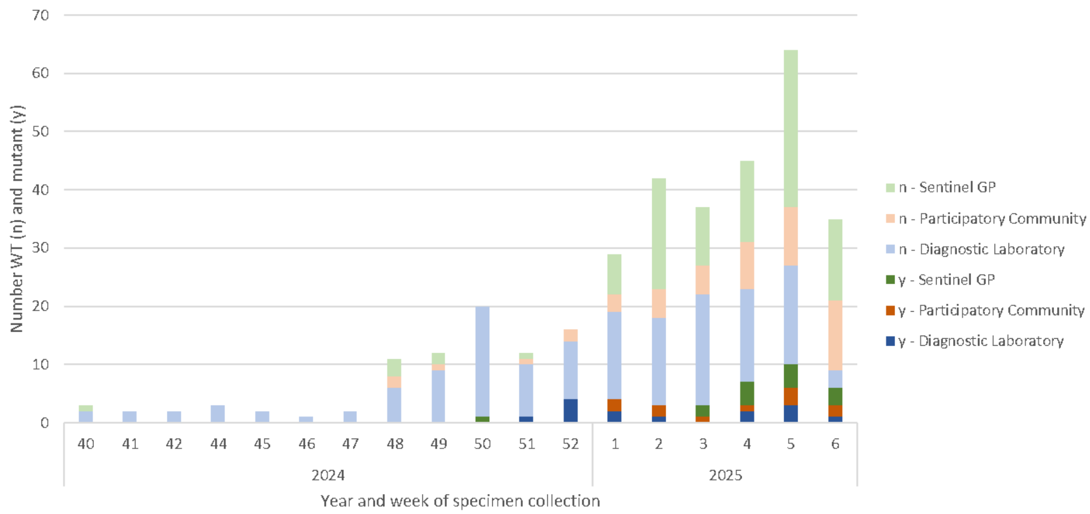
Convergent evolution was observed with several sequences in separate clades characterised by substitutions such as S124N, S145N and fewer viruses with substitutions N158K and K189R. A few clades had substitutions involving loss of potential N-glycosylation sites, such as T135K, T128A, N63K, T65K, N94K, S96I and N8D.

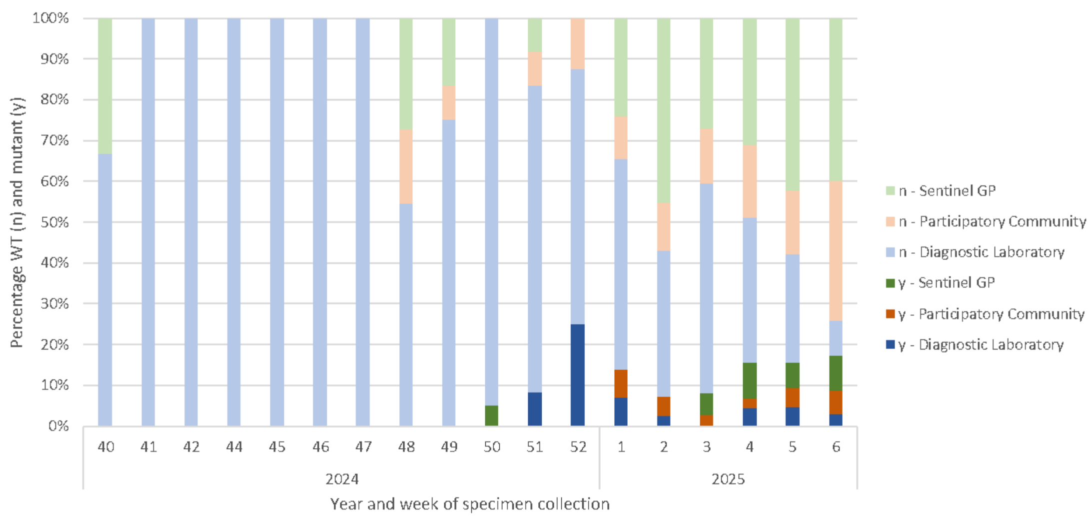
Antigenic analyses

HI titres show that the SH 2024 and NH 2024-25 vaccine strains, egg-based A/Thailand/08/2022 (2a.3a.1 (J)) and cell-based A/Massachusetts/18/2022 (2a.3a.1 (J)), and the SH 2025 vaccine strain, egg-based A/Croatia/10136/RV/2023 showed four- and eight-fold drops against many J.2, J.2.1 and J.2.2 subclade viruses.

MN titres show that both egg-based SH 2024 and NH 2024-25 vaccine strain A/Thailand/08/2022 (2a.3a.1 (J)) and the SH 2025 vaccine strain A/Croatia/10136/RV/2023 (2a.3a.1 (J.2)) showed reduced reactivity to the majority of viruses tested. The cell-based SH 2024 and NH 2024-25 vaccine strain A/Massachusetts/18/2022 (2a.3a.1 (J)) and the SH 2025 vaccine strain A/District Of Columbia/27/2023 recognised most viruses tested within 2-fold, although the former showed significant drop in titre to a few J.2 viruses tested to date.

Detection of emergent J.2 clade with HA substitutions N158K and K189R in the Netherlands





Influenza B

Genetic analyses: B/Victoria

Global distribution of B/Victoria subclades

Global geographical distribution of influenza B/Victoria genetic clades (subclades) viruses with collection dates from 1st November 2024 to 7th February 2025, obtained with full HA sequences as deposited in GISAID, downloaded on 14th February 2025 and classified using Nextclade. Map prepared with Microreact. Geographic markers scaled to detection proportions.

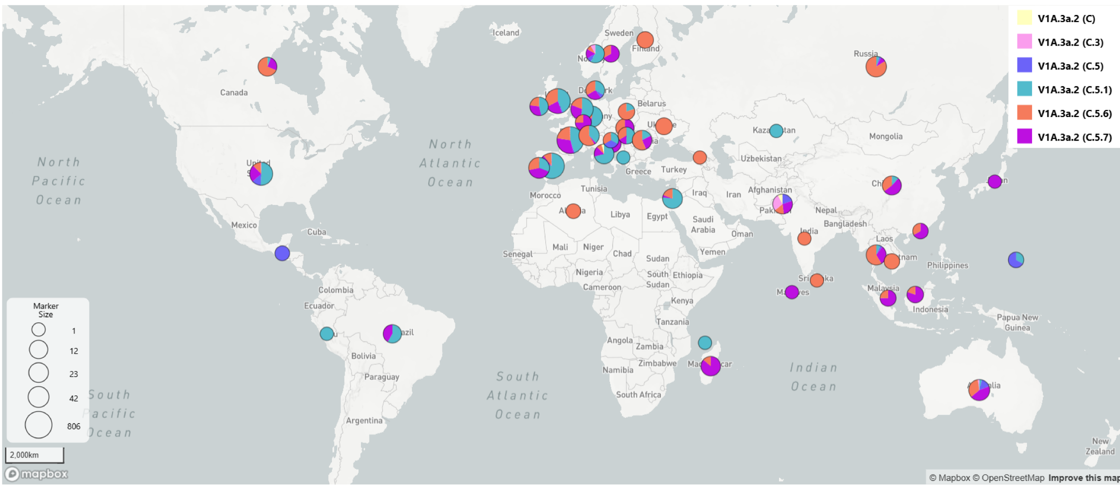


Figure 14: Global distribution of B/Victoria genetic subclades

Global time-dependent variation of B/Victoria subclades

Global time-dependent variation in frequencies of genetic clades-subclades of B/Victoria viruses collected from 1st September 2024 to 7th February 2025.

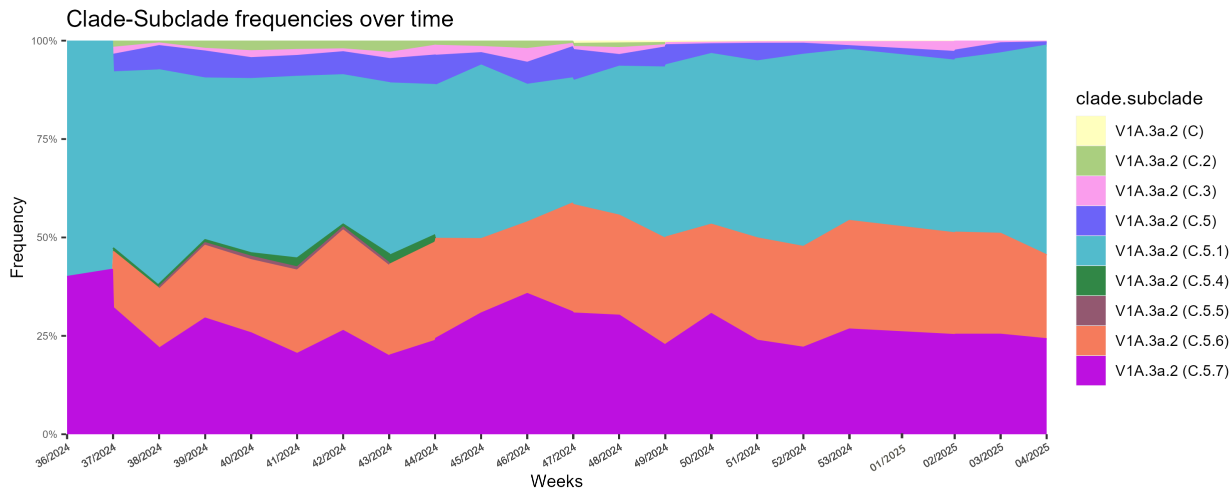


Figure 15: B/Victoria clade-subclade frequencies over time (weeks 36/2024 to 4/2025)

Phylogenetic analysis of B/Victoria sequences

Maximum likelihood phylogenetic tree inferred using IQTree2 with HA sequence data obtained from GISAID, collection dates from 1st November 2024 to 7th February 2025, manually curated and then downsampled using Treemmer to retain a representative tree topology of 200 sequences, with emphasis on sequences generated at the WIC. Amino acid signatures annotated with Treetime. References and CVVs are marked as Cell or Egg.

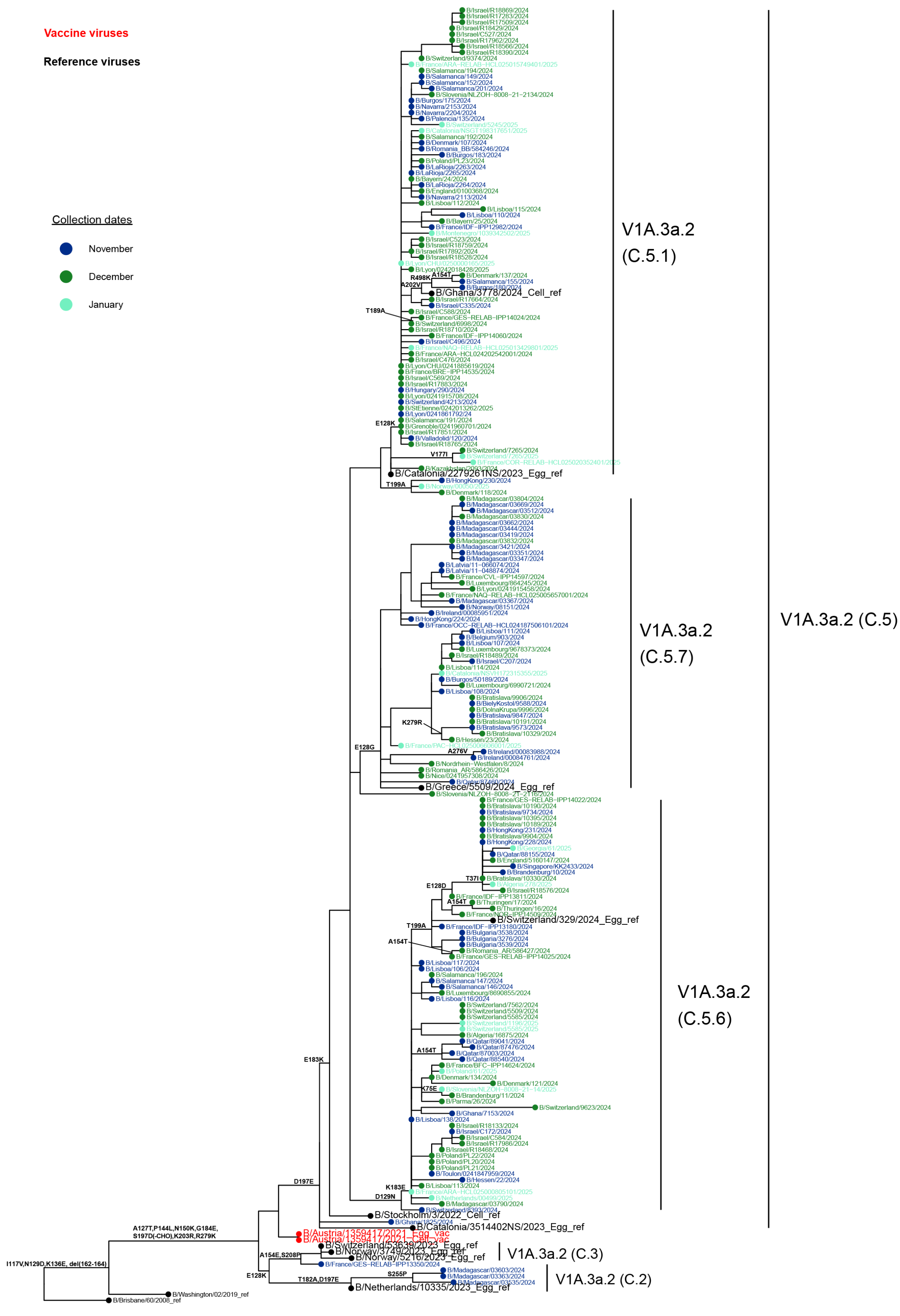


Figure 16: B/Victoria Maximum likelihood phylogenetic tree, HA segment

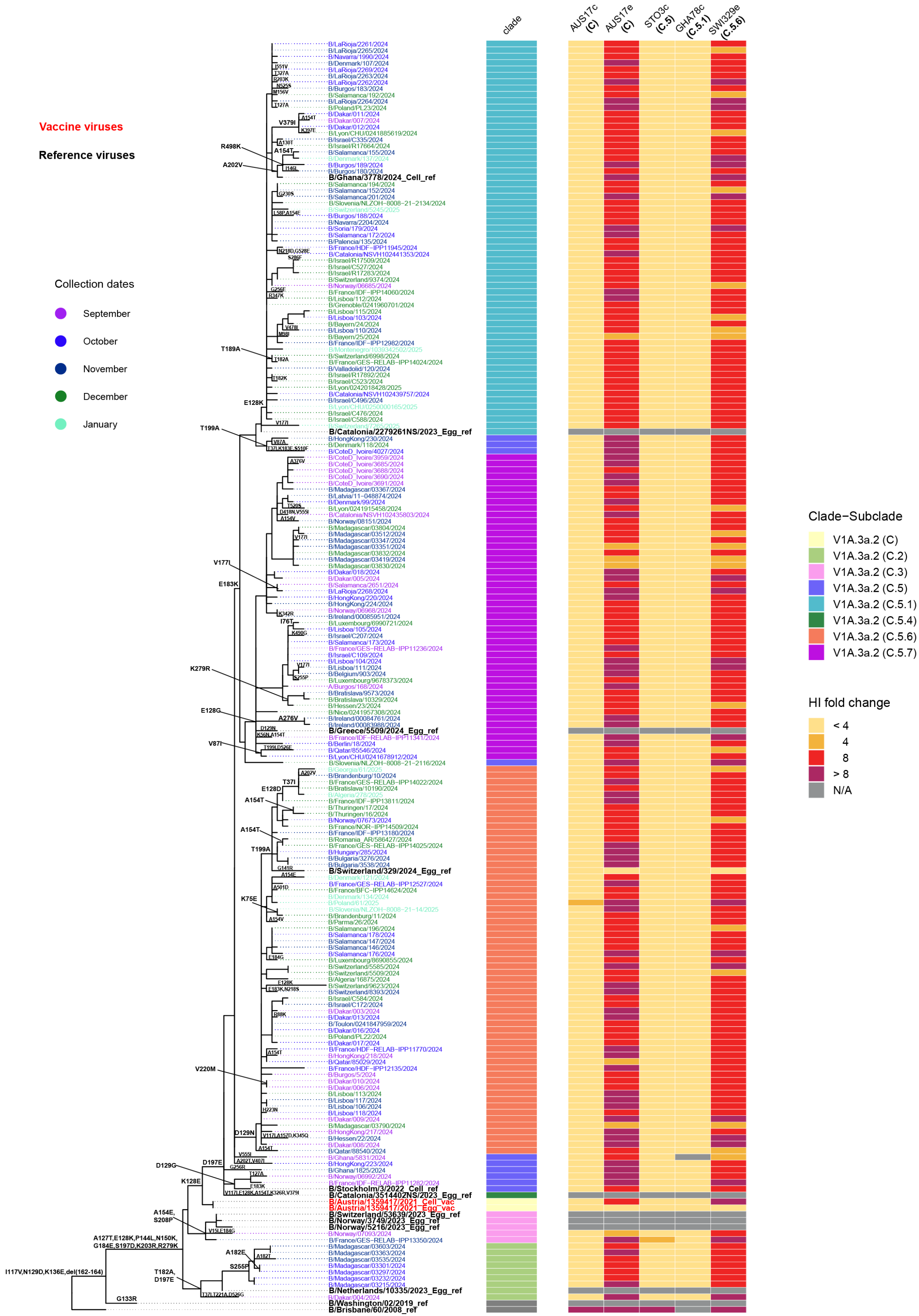


Figure 17: ML phylogenetic tree with combined heatmap representing antigenic reactivities by HI

Antigenic analyses: B/Victoria

Haemagglutination inhibition tables: B/Victoria

Table BV-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-10-16

						Haemagglutination inhibition titre					
						Post-infection ferret antiserum					
	Viruses	Other information			Passage	B/Bris /60/08 Egg	B/Austria /1359417/21 MDCK	B/Austria /1359417/21 Egg G141	B/Austria /1359417/21 Egg G141R	B/Stock /3/22 MDCK	B/Switzerland /329/24 Egg
					Ferret number	Sh 539, 540, 543, 544, 570, 571, 5741	NIB F01/21	F40/21	F44/21	F28/22	F33/24
			Genetic group	Collection Date	Genetic group	V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	V1A.3a.2 (C.5.6)
	REFERENCE VIRUSES										
	B/Brisbane/60/2008		V1A	2008-08-04	E4/E3	1280	40	<40	<40	<40	<40
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	320	1280	640	320	640	160
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	320	1280	640	320	640	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	640	2560	640	2560
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	320	640	320	320	640	320
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4(Am1Al3)	80	320	80	640	160	1280
	TEST VIRUSES										
250014	B/Ghana/3973/2024		V1A.3a.2 (C.5)	2024-07-17	MDCK1	640	640	320	320	640	320
250011	B/Ghana/4921/2024		V1A.3a.2 (C.5)	2024-08-13	MDCK1	320	640	640	320	1280	320
250008	B/Ghana/5831/2024		V1A.3a.2 (C.5)	2024-09-18	MDCK1	640	640	320	320	640	320
250015	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK1	160	640	160	160	640	320
250012	B/Ghana/1216/2024		V1A.3a.2 (C.5.1)	2024-08-01	MDCK1	320	1280	320	320	1280	320
250013	B/Ghana/4172/2024		no sequence	2024-07-23	MDCK1	320	1280	640	320	1280	320
							Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		

< relates to the lowest dilution of antiserum used
[1] hyperimmune sheep serum; ND =

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table BV-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-11-22

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre					
						Post-infection ferret antiserum					
					Passage	B/Austria /1359417/21 MDCK	B/Austria /1359417/21 Egg G141	B/Austria /1359417/21 Egg G141R	B/Stock /3/22 MDCK	B/Switzerland /329/24 Egg	B/Ghana /3778/2024 MDCK
					Ferret number	NIB F01/21	F40/21	F44/21	F28/22	F33/24	F42/24
					Genetic group	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	V1A.3a.2 (C.5.6)	V1A.3a.2 (C.5.1)
	REFERENCE VIRUSES										
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	1280	640	320	640	160	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	640	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	1280	640	2560	640	2560	320
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	1280	640	320	640	160	320
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4/E1	640	160	1280	320	2560	320
250015	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK2	640	320	160	640	320	640
	TEST VIRUSES										
250063	B/Catalonia/NSVH102439757/2024		V1A.3a.2 (C.5.1)	2024-10-04	MDCK1	1280	320	320	640	320	320
250061	B/Catalonia/NSVH102441353/2024		V1A.3a.2 (C.5.1)	2024-10-07	MDCK2	1280	320	160	640	320	320
250035	B/SaintPetersburg/RII-59/2024		V1A.3a.2 (C.5.6)	2024-05-20	MDCK2/MDCK1	640	320	160	640	160	320
250044	B/Yekaterinburg/37/2024		V1A.3a.2 (C.5.7)	2024-03-26	MDCK3/MDCK1	640	320	160	320	160	320
250043	B/Yekaterinburg/43/2024		V1A.3a.2 (C.5.7)	2024-03-26	MDCK3/MDCK1	640	320	160	640	160	320
250042	B/Novorossiysk/RII-01/2024		V1A.3a.2 (C.5.7)	2024-04-10	MDCK3/MDCK1	640	320	160	640	320	320
250041	B/SaintPetersburg/RII-51/2024		V1A.3a.2 (C.5.7)	2024-04-23	MDCK1/MDCK1	640	320	160	640	320	640
250040	B/SaintPetersburg/RII-53/2024		V1A.3a.2 (C.5.7)	2024-04-23	MDCK1/MDCK1	640	320	160	320	160	320
250039	B/RepublicOfTatarstan/RII-01/2024		V1A.3a.2 (C.5.7)	2024-04-24	MDCK2/MDCK1	1280	640	320	640	320	640
250038	B/SaintPetersburg/RII-52/2024		V1A.3a.2 (C.5.7)	2024-04-24	MDCK1/MDCK1	1280	320	160	640	320	320
250037	B/SaintPetersburg/RII-55/2024		V1A.3a.2 (C.5.7)	2024-04-24	MDCK1/MDCK1	640	320	160	1280	320	320
250036	B/Kazan/RII-01/2024		V1A.3a.2 (C.5.7)	2024-04-27	MDCK2/MDCK1	640	320	160	640	160	320
250064	B/Catalonia/NSVH102435803/2024		V1A.3a.2 (C.5.7)	2024-09-29	MDCK2	640	320	160	640	320	320
						Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025			Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		

< relates to the lowest dilution of antiserum used

[1] hyperimmune sheep serum: ND =

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table BV-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-11-29

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre Post-infection ferret antiserum					
					Passage	B/Austria /1359417/21 MDCK	B/Austria /1359417/21 Egg G141	B/Austria /1359417/21 Egg G141R	B/Stock /3/22 MDCK	3/Switzerland /329/24 Egg	B/Ghana /3778/2024 MDCK
					Ferret number	NIB F01/21	F40/21	F44/21	F28/22	F33/24	F42/24
					Genetic group	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	V1A.3a.2 (C.5.6)	V1A.3a.2 (C.5.1)
	REFERENCE VIRUSES										
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	IAT1/MDCK4	1280	640	320	640	160	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	640	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	2560	640	2560	640	2560	320
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	IAT1/MDCK3	640	320	160	640	320	320
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4/E1	640	160	1280	640	2560	320
250015	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK2	640	320	160	640	320	320
	TEST VIRUSES										
250097	B/Norway/07093/2024		V1A.3a.2 (C.3)	2024-09-27	MDCK1	2560	640	640	640	320	640
250098	B/Norway/06992/2024		V1A.3a.2 (C.5)	2024-09-23	MDCK1	1280	640	160	640	320	640
250100	B/Norway/06685/2024		V1A.3a.2 (C.5.1)	2024-09-11	MDCK1	1280	640	320	1280	640	640
250096	B/Norway/07673/2024		V1A.3a.2 (C.5.6)	2024-10-17	MDCK1	1280	320	320	640	640	640
250099	B/Norway/06968/2024		V1A.3a.2 (C.5.7)	2024-09-23	MDCK1	1280	640	320	1280	320	640
250095	B/Norway/07738/2024		V1A.3a.2 (C.5.7)	2024-10-21	MDCK2	1280	640	320	1280	640	640
250093	B/Norway/08151/2024		V1A.3a.2 (C.5.7)	2024-11-01	MDCK1	1280	640	320	1280	320	640
						Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025			

< relates to the lowest dilution of antiserum used

[1] hyperimmune sheep serum: ND =

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table BV-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-12-10

	Viruses	Other information	Genetic group	Collection Date	Passage	Haemagglutination inhibition titre – Post-infection ferret antiserum					
					Passage	B/Austria /1359417/21 MDCK	B/Austria /1359417/21 Egg G141	B/Austria /1359417/21 Egg G141R	B/Stock /3/22 MDCK	B/Ghana /3778/2024 MDCK	B/Switzerland /329/24 Egg
					Ferret number	NIB F01/21	F40/21	F44/21	F28/22	F42/24	F33/24
					Genetic group	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	V1A.3a.2 (C.5.1)	V1A.3a.2 (C.5.6)
	REFERENCE VIRUSES										
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	1280	640	320	640	320	160
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	320	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	1280	640	1280	640	320	1280
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	640	320	160	640	320	320
	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK2	640	320	160	320	320	160
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4/E1	640	320	2560	640	320	2560
	TEST VIRUSES										
250162	B/Salalah/52423023/2024		V1A.3a.2 (C.5.6)	2024-08-12	MDCK1	640	320	160	640	320	320
250161	B/Salalah/52423024/2024		V1A.3a.2 (C.5.6)	2024-08-13	MDCK1	640	320	160	640	320	320
						Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025			

< relates to the lowest dilution of antiserum used
[1] hyperimmune sheep serum; ND =

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table BV-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-12-18

						Haemagglutination inhibition titre					
						Post-infection ferret antiserum					
	Viruses	Other information				B/Austria /1359417/21	B/Austria /1359417/21	B/Austria /1359417/21	B/Stock /3/22	B/Ghana /3778/2024	B/Switzerland /329/24
					Passage	MDCK	Egg G141	Egg G141R	MDCK	MDCK	Egg
					Ferret number	NIB F01/21	F40/21	F44/21	F28/22	F42/24	F33/24
			Genetic group	Collection Date	Genetic group	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	V1A.3a.2 (C.5.1)	V1A.3a.2 (C.5.6)
	REFERENCE VIRUSES										
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	IAT1/MDCK4	1280	640	320	640	320	160
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	320	320	320	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	2560	640	2560	640	320	2560
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	IAT1/MDCK3	640	320	160	640	320	320
	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK2	640	320	320	320	320	320
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4/E1	640	160	1280	320	320	2560
	TEST VIRUSES										
250179	B/Hessen/22/2024		V1A.3a.2 (C.5.6)	2024-11-08	P1/MDCK1	640	160	160	640	320	160
250180	B/Berlin/18/2024		V1A.3a.2 (C.5.7)	2024-10-31	P2/MDCK1	640	320	160	640	320	320
	< relates to the lowest dilution of antiserum used					Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025			

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table BV-6. Antigenic analyses of influenza B viruses (Victoria lineage) 2025-01-08

	Viruses	Other information	Genetic group	Collection Date	Passage / Ferret number / Genetic group	Haemagglutination inhibition titre, Post-infection ferret antiserum: B/Austria /1359417/21 MDCK	B/Austria /1359417/21 Egg G141	B/Austria /1359417/21 Egg G141R	B/Stock /3/22 MDCK	B/Ghana /3778/2024 MDCK	B/Switzerland /329/24 Egg
					Ferret number	NIB F01/21	F40/21	F44/21	F28/22	F42/24	F33/24
					Genetic group	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	V1A.3a.2 (C.5.1)	V1A.3a.2 (C.5.6)
	REFERENCE VIRUSES										
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	1280	320	320	640	320	160
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	640	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	1280	320	2560	640	320	2560
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	640	320	160	640	320	320
	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK2	640	320	160	640	320	160
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4/E1	640	160	1280	320	320	2560
	TEST VIRUSES										
250583	B/France/IDF-RELAB-IPP11282/2024		V1A.3a.2 (C.5)	2024-09-24	C1/MDCK1	640	320	160	320	320	160
250379	B/HongKong/223/2024		V1A.3a.2 (C.5)	2024-10-23	MDCK1/MDCK1	640	320	160	640	320	320
250373	B/HongKong/230/2024		V1A.3a.2 (C.5)	2024-11-16	MDCK1/MDCK1	1280	320	160	640	640	320
250579	B/France/HDF-IPP11945/2024		V1A.3a.2 (C.5.1)	2024-10-15	C1/MDCK1	640	320	320	640	320	320
250539	B/Lisboa/103/2024		V1A.3a.2 (C.5.1)	2024-10-21	MDCK1/MDCK1	1280	640	320	640	640	640
250430	B/Hungary/290/2024		V1A.3a.2 (C.5.1)	2024-11-18	MDCK1/MDCK1	640	320	160	640	320	320
250524	B/Lisboa/110/2024		V1A.3a.2 (C.5.1)	2024-11-29	MDCK1/MDCK1	1280	640	320	640	640	640
250520	B/Lisboa/112/2024		V1A.3a.2 (C.5.1)	2024-12-02	MDCK1/MDCK1	640	160	160	320	160	320
250389	B/HongKong/217/2024		V1A.3a.2 (C.5.6)	2024-09-29	MDCK1/MDCK1	640	320	160	640	320	320
250387	B/HongKong/218/2024		V1A.3a.2 (C.5.6)	2024-09-29	MDCK1/MDCK1	640	320	160	640	320	320
250577	B/France/HDF-IPP12135/2024		V1A.3a.2 (C.5.6)	2024-10-23	C1/MDCK1	640	320	160	640	320	320
250431	B/Hungary/285/2024		V1A.3a.2 (C.5.6)	2024-10-28	MDCK1/MDCK1	640	320	160	640	320	320
250536	B/Lisboa/118/2024		V1A.3a.2 (C.5.6)	2024-10-28	MDCK1/MDCK1	1280	320	320	640	320	320
250375	B/HongKong/228/2024		V1A.3a.2 (C.5.6)	2024-11-12	MDCK1/MDCK1	640	320	160	640	320	320
250532	B/Lisboa/106/2024		V1A.3a.2 (C.5.6)	2024-11-14	MDCK1/MDCK1	1280	320	160	640	640	320
250371	B/HongKong/231/2024		V1A.3a.2 (C.5.6)	2024-11-19	MDCK1/MDCK1	640	320	160	320	320	320
250522	B/Lisboa/113/2024		V1A.3a.2 (C.5.6)	2024-12-02	MDCK1/MDCK1	640	320	160	640	320	320
250581	B/France/GES-RELAB-IPP11236/2024		V1A.3a.2 (C.5.7)	2024-09-28	C1/MDCK1	640	320	160	640	320	320
250385	B/HongKong/219/2024		V1A.3a.2 (C.5.7)	2024-10-10	MDCK1/MDCK1	640	320	80	320	320	160
250383	B/HongKong/220/2024		V1A.3a.2 (C.5.7)	2024-10-12	MDCK1/MDCK1	1280	320	160	640	640	320
250381	B/HongKong/222/2024		V1A.3a.2 (C.5.7)	2024-10-21	MDCK1/MDCK1	640	320	160	640	320	320
250576	B/France/IDF-RELAB-IPP12446/2024		V1A.3a.2 (C.5.7)	2024-10-24	C1/MDCK1	1280	320	160	640	320	320
250534	B/Lisboa/105/2024		V1A.3a.2 (C.5.7)	2024-10-30	MDCK1/MDCK1	1280	320	320	640	640	320
250377	B/HongKong/224/2024		V1A.3a.2 (C.5.7)	2024-11-06	MDCK1/MDCK1	1280	320	320	1280	640	320
250507	B/Latvia/11-048874/2024		V1A.3a.2 (C.5.7)	2024-11-12	P2/MDCK1	1280	640	320	640	640	320
250506	B/Latvia/11-066074/2024		V1A.3a.2 (C.5.7)	2024-11-19	P2/MDCK1	1280	320	320	640	640	320
250526	B/Lisboa/111/2024		V1A.3a.2 (C.5.7)	2024-11-27	MDCK1/MDCK1	640	160	80	320	160	160
250508	B/Latvia/09-093738/2024		no sequence	2024-09-23	P2/MDCK1	640	320	160	640	320	320
						Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025			

< relates to the lowest dilution of antiserum used

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table BV-7. Antigenic analyses of influenza B viruses (Victoria lineage) 2025-01-15

	Viruses	Other information	Genetic group	Collection Date	Passage	B/Austria /1359417/21 MDCK	B/Austria /1359417/21 Egg G141	B/Austria /1359417/21 Egg G141R	B/Stock /3/22 MDCK	B/Ghana /3778/2024 MDCK	B/Switzerland /329/24 Egg
						Haemagglutination inhibition titre					
						Post-infection ferret antiserum					
					Ferret number	NIB F01/21	F40/21	F44/21	F28/22	F42/24	F33/24
					Genetic group	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	V1A.3a.2 (C.5.1)	V1A.3a.2 (C.5.6)
	REFERENCE VIRUSES										
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	IAT1/MDCK4	1280	320	160	320	320	160
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	640	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	2560	640	2560	640	320	2560
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	IAT1/MDCK3	640	320	160	640	320	160
	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK2	640	320	160	320	320	160
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4/E1	640	160	1280	320	160	2560
	TEST VIRUSES										
251016	B/Dakar/004/2024		V1A.3a.2 (C.2)	2024-09-18	CK1/MDCK1	640	320	160	640	320	160
250639	B/CoteD'Ivoire/4027/2024		V1A.3a.2 (C.5)	2024-10-03	MDCK1	640	320	160	640	320	320
250331	B/Ghana/1825/2024		V1A.3a.2 (C.5)	2024-11-06	MDCK2	640	320	160	640	320	320
250931	B/Soria/154/2024		V1A.3a.2 (C.5.1)	2024-08-22	CKx/MDCK1	640	320	160	640	320	320
250926	B/Salamanca/157/2024		V1A.3a.2 (C.5.1)	2024-09-19	CKx/MDCK1	640	320	160	640	320	320
251013	B/Dakar/007/2024		V1A.3a.2 (C.5.1)	2024-09-21	CK1/MDCK1	1280	640	320	640	320	320
250923	B/Salamanca/172/2024		V1A.3a.2 (C.5.1)	2024-10-01	CKx/MDCK1	1280	320	160	640	640	320
250921	B/Soria/179/2024		V1A.3a.2 (C.5.1)	2024-10-02	CKx/MDCK1	640	320	160	640	320	160
251009	B/Dakar/011/2024		V1A.3a.2 (C.5.1)	2024-10-03	CK2/MDCK1	1280	320	320	640	320	320
251008	B/Dakar/012/2024		V1A.3a.2 (C.5.1)	2024-10-03	CK1/MDCK1	1280	640	320	1280	640	320
250916	B/Burgos/189/2024		V1A.3a.2 (C.5.1)	2024-10-07	CKx/MDCK1	640	320	160	320	320	160
250335	B/Ghana/6346/2024		V1A.3a.2 (C.5.1)	2024-10-09	MDCK1	1280	320	160	640	320	320
250910	B/Salamanca/180/2024		V1A.3a.2 (C.5.1)	2024-10-13	CKx/MDCK1	1280	320	320	640	640	320
250908	B/Salamanca/181/2024		V1A.3a.2 (C.5.1)	2024-10-15	CKx/MDCK1	640	320	160	320	320	160
250906	B/Burgos/188/2024		V1A.3a.2 (C.5.1)	2024-10-23	CKx/MDCK1	1280	320	320	640	640	320
250901	B/Salamanca/201/2024		V1A.3a.2 (C.5.1)	2024-11-13	CKx/MDCK1	640	320	160	640	320	160
250717	B/Denmark/107/2024		V1A.3a.2 (C.5.1)	2024-11-15	IAT3/MDCK1	640	320	160	640	320	320
250787	B/Switzerland/4213/2024		V1A.3a.2 (C.5.1)	2024-11-28	MDCK1	1280	320	160	1280	640	320
250517	B/Lisboa/115/2024		V1A.3a.2 (C.5.1)	2024-12-04	MDCK1	1280	320	320	1280	640	320
250784	B/Switzerland/7265/2024		V1A.3a.2 (C.5.1)	2024-12-09	MDCK1	640	320	160	640	320	320
251017	B/Dakar/003/2024		V1A.3a.2 (C.5.6)	2024-09-09	CK1/MDCK1	640	320	160	640	320	320
251014	B/Dakar/006/2024		V1A.3a.2 (C.5.6)	2024-09-19	CK1/MDCK1	640	640	320	640	320	320
251012	B/Dakar/008/2024		V1A.3a.2 (C.5.6)	2024-09-23	CK1/MDCK1	640	320	160	640	320	160
251011	B/Dakar/009/2024		V1A.3a.2 (C.5.6)	2024-09-23	CK1/MDCK1	1280	640	160	640	320	160
251010	B/Dakar/010/2024		V1A.3a.2 (C.5.6)	2024-09-23	CK1/MDCK1	1280	640	320	1280	640	320
251007	B/Dakar/013/2024		V1A.3a.2 (C.5.6)	2024-10-08	CK2/MDCK1	640	640	160	640	320	320
250914	B/Salamanca/176/2024		V1A.3a.2 (C.5.6)	2024-10-08	CKx/MDCK1	640	320	160	640	320	160
251006	B/Dakar/014/2024		V1A.3a.2 (C.5.6)	2024-10-09	CK1/MDCK1	1280	640	320	640	320	320
251005	B/Dakar/015/2024		V1A.3a.2 (C.5.6)	2024-10-09	CK1/MDCK1	1280	640	320	640	640	320
251004	B/Dakar/016/2024		V1A.3a.2 (C.5.6)	2024-10-10	CK1/MDCK1	1280	640	320	640	320	320
250912	B/Salamanca/178/2024		V1A.3a.2 (C.5.6)	2024-10-10	CKx/MDCK1	640	640	320	640	640	320
251003	B/Dakar/017/2024		V1A.3a.2 (C.5.6)	2024-10-14	CK1/MDCK1	1280	320	320	640	640	320
250548	B/Bulgaria/3276/2024		V1A.3a.2 (C.5.6)	2024-11-12	MDCK1	640	320	160	640	320	320
250788	B/Switzerland/8393/2024		V1A.3a.2 (C.5.6)	2024-11-20	MDCK1	640	320	160	640	320	320
250516	B/Lisboa/117/2024		V1A.3a.2 (C.5.6)	2024-11-26	MDCK2	1280	320	160	640	640	320
250547	B/Bulgaria/3538/2024		V1A.3a.2 (C.5.6)	2024-11-29	MDCK1	640	320	160	640	320	320
250546	B/Bulgaria/3539/2024		V1A.3a.2 (C.5.6)	2024-11-29	MDCK1	640	320	160	640	320	320
250785	B/Switzerland/9623/2024		V1A.3a.2 (C.5.6)	2024-12-05	MDCK1	640	320	160	640	320	320
250782	B/Switzerland/5585/2024		V1A.3a.2 (C.5.6)	2024-12-10	MDCK1	640	320	160	640	320	160
250645	B/CoteD'Ivoire/3685/2024		V1A.3a.2 (C.5.7)	2024-09-02	MDCK1	1280	320	160	640	640	320
250644	B/CoteD'Ivoire/3688/2024		V1A.3a.2 (C.5.7)	2024-09-04	MDCK1	640	320	320	640	640	320
250643	B/CoteD'Ivoire/3690/2024		V1A.3a.2 (C.5.7)	2024-09-04	MDCK1	640	320	160	640	320	320
250642	B/CoteD'Ivoire/3691/2024		V1A.3a.2 (C.5.7)	2024-09-04	MDCK1	640	320	160	640	640	320
250928	A/Burgos/168/2024		V1A.3a.2 (C.5.7)	2024-09-05	CKx/MDCK2	640	320	160	640	640	320
250641	B/CoteD'Ivoire/3687/2024		V1A.3a.2 (C.5.7)	2024-09-05	MDCK1	1280	320	320	640	640	320
251015	B/Dakar/005/2024		V1A.3a.2 (C.5.7)	2024-09-18	CK1/MDCK1	640	320	160	640	320	160
250640	B/CoteD'Ivoire/3959/2024		V1A.3a.2 (C.5.7)	2024-09-30	MDCK1	1280	320	160	640	320	320
250918	B/Salamanca/173/2024		V1A.3a.2 (C.5.7)	2024-10-07	CKx/MDCK2	640	320	320	640	640	320
251002	B/Dakar/018/2024		V1A.3a.2 (C.5.7)	2024-10-22	CK1/MDCK1	640	320	160	640	320	320
250537	B/Lisboa/104/2024		V1A.3a.2 (C.5.7)	2024-10-23	MDCK1	640	320	160	640	320	160
250352	B/Ireland/00078114/2024		V1A.3a.2 (C.5.7)	2024-10-24	MDCK1	1280	320	320	640	320	320
250718	B/Denmark/99/2024		V1A.3a.2 (C.5.7)	2024-10-25	IAT2/MDCK1	640	320	160	640	320	320
250351	B/Ireland/00083988/2024		V1A.3a.2 (C.5.7)	2024-11-18	MDCK1	640	320	160	640	320	320
250530	B/Lisboa/108/2024		V1A.3a.2 (C.5.7)	2024-11-19	MDCK1	640	320	160	640	320	320
250801	B/Belgium/903/2024		V1A.3a.2 (C.5.7)	2024-11-20	MDCK1	1280	320	160	640	640	320
250350	B/Ireland/00084761/2024		V1A.3a.2 (C.5.7)	2024-11-20	MDCK1	640	320	160	640	320	320
250518	B/Lisboa/114/2024		V1A.3a.2 (C.5.7)	2024-12-03	MDCK1	640	320	160	640	320	160
						Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025			

< relates to the lowest dilution of antiserum used

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table BV-8. Antigenic analyses of influenza B viruses (Victoria lineage) 2025-01-21

	Viruses	Other information	Genetic group	Collection Date	Passage	B/Austria /1359417/21 MDCK	B/Austria /1359417/21 Egg G141	B/Austria /1359417/21 Egg G141R	B/Stock /3/22 MDCK	B/Ghana /3778/2024 MDCK	B/Switzerland /329/24 Egg
					Ferret number	NIB F01/21	F40/21	F44/21	F28/22	F42/24	F33/24
					Genetic group	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	V1A.3a.2 (C.5.1)	V1A.3a.2 (C.5.6)
	REFERENCE VIRUSES										
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	IAT1/MDCK4	1280	640	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	640	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	2560	640	2560	1280	320	2560
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	IAT1/MDCK3	1280	640	320	1280	320	320
	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK2	1280	640	320	320	320	320
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4/E1	640	160	2560	640	320	2560
	TEST VIRUSES										
250961	B/Madagascar/03535/2024		V1A.3a.2 (C.2)	2024-11-11	MDCK1	1280	640	320	640	320	320
250957	B/Madagascar/03603/2024		V1A.3a.2 (C.2)	2024-11-20	MDCK1	1280	640	320	1280	640	320
251033	B/Stockholm/SE24-54631/2024		V1A.3a.2 (C.5)	2024-11-01	IAT1/MDCK1	1280	640	160	640	320	320
250904	B/Palencia/135/2024		V1A.3a.2 (C.5.1)	2024-11-02	MDCK1	1280	640	320	1280	640	320
250832	B/LaRioja/2265/2024		V1A.3a.2 (C.5.1)	2024-11-05	MDCK1	1280	640	320	1280	640	640
250830	B/LaRioja/2264/2024		V1A.3a.2 (C.5.1)	2024-11-06	MDCK1	1280	320	160	640	320	160
250902	B/Burgos/183/2024		V1A.3a.2 (C.5.1)	2024-11-07	MDCK1	1280	640	320	1280	640	320
250828	B/LaRioja/2263/2024		V1A.3a.2 (C.5.1)	2024-11-07	MDCK1	1280	640	320	1280	640	320
250826	B/Navarra/2204/2024		V1A.3a.2 (C.5.1)	2024-11-14	MDCK1	1280	640	320	640	640	320
250898	B/Valladolid/120/2024		V1A.3a.2 (C.5.1)	2024-11-20	MDCK1	1280	640	320	1280	640	320
250895	B/Salamanca/149/2024		V1A.3a.2 (C.5.1)	2024-11-24	MDCK1	1280	640	320	1280	640	640
250894	B/Salamanca/152/2024		V1A.3a.2 (C.5.1)	2024-11-26	MDCK1	1280	640	320	1280	640	640
250890	B/Burgos/175/2024		V1A.3a.2 (C.5.1)	2024-11-27	MDCK1	1280	640	320	1280	640	320
250891	B/Salamanca/155/2024		V1A.3a.2 (C.5.1)	2024-11-27	MDCK1	1280	640	320	1280	640	320
250888	B/Burgos/180/2024		V1A.3a.2 (C.5.1)	2024-11-29	MDCK1	1280	640	320	640	640	320
250786	B/Switzerland/6998/2024		V1A.3a.2 (C.5.1)	2024-12-03	MDCK2	1280	640	320	1280	640	320
251110	B/Bayern/24/2024		V1A.3a.2 (C.5.1)	2024-12-04	P1/MDCK1	1280	640	320	1280	640	320
250887	B/Salamanca/191/2024		V1A.3a.2 (C.5.1)	2024-12-06	MDCK1	1280	640	640	1280	640	640
250886	B/Salamanca/194/2024		V1A.3a.2 (C.5.1)	2024-12-06	MDCK1	1280	640	320	1280	640	320
251109	B/Nordrhein-Westfalen/8/2024		V1A.3a.2 (C.5.1)	2024-12-06	P1/MDCK1	1280	640	320	1280	640	320
250884	B/Salamanca/192/2024		V1A.3a.2 (C.5.1)	2024-12-07	MDCK1	1280	640	320	1280	640	640
251107	B/Bayern/25/2024		V1A.3a.2 (C.5.1)	2024-12-12	P1/MDCK1	1280	640	640	1280	640	640
250189	B/Qatar/85029/2024		V1A.3a.2 (C.5.6)	2024-10-28	MDCK1	1280	1280	640	1280	640	320
250280	B/Qatar/88540/2024		V1A.3a.2 (C.5.6)	2024-11-08	MDCK2	1280	640	320	1280	640	640
250897	B/Salamanca/146/2024		V1A.3a.2 (C.5.6)	2024-11-23	MDCK1	1280	640	320	1280	640	320
250896	B/Salamanca/147/2024		V1A.3a.2 (C.5.6)	2024-11-24	MDCK1	1280	640	320	1280	640	320
251111	B/Brandenburg/10/2024		V1A.3a.2 (C.5.6)	2024-11-26	P1/MDCK1	1280	640	320	640	640	320
250955	B/Madagascar/03790/2024		V1A.3a.2 (C.5.6)	2024-12-05	MDCK1	1280	1280	640	1280	640	640
250885	B/Salamanca/196/2024		V1A.3a.2 (C.5.6)	2024-12-06	MDCK1	1280	640	320	1280	640	640
251108	B/Brandenburg/11/2024		V1A.3a.2 (C.5.6)	2024-12-09	P2/MDCK1	1280	640	320	1280	640	320
250783	B/Switzerland/5509/2024		V1A.3a.2 (C.5.6)	2024-12-10	MDCK2	1280	640	320	1280	640	640
251106	B/Thuringen/16/2024		V1A.3a.2 (C.5.6)	2024-12-16	P1/MDCK1	1280	640	320	640	320	320
251104	B/Thuringen/17/2024		V1A.3a.2 (C.5.6)	2024-12-18	P1/MDCK1	1280	640	320	640	640	320
250187	B/Qatar/85546/2024		V1A.3a.2 (C.5.7)	2024-10-30	MDCK2	1280	640	320	1280	640	640
250963	B/Madagascar/03351/2024		V1A.3a.2 (C.5.7)	2024-11-07	MDCK1	1280	1280	640	1280	640	640
250962	B/Madagascar/03419/2024		V1A.3a.2 (C.5.7)	2024-11-08	MDCK1	1280	1280	640	1280	640	640
250960	B/Madagascar/03444/2024		V1A.3a.2 (C.5.7)	2024-11-12	MDCK1	1280	640	320	1280	640	320
250959	B/Madagascar/03512/2024		V1A.3a.2 (C.5.7)	2024-11-14	MDCK1	1280	640	320	1280	640	640
250899	B/Burgos/50189/2024		V1A.3a.2 (C.5.7)	2024-11-17	MDCK1	1280	640	320	1280	640	320
250958	B/Madagascar/03669/2024		V1A.3a.2 (C.5.7)	2024-11-25	MDCK1	1280	640	640	1280	1280	640
250953	B/Madagascar/03832/2024		V1A.3a.2 (C.5.7)	2024-12-06	MDCK1	1280	640	320	1280	640	320
250952	B/Madagascar/03830/2024		V1A.3a.2 (C.5.7)	2024-12-09	MDCK1	2560	1280	640	2560	1280	640
251105	B/Hessen/23/2024		V1A.3a.2 (C.5.7)	2024-12-16	P1/MDCK1	1280	640	320	1280	640	640

Haemagglutination inhibition titre — Post-infection ferret antiserum

< relates to the lowest dilution of antiserum used

Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025 (B/Austria/1359417/21 MDCK, NIB F01/21)

Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025 (B/Austria/1359417/21 Egg G141R, F44/21)

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table BV-9. Antigenic analyses of influenza B viruses (Victoria lineage) 2025-01-28

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre: Post-infection ferret antiserum					
					Passage	B/Austria /1359417/21 MDCK	B/Austria /1359417/21 Egg G141	B/Austria /1359417/21 Egg G141R	B/Stock /3/22 MDCK	B/Ghana /3778/2024 MDCK	B/Switzerland /329/24 Egg
					Ferret number	NIB F01/21	F40/21	F44/21	F28/22	F42/24	F33/24
					Genetic group	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	V1A.3a.2 (C.5.1)	V1A.3a.2 (C.5.6)
	REFERENCE VIRUSES										
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	1280	640	320	640	320	160
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	640	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	1280	640	2560	640	320	2560
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	1280	640	320	1280	640	320
	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK2	1280	320	320	640	320	320
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4/E1	640	160	1280	320	160	2560
	TEST VIRUSES										
250933	B/Salamanca/2654/2024		V1A.3a.2 (C.5.1)	2024-08-06	MDCK1	1280	640	320	1280	640	320
250932	B/Salamanca/2655/2024		V1A.3a.2 (C.5.1)	2024-08-13	MDCK1	1280	640	320	1280	640	320
251163	B/Israel/C38/2024		V1A.3a.2 (C.5.1)	2024-10-08	Px/MDCK1	640	640	320	640	640	320
251202	B/Lyon/CHU/0241678912/2024		V1A.3a.2 (C.5.1)	2024-10-28	MDCK2/MDCK1	1280	640	320	1280	320	320
250835	B/LaRioja/2261/2024		V1A.3a.2 (C.5.1)	2024-10-29	MDCK1	1280	640	320	1280	640	320
251161	B/Israel/C496/2024		V1A.3a.2 (C.5.1)	2024-11-05	Px/MDCK1	640	320	320	640	320	320
251158	B/Israel/C335/2024		V1A.3a.2 (C.5.1)	2024-11-18	Px/MDCK1	1280	640	320	640	320	320
251201	B/Lyon/0241861792/24		V1A.3a.2 (C.5.1)	2024-11-25	MDCK2/MDCK1	1280	640	320	1280	640	320
251198	B/Lyon/0241915708/2024		V1A.3a.2 (C.5.1)	2024-12-02	MDCK2/MDCK1	1280	320	320	1280	640	320
251197	B/StEtienne/0242013262/2025		V1A.3a.2 (C.5.1)	2024-12-02	MDCK2/MDCK1	1280	640	320	1280	640	640
251157	B/Israel/C476/2024		V1A.3a.2 (C.5.1)	2024-12-03	Px/MDCK1	1280	640	320	640	320	320
251196	B/Lyon/CHU/0241885619/2024		V1A.3a.2 (C.5.1)	2024-12-03	MDCK2/MDCK1	1280	640	320	1280	640	640
251156	B/Israel/R17283/2024		V1A.3a.2 (C.5.1)	2024-12-08	Px/MDCK1	1280	640	320	1280	640	320
251195	B/Grenoble/0241960701/2024		V1A.3a.2 (C.5.1)	2024-12-11	MDCK2/MDCK2	1280	640	320	1280	640	320
251154	B/Israel/C523/2024		V1A.3a.2 (C.5.1)	2024-12-11	Px/MDCK1	1280	640	320	640	640	320
251155	B/Israel/C588/2024		V1A.3a.2 (C.5.1)	2024-12-11	Px/MDCK1	1280	640	320	1280	640	320
251153	B/Israel/R17509/2024		V1A.3a.2 (C.5.1)	2024-12-11	Px/MDCK1	1280	640	320	1280	640	320
251150	B/Israel/C527/2024		V1A.3a.2 (C.5.1)	2024-12-15	Px/MDCK1	1280	640	320	640	320	320
251151	B/Israel/C569/2024		V1A.3a.2 (C.5.1)	2024-12-15	Px/MDCK1	1280	640	320	640	320	320
251149	B/Israel/R17664/2024		V1A.3a.2 (C.5.1)	2024-12-15	Px/MDCK1	640	640	320	640	320	320
251193	B/Lyon/0242018428/2025		V1A.3a.2 (C.5.1)	2024-12-23	MDCK2/MDCK1	1280	640	320	640	640	320
251192	B/Lyon/CHU/0250000165/2025		V1A.3a.2 (C.5.1)	2025-01-01	MDCK2/MDCK1	1280	640	320	640	320	320
250924	B/Burgos/5/2024		V1A.3a.2 (C.5.6)	2024-09-22	MDCK1	1280	640	320	1280	640	320
251160	B/Israel/C172/2024		V1A.3a.2 (C.5.6)	2024-11-05	Px/MDCK1	640	640	320	1280	320	320
251100	B/Bratislava/9734/2024		V1A.3a.2 (C.5.6)	2024-11-25	MDCKx/MDCK1	1280	640	320	640	640	640
251200	B/Toulon/0241847959/2024		V1A.3a.2 (C.5.6)	2024-11-25	MDCK2/MDCK1	1280	640	320	640	320	320
251098	B/Bratislava/9904/2024		V1A.3a.2 (C.5.6)	2024-12-02	MDCKx/MDCK1	1280	640	320	640	640	320
251094	B/Bratislava/10189/2024		V1A.3a.2 (C.5.6)	2024-12-11	MDCKx/MDCK1	1280	640	320	640	640	640
251093	B/Bratislava/10190/2024		V1A.3a.2 (C.5.6)	2024-12-11	MDCKx/MDCK1	1280	640	320	640	640	320
251152	B/Israel/C584/2024		V1A.3a.2 (C.5.6)	2024-12-15	Px/MDCK1	1280	640	320	1280	640	320
251087	B/Bratislava/10330/2024		V1A.3a.2 (C.5.6)	2024-12-17	MDCK1	1280	640	320	640	640	640
251086	B/Bratislava/10395/2024		V1A.3a.2 (C.5.6)	2024-12-18	MDCK1	1280	640	320	640	320	320
250929	B/Salamanca/2651/2024		V1A.3a.2 (C.5.7)	2024-09-05	MDCK1	1280	640	320	1280	640	320
251203	B/Lyon/CHU 0241454186/2024		V1A.3a.2 (C.5.7)	2024-09-18	MDCK2/MDCK1	1280	640	320	1280	640	320
251162	B/Israel/C109/2024		V1A.3a.2 (C.5.7)	2024-10-29	Px/MDCK1	1280	640	320	1280	640	320
251159	B/Israel/C207/2024		V1A.3a.2 (C.5.7)	2024-11-07	Px/MDCK1	1280	640	320	1280	640	320
251101	B/Bratislava/9573/2024		V1A.3a.2 (C.5.7)	2024-11-19	MDCKx/MDCK1	1280	640	320	1280	640	320
251099	B/Bratislava/9847/2024		V1A.3a.2 (C.5.7)	2024-11-27	MDCKx/MDCK1	1280	640	320	1280	640	320
251096	B/Bratislava/9906/2024		V1A.3a.2 (C.5.7)	2024-12-02	MDCKx/MDCK1	1280	640	320	1280	640	320
251199	B/Lyon/0241915458/2024		V1A.3a.2 (C.5.7)	2024-12-02	MDCK2/MDCK1	1280	640	320	1280	640	640
250954	B/Madagascar/03804/2024		V1A.3a.2 (C.5.7)	2024-12-06	MDCK3	1280	640	320	1280	640	320
251092	B/Bratislava/10191/2024		V1A.3a.2 (C.5.7)	2024-12-11	MDCKx/MDCK1	1280	640	320	1280	640	320
251194	B/Nice/0241957308/2024		V1A.3a.2 (C.5.7)	2024-12-12	MDCK2/MDCK1	1280	640	320	640	320	320
251088	B/Bratislava/10329/2024		V1A.3a.2 (C.5.7)	2024-12-17	MDCK1	1280	640	320	1280	640	320
	< relates to the lowest dilution of antiserum used					Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025			

Legend: < 4-fold; 4-fold; 8-fold; > 8-fold; < not recognised by the antiserum

Table BV-10. Antigenic analyses of influenza B viruses (Victoria lineage) 2025-02-04

	Viruses	Other information	Genetic group	Collection Date	Passage / Ferret number / Genetic group	B/Austria /1359417/21 MDCK NIB F01/21 V1A.3a.2 (C)	B/Austria /1359417/21 Egg G141 F40/21 V1A.3a.2 (C)	B/Austria /1359417/21 Egg G141R F44/21 V1A.3a.2 (C)	B/Stock /3/22 MDCK F28/22 V1A.3a.2 (C.5)	B/Ghana /3778/2024 MDCK F42/24 V1A.3a.2 (C.5.1)	B/Switzerland /329/24 Egg F33/24 V1A.3a.2 (C.5.6)
						Haemagglutination inhibition titre — Post-infection ferret antiserum					
	REFERENCE VIRUSES										
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	IAT1/MDCK4	1280	640	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	640	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	1280	320	2560	640	320	2560
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	IAT1/MDCK3	640	640	320	1280	640	320
	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK2	640	320	160	640	320	160
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4/E1	640	160	1280	640	160	2560
	TEST VIRUSES										
251584	B/France/GES-RELAB-IPP13350/2024		V1A.3a.2 (C.3)	2024-11-19	)CK1/MDCK1	1280	320	160	320	160	160
251372	B/Slovenia/NLZOH-8008-21-2116/2024		V1A.3a.2 (C.5)	2024-12-11	)CKx/MDCK1	640	320	160	640	320	160
251558	B/Denmark/118/2024		V1A.3a.2 (C.5)	2024-12-13	IAT2/MDCK1	640	320	160	640	320	320
251586	B/France/IDF-IPP12982/2024		V1A.3a.2 (C.5.1)	2024-11-18	)CK1/MDCK1	640	640	320	640	320	320
251577	B/France/GES-RELAB-IPP14024/2024		V1A.3a.2 (C.5.1)	2024-12-03	)CK2/MDCK1	1280	320	320	640	320	320
251573	B/France/IDF-IPP14060/2024		V1A.3a.2 (C.5.1)	2024-12-09	)CK1/MDCK1	640	320	160	640	320	320
251370	B/Slovenia/NLZOH-8008-21-2134/2024		V1A.3a.2 (C.5.1)	2024-12-12	)CK1/MDCK1	1280	640	320	640	640	320
251351	B/Switzerland/9374/2024		V1A.3a.2 (C.5.1)	2024-12-12	MDCK1	1280	640	320	1280	640	320
251569	B/France/BRE-IPP14535/2024		V1A.3a.2 (C.5.1)	2024-12-16	)CK1/MDCK1	640	320	320	640	320	320
251146	B/Israel/R17892/2024		V1A.3a.2 (C.5.1)	2024-12-18	Px/MDCK1	640	320	320	640	320	320
251349	B/Switzerland/5245/2025		V1A.3a.2 (C.5.1)	2025-01-06	MDCK1	1280	640	320	640	640	320
251347	B/Switzerland/7265/2025		V1A.3a.2 (C.5.1)	2025-01-09	MDCK1	640	640	320	640	640	320
251555	B/Denmark/137/2024		V1A.3a.2 (C.5.1)	2025-01-15	IAT2/MDCK1	1280	320	320	640	320	160
251589	B/France/HDF-RELAB-IPP11770/2024		V1A.3a.2 (C.5.6)	2024-10-11	)CK1/MDCK1	640	320	160	640	320	320
251588	B/France/GES-RELAB-IPP12527/2024		V1A.3a.2 (C.5.6)	2024-10-29	)CK1/MDCK1	640	320	160	640	320	320
251583	B/France/IDF-IPP13180/2024		V1A.3a.2 (C.5.6)	2024-11-22	)CK1/MDCK1	640	640	320	640	320	320
251581	B/France/IDF-IPP13811/2024		V1A.3a.2 (C.5.6)	2024-12-02	)CK1/MDCK1	640	320	160	640	320	320
251579	B/France/GES-RELAB-IPP14022/2024		V1A.3a.2 (C.5.6)	2024-12-03	)CK1/MDCK1	640	320	160	640	320	320
251575	B/France/GES-RELAB-IPP14025/2024		V1A.3a.2 (C.5.6)	2024-12-03	)CK1/MDCK1	640	320	160	640	320	320
251426	B/Parma/26/2024		V1A.3a.2 (C.5.6)	2024-12-04	)CK3/MDCK1	640	320	320	640	320	320
251571	B/France/NOR-IPP14509/2024		V1A.3a.2 (C.5.6)	2024-12-14	)CK1/MDCK1	640	640	320	640	320	320
251565	B/France/BFC-IPP14624/2024		V1A.3a.2 (C.5.6)	2024-12-18	)CK1/MDCK1	1280	640	320	1280	320	320
251337	B/Algeria/16875/2024		V1A.3a.2 (C.5.6)	2024-12-22	MDCK1	1280	640	320	640	640	320
251242	B/Luxembourg/8690855/2024		V1A.3a.2 (C.5.6)	2024-12-23	MDCK1	640	320	320	640	320	320
251350	B/Switzerland/7562/2024		V1A.3a.2 (C.5.6)	2024-12-23	MDCK1	1280	640	320	1280	640	320
251336	B/Algeria/278/2025		V1A.3a.2 (C.5.6)	2025-01-05	MDCK1	640	320	160	640	320	320
251368	B/Slovenia/NLZOH-8008-21-14/2025		V1A.3a.2 (C.5.6)	2025-01-06	)CKx/MDCK1	640	320	160	640	320	320
251348	B/Switzerland/5585/2025		V1A.3a.2 (C.5.6)	2025-01-09	MDCK1	1280	640	320	1280	640	320
251346	B/Switzerland/1196/2025		V1A.3a.2 (C.5.6)	2025-01-10	MDCK1	1280	640	320	1280	640	320
251557	B/Denmark/121/2024		V1A.3a.2 (C.5.6)	2025-01-15	IAT2/MDCK1	1280	640	320	1280	640	320
251556	B/Denmark/134/2024		V1A.3a.2 (C.5.6)	2025-01-15	IAT2/MDCK1	1280	640	320	1280	320	320
251591	B/France/IDF-RELAB-IPP11341/2024		V1A.3a.2 (C.5.7)	2024-09-24	)CK1/MDCK1	640	320	160	640	320	160
251387	B/Ireland/00085951/2024		V1A.3a.2 (C.5.7)	2024-11-17	MDCK1	1280	640	320	640	320	320
251247	B/Luxembourg/9678373/2024		V1A.3a.2 (C.5.7)	2024-12-02	MDCK1	1280	640	320	640	640	320
251244	B/Luxembourg/6990721/2024		V1A.3a.2 (C.5.7)	2024-12-16	MDCK1	640	320	320	640	320	320
251567	B/France/CVL-IPP14597/2024		V1A.3a.2 (C.5.7)	2024-12-17	)CK1/MDCK1	640	640	160	640	320	320
	< relates to the lowest dilution of antiserum used					Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025			

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum

Table BV-11. Antigenic analyses of influenza B viruses (Victoria lineage) 2025-02-12

	Viruses	Other information	Genetic group	Collection Date		Haemagglutination inhibition titre / Post-infection ferret antiserum					
					Passage	B/Austria /1359417/21 MDCK	B/Austria /1359417/21 Egg G141	B/Austria /1359417/21 Egg G141R	B/Stock /3/22 MDCK	B/Ghana /3778/2024 MDCK	3/Switzerland /329/24 Egg
					Ferret number	NIB F01/21	F40/21	F44/21	F28/22	F42/24	F33/24
					Genetic group	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	V1A.3a.2 (C.5.1)	V1A.3a.2 (C.5.6)
	REFERENCE VIRUSES										
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	iAT1/MDCK4	1280	640	320	640	320	160
	B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	640	320	640	320	320
	B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E4	1280	320	2560	640	160	1280
	B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	iAT1/MDCK3	640	640	320	640	640	320
	B/Ghana/3778/2024		V1A.3a.2 (C.5.1)	2024-07-11	MDCK2	640	320	320	640	320	160
	B/Switzerland/329/2024		V1A.3a.2 (C.5.6)	2024-01-16	E4/E1	640	320	2560	640	320	2560
	TEST VIRUSES										
250972	B/Madagascar/03215/2024		V1A.3a.2 (C.2)	2024-10-21	MDCK1	640	320	160	640	320	320
250971	B/Madagascar/03232/2024		V1A.3a.2 (C.2)	2024-10-25	MDCK1	1280	640	320	640	640	320
250970	B/Madagascar/03301/2024		V1A.3a.2 (C.2)	2024-10-28	MDCK1	1280	640	320	640	640	320
250969	B/Madagascar/03297/2024		V1A.3a.2 (C.2)	2024-10-29	MDCK1	1280	640	320	640	640	320
250966	B/Madagascar/03363/2024		V1A.3a.2 (C.2)	2024-11-04	MDCK1	640	320	160	640	320	320
250855	B/LaRioja/2270/2024		V1A.3a.2 (C.5.1)	2024-10-01	MDCK1	640	320	160	640	320	160
250853	B/LaRioja/2269/2024		V1A.3a.2 (C.5.1)	2024-10-03	MDCK1	1280	640	320	640	640	320
250851	B/LaRioja/2267/2024		V1A.3a.2 (C.5.1)	2024-10-05	MDCK1	640	320	160	640	320	320
250850	B/Navarra/2111/2024		V1A.3a.2 (C.5.1)	2024-10-09	MDCK1	640	320	160	640	320	320
250847	B/Navarra/1990/2024		V1A.3a.2 (C.5.1)	2024-10-18	MDCK1	1280	640	320	640	640	320
250843	B/LaRioja/2266/2024		V1A.3a.2 (C.5.1)	2024-10-22	MDCK1	640	320	160	640	320	160
250839	B/Andalucia/2231/2024		V1A.3a.2 (C.5.1)	2024-10-24	MDCK1	640	640	320	640	320	320
250838	B/LaRioja/2262/2024		V1A.3a.2 (C.5.1)	2024-10-24	MDCK1	640	320	160	320	160	160
251486	B/Poland/PL23/2024		V1A.3a.2 (C.5.1)	2024-12-16	MDCK1	640	320	160	640	320	160
251458	B/Montenegro/1039342502/2025		V1A.3a.2 (C.5.1)	2025-01-15	MDCK1	1280	640	320	1280	640	320
251489	B/Poland/PL20/2024		V1A.3a.2 (C.5.6)	2024-12-11	MDCK1	1280	1280	320	1280	640	640
251488	B/Poland/PL21/2024		V1A.3a.2 (C.5.6)	2024-12-11	MDCK1	1280	640	320	1280	640	320
251487	B/Poland/PL22/2024		V1A.3a.2 (C.5.6)	2024-12-11	MDCK1	1280	640	320	1280	640	320
251735	B/Romania_AR/586427/2024		V1A.3a.2 (C.5.6)	2024-12-23	iAT1/MDCK1	640	640	320	640	320	320
251483	B/Poland/61/2025		V1A.3a.2 (C.5.6)	2025-01-08	MDCK1	320	320	80	320	160	160
251440	B/Georgia/61/2025		V1A.3a.2 (C.5.6)	2025-01-13	MDCK1	1280	640	320	640	640	640
250852	B/LaRioja/2268/2024		V1A.3a.2 (C.5.7)	2024-10-04	MDCK1	640	320	160	640	320	160
250968	B/Madagascar/03303/2024		V1A.3a.2 (C.5.7)	2024-10-31	MDCK1	640	640	160	640	320	320
250967	B/Madagascar/03347/2024		V1A.3a.2 (C.5.7)	2024-11-04	MDCK1	1280	640	320	1280	640	320
250965	B/Madagascar/03367/2024		V1A.3a.2 (C.5.7)	2024-11-05	MDCK1	1280	640	320	640	640	320
250964	B/Madagascar/03422/2024		V1A.3a.2 (C.5.7)	2024-11-06	MDCK1	1280	640	320	1280	640	320
251737	B/Romania_BB/584246/2024		V1A.3a.2 (C.5.7)	2024-11-11	iAT1/MDCK1	1280	640	320	640	640	320
250956	B/Madagascar/03662/2024		V1A.3a.2 (C.5.7)	2024-11-26	MDCK1	1280	640	320	1280	640	320
251736	B/Romania_AR/586426/2024		V1A.3a.2 (C.5.7)	2024-12-23	iAT1/MDCK1	1280	640	320	1280	320	320
	< relates to the lowest dilution of antiserum used					Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025		Vaccine NH 2023-24 SH 2024 NH 2024-25 SH 2025			

Legend: < 4-fold; 4-fold; 8-fold; > 8-fold; < not recognised by the antiserum

Fold-reduction table: B/Victoria

		<4-fold difference		4-fold difference		>4-fold difference		
Reference Virus	Clade	Number	Percentage	Number	Percentage	Number	Percentage	Total
B/Austria/1359417/2021 Cell	V1A.3a.2 (C)	252	99.6	1	0.4	0	0.0	253
B/Austria/1359417/2021 G141 Egg	V1A.3a.2 (C)	251	99.2	2	0.8	0	0.0	253
B/Austria/1359417/2021 G141R Egg	V1A.3a.2 (C)	0	0.0	9	3.6	244	96.4	253
B/Stockholm/3/2022 Cell	V1A.3a.2 (C.5)	252	99.6	1	0.4	0	0.0	253
B/Ghana/3778/2024 Cell	V1A.3a.2 (C.5.1)	252	100.0	0	0.0	0	0.0	252
B/Switzerland/329/2024 Egg	V1A.3a.2 (C.5.6)	0	0.0	31	12.3	222	87.7	253

B/Victoria: Egg isolates

Virus	Status	Genetic group (subclade) [Additional subst.]	Comment
B/Catalonia/3514402NS/2023	BX-129 (1-way fail)	V1A.3a.2 (C.5) [D197E, E128K, A154T, V117I, K326R]	Egg adaptation: (no change)
B/Catalonia/2279261NS/2023	BX-123 (1-way pass, 2-way fail)	V1A.3a.2 (C.5.1) [E183K]	Egg adaptation: (no change)
B/Netherlands/10335/2023	BX-127 (1-way pass)	V1A.3a.2 (C.2) [E128K, A221T, T182A, D197E, T221A]	Egg adaptation: (no change)
B/Norway/5216/2023	available to send out	V1A.3a.2 (C.3) [E128K, A154E, S208P]	Egg adaptation: (no change)
B/Norway/3749/2023	available to send out	V1A.3a.2 (C.3) [E128K, A154E, S208P]	Egg adaptation: (no change)
B/Switzerland/53639/2023	available to send out	V1A.3a.2 (C.3) [E128K, A154E, S208P]	Egg adaptation: (no change)
B/Greece/5509/2024	available to send out	V1A.3a.2 (C.5.7) [E128G, D129N, E183K]	Egg adaptation: (no change)
B/Switzerland/329/2024	sent out recently	V1A.3a.2 (C.5.6) [D129N, T199A]	Egg adaptation: (G141R)

B/Victoria: Reagents and references

Virus	Genetic group	Virus passage	Ferret ID
B/Austria/1359417/2021	V1A.3a.2 (C)	SIAT1/MDCK4	NIB F01/21
B/Austria/1359417/2021 G141	V1A.3a.2 (C)	E3/E4	F40/21
B/Austria/1359417/2021 G141R	V1A.3a.2 (C)	E3/E5	F44/21
B/Stockholm/3/2022	V1A.3a.2 (C.5)	SIAT1/MDCK3	F28/22
B/Ghana/3778/2024	V1A.3a.2 (C.5.1)	MDCK2	F42/242
B/Switzerland/329/2024	V1A.3a.2 (C.5.6)	E4/E1	F33/24

Summary: B/Victoria

Genetic analyses

The SH 2024 winter season was characterised by cocirculation of subclades V1A.3a.2 C.5.1, C.5.4, C.5.6, C.5.7 and basal C.5, with predominance of C.5.7 and C.5.1 towards the end of the season. During this reporting period, only a minority of B/Victoria viruses were detected and characterised in Europe. Since September 2024, 90% of B/Victoria viruses detected belong to subclades C.5.1, C.5.6 or C.5.7.

Within V1A.3a.2 subclade C.5 (characterised by D197E), subclade C.5.1 with E183K represented by B/Catalonia/2279261NS/2023 is the dominant subclade (~40%) predominating in the Americas, Spain, France, UK, Italy, Germany, the Netherlands and Israel. The second most frequent clades are C.5.6 with D129N (reference B/Switzerland/329/2024) predominating in Canada, Colombia, Senegal, South Africa, Poland, Finland, Ukraine, Switzerland, Russian Federation, the Middle East, India and Japan; and C.5.7 with E183K and E128G represented by B/Guangxi-Beiliu/2298/2023 predominating in Ireland, Portugal, Sweden, Madagascar, Ivory Coast, Zambia, South East Asia, China and Australia.
Outside of C.5 viruses, other subclades were detected with very low frequency, such as C.3 (E128K, A154E, S208P, B/Moldova/2030521/2023) in Norway and Pakistan and C.2 with T182A, D197E, T221A (reference B/Netherlands/10335/2023) only in viruses from Madagascar.

No B/Yamagata lineage viruses have been detected since March 2020, except for a reported detection in the Netherlands which has been unable to be confirmed by lineage-specific RT-PCR or sequencing.

Antigenic analyses

All V1A.3a.2 viruses tested were well-recognised by antisera raised against B/Austria/1359417/2021 and -like viruses. Antisera against egg-based B/Switzerland/329/2024 reference, a representative of subclade C.5.6, showed significant reduced reactivity against the majority of samples tested.

Susceptibility to antivirals

Table AV1. Antiviral (NAI) testing of isolates with collection dates after 2024-08-31*

Month of Collection	Number of viruses tested for phenotype[1]								
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	Totals
2024									
September	6		12		5				23
October	29	1	11		8				49
November	19	1	22		12				54
December	12		7		8				27
2025									
January	0		0		0				0
Totals	66	2	52	0	33	0	0	0	153

[1] Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2025-02-14

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
A/Norway/07557/2024	H1N1pdm	07-Oct-24	566.7		H275Y
A/HongKong/2860/2024	H1N1pdm	06-Nov-24	45.55		H275Y MIX

Table AV2. Antiviral (BXM) testing of isolates with collection dates after 2024-08-31*

Month of Collection	Number of viruses tested for phenotype[1]								
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	Totals
2024									
September	7		8		4				19
October	6		4		4				14
November	1		2		2				5
December	0		0		0				0
2025									
January	0		0		0				0
Totals	14	0	14	0	10	0	0	0	38

[1] Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2025-02-14

Influenza viruses received at the WIC with collection dates from 1st September 2024 to 7th February 2025 were assessed for phenotypic and/or genotypic susceptibility to antivirals. Of these, 68 A/H1N1, 52 A/H3N2 and 33 B/Victoria viruses were phenotypically assessed against oseltamivir and zanamivir. Most viruses showed Normal Inhibition (NI) by both NAIs except for two A/H1N1 showing HRI with substitution H275Y alone or an H275H/Y mixture.

Phenotypic testing for susceptibility to baloxavir marboxil was performed for 14 A/H1N1, 14 A/H3N2 viruses and 10 B/Victoria viruses, with all of them showing Normal Inhibition.

Genotypic assessment of 168 A/H1N1, 211 A/H3N2 and 197 B/Victoria neuraminidase (NA) gene sequences found some viruses with markers associated with reduced susceptibility to NAI: two A/H1N1 viruses with substitution H275Y, two with D199N, two with I223V and five A/H1N1 viruses with S247N (virus isolation and phenotypic tests in progress).

For 150 A/H1N1, 206 A/H3N2 and 188 B/Victoria viruses where PA gene sequencing was successful, no markers associated with reduced inhibition by baloxavir marboxil were identified, except for two A/H1N1 viruses: one with substitution E23G and another virus with E199G (virus isolation and phenotypic test in progress).